Some Linear Algebra

In this class we'll study functions $f : \mathbb{R}^n \to \mathbb{R}^m$. In general, this is harder than the case for $n > 1$ because the limit of a function at a point can (and should) be taken from infinitely many directions, and not just two.

Review

The space $\mathbb{R}^n = \{(x_1, \ldots, x_n) : x_i \in \mathbb{R}\}$ is an n-dimensional real vector space, with pointwise addition and scalar multiplication. It has a standard basis given by vectors

$$e_i = (0, \ldots, 0, \underbrace{1}_{i\text{th position}}, 0, \ldots, 0), \quad i = 1, \ldots, n$$

so every vector $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$ can be represented uniquely by $x = \sum_{i=1}^n x_i e_i$. Choosing bases for $\mathbb{R}^n$ and $\mathbb{R}^m$ is in general an arbitrary choice, though picking them appropriately for the domain and codomain of a linear map can make their geometry more apparent.

Dot Product and Norms

In general for this course, everything will work for finite dimensional real vector spaces with norms (and possibly inner products), not just $\mathbb{R}^n$.

Definition

$\mathbb{R}^n$ comes equipped with a standard (positive definite) inner product, the dot product, given by $x \cdot y = \sum_{k=1}^n y_k x_k$ for $x = (x_1, \ldots, x_n), y = (y_1, \ldots, y_n) \in \mathbb{R}^n$. This satisfies, for all $x, y, z \in \mathbb{R}^n, t \in \mathbb{R}$,
(1) $x \cdot y = y \cdot x$ (symmetry)
(2) $(x + y) \cdot z = x \cdot z + y \cdot z$, $x \cdot (y + z) = x \cdot y + x \cdot z$, $(tx) \cdot y = x \cdot (ty) = t(x \cdot y)$ (bilinearity)
(3) $x \cdot x \geq 0$ with equality if and only if $x = 0$ (positive definiteness)
We define the Euclidean norm $||x|| = \sqrt{x \cdot x}$. This satisfies
(1) Expanding the definitions, $||tx|| = |t| \; ||x||$.
(2) By (3), $||x|| \geq 0$ with equality if and only if $x = 0$.

Theorem (Cauchy-Schwarz)

For $x, y \in \mathbb{R}^n$, $|x \cdot y| \leq ||x|| \; ||y||$ with equality if and only if x and y are linearly dependent.
Proof:
The theorem is clear if $x = 0$ or $y = 0$, so assume they are nonzero. By (2) and (3),

$$0 \leq ||x - ty||^2 = (x - ty) \cdot (x - ty) = ||x||^2 - 2t(x \cdot y) + t^2 ||y||^2$$

This is a parabola in t opening upward, so by high school algebra, its discriminant is non-positive, which expanded out tells us $4(x \cdot y)^2 \leq 4||x||^2||y||^2$, which gives the result after dividing by 4 and taking square roots. To get equality, we have $||x - ty||^2 = 0$ if and only if $x = ty$, i.e. they are linearly dependent. ■

Corollary (Triangle Inequality)

For all $x, y \in \mathbb{R}^n$, $||x + y|| \leq ||x|| + ||y||$.
Proof:

$$||x + y||^2 = (x + y) \cdot (x + y) = ||x||^2 + 2(x \cdot y) + ||y||^2 \overbrace{\leq}^{\text{CS}} ||x||^2 + 2||x|| ||y|| + ||y||^2 = (||x|| + ||y||)^2 \quad ■$$

Corollary (Angle)

Let $x, y \in \mathbb{R}^n$. Then $-1 \leq \frac{x \cdot y}{||x|| \; ||y||} \leq 1$, so there exists a unique $\theta \in [0, \pi]$ such that $x \cdot y = ||x|| \; ||y|| \cos \theta$, called the angle between x and y. ■

Example
If $\theta = 0$, x is a positive multiple of y, and if $\theta = \pi$, then x is a negative multiple of y, where we see they are orthogonal in these cases. Otherwise, $\text{Span}\{x, y\}$ is a 2-dimensional subspace of $\mathbb{R}^n$, so thinking of it as a copy of $\mathbb{R}^2$, we recover the law of cosines:

$$||x-y||^2 = (x-y)\cdot(x-y) = ||x||^2 + ||y||^2 - 2(x\cdot y) = ||x||^2 + ||y||^2 - 2||x||\,||y||\cos\theta$$

which if we think of $a = ||x||$, $b = ||y||$ as legs of a triangle, with $c = ||x-y||$ the length of the other side, we see $c^2 = a^2 + b^2 - 2ab\cos\theta$.

Example
As an application of the angle function, let $a, b \in \mathbb{R}^n$. The hyperplane P passing through a and orthogonal to b is the set $P = \{x \in \mathbb{R}^n : (x-a)\cdot b = 0\}$, which if $a = 0$ (or $0 \in P$) this is the $(n-1)$-dimensional subspace called the orthogonal complement to P.

Definition
A norm $||\cdot||$ on $\mathbb{R}^n$ is a map $||\cdot|| : \mathbb{R}^n \to \mathbb{R}^{\geq 0}$ satisfying, for all $x, y \in \mathbb{R}^n, t \in \mathbb{R}$
(1) $||x|| = 0$ if and only if $x = 0$
(2) $||tx|| = |t|\,||x||$
(3) $||x+y|| \leq ||x|| + ||y||$ (triangle inequality)

Example
The ℓ^1 norm $||x||_1 = \sum_{k=1}^n |x_k|$, ℓ^p norms $(1 < p < \infty)$ $||x_p|| = (\sum_{k=1}^n |x_k|^p)^{1/p}$, and ℓ^∞ norm (also called the sup norm) $||x||_\infty = \max_{k=1}^n |x_k|$ are all norms on $\mathbb{R}^n$. These are all the same if $n = 1$, but different otherwise. The fact the ℓ^1 norm is a norm can be checked easily, the triangle inequality of the ℓ^p norms is not obvious, and the ℓ^∞ norm is called that because it is the limit as $p \to \infty$ of the ℓ^p norms.

Linear Maps and Matrices

Since $\mathbb{R}^n$ is a vector space (and so is $\mathbb{R}^m$), there is a natural class of maps from $\mathbb{R}^n$ to $\mathbb{R}^m$, namely the linear maps (those preserving linear combinations): the maps $T : \mathbb{R}^n \to \mathbb{R}^m$ such that $T(\sum_{i=1}^k c_i x_i) = \sum_{i=1}^k c_i T(x_i)$ for all $c_1, \ldots, c_k \in \mathbb{R}$, $x_1, \ldots, x_k \in \mathbb{R}^n$.

Matrices
Let $(e_1, \ldots, e_n)$ be the standard basis of $\mathbb{R}^n$, and let $(f_1, \ldots, f_m)$ be the standard basis of $\mathbb{R}^m$. Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be linear, with $x \in \mathbb{R}^n$ and $y = T(x) \in \mathbb{R}^m$. Then we can write $y = \sum_{j=1}^m y_j f_j = T(\sum_{i=1}^n x_i e_i) = \sum_{i=1}^n x_i T(e_i)$ by linearity. Since $T(e_i) \in \mathbb{R}^m$, there exist unique scalars $A_{1i}, \ldots, A_{mi}$ such that $T(e_i) = \sum_{j=1}^m A_{ji} f_j$ for all $1 \leq i \leq n$, so $\sum_{j=1}^m y_j f_j = \sum_{i=1}^n x_i (\sum_{j=1}^m A_{ji} f_j)$, so by uniqueness of the basis representation, $y_j = \sum_{i=1}^n A_{ji} x_i$ for $1 \leq j \leq m$. Putting these into a matrix, $A = \{A_{ji}\}_{1 \leq j \leq m, 1 \leq i \leq n}$ is a $m \times n$ matrix. If we represent $y = (y_1, \ldots, y_m) \in \mathbb{R}^m$ as a $m \times 1$ matrix, and $x = (x_1, \ldots, x_n)$ as an $n \times 1$ matrix, then we see this means $y = T(x)$ is equivalent to

$$\underbrace{A}_{m\times n}\underbrace{x}_{n\times 1} = \underbrace{y}_{m\times 1}$$

A is called the matrix representation of T with respect to the standard bases of $\mathbb{R}^n$ and $\mathbb{R}^m$.

Remark
If we choose other bases $\{\widetilde{e}_1, \ldots, \widetilde{e}_n\}$ and $\{\widetilde{f}_1, \ldots, \widetilde{f}_m\}$ for $\mathbb{R}^n$ and $\mathbb{R}^m$, we get a different $m \times n$ matrix for T, defined by $\sum_{j=1}^m \widetilde{A}_{ji} \widetilde{f}_j = T(\widetilde{e}_i)$ for $1 \leq i \leq n$.

Matrix Multiplication
The rule for matrix multiplication is forced on us by the composition of linear maps. Let $T : \mathbb{R}^n \to \mathbb{R}^m$

and $S : \mathbb{R}^m \to \mathbb{R}^p$ be linear maps, with $T(x) = Ax, S(y) = By$ for all $x \in \mathbb{R}^n, y \in \mathbb{R}^m$ for A a unique $m \times n$ matrix, B a unique $p \times m$ matrix. Then $S \circ T : \mathbb{R}^n \to \mathbb{R}^p$ is a linear map, say $(S \circ T)(x) = Cx$ for C a unique $p \times n$ matrix. Let $x = (x_1, \dots, x_n) \in \mathbb{R}^n$. Then,

$$Cx = (S \circ T)(x) = S(T(x)) = S(Ax) = B(Ax)$$

We want $B(Ax) = (BA)x$. Write $y = Ax = (y_1, \dots, y_m) \in \mathbb{R}^m$, and let $w = (S \circ T)(x) = BAx = By$. Then,

$$\sum_{k=1}^{n} C_{\ell k} = w_\ell = \sum_{j=1}^{m} B_{\ell j} y_j = \sum_{j=1}^{m} B_{\ell j} (\sum_{k=1}^{n} A_{jk} x_k) = \sum_{k=1}^{n} (\sum_{j=1}^{m} B_{\ell j} A_{jk}) x_k$$

So by uniqueness $C_{\ell k} = \sum_{j=1}^{m} B_{\ell j} A_{jk}$, which is just matrix multiplication.

Definition
Let $M_{m \times n}$ denote the real $m \times n$ matrices. These form an mn-dimensional real vector space, with the natural inner product and induced norm given by $\langle A, B \rangle = \sum_{i=1}^{n} \sum_{j=1}^{m} A_{ij} B_{ij} = \operatorname{tr}(A^T B)$, with $||A||^2 = \langle A, A \rangle$ for $A, B \in M_{m \times n}$.

Proposition
Let $A \in M_{m \times n}, x \in \mathbb{R}^n$. Then $\underbrace{||Ax||}_{\mathbb{R}^m} \leq \underbrace{||A||}_{M_{m \times n}} \underbrace{||x||}_{\mathbb{R}^n}$.
Proof:
Write the rows of A as $A_1, \dots, A_m \in \mathbb{R}^n$. Then $Ax = (A_1 \cdot x, \dots, A_m \cdot x)$, so

$$||Ax||^2 = \sum_{i=1}^{m} (Ax)_i^2 = \sum_{i=1}^{m} (A_i \cdot x)^2 \overbrace{\leq}^{\text{CS}} \sum_{i=1}^{m} ||A_i||^2 \ ||x||^2 = ||x||^2 (\sum_{i=1}^{m} \sum_{j=1}^{n} A_{ij}^2) = ||A||^2 ||x||^2 \quad \blacksquare$$

We've shown $|| \cdot ||$ is a norm on $M_{m \times n}$ satisfying $||Ax|| \leq ||A|| \ ||x||$ for all $x \in \mathbb{R}^n$. There is another natural norm on $M_{m \times n}$, using the fact that $M_{m \times n}$ is really the space of linear maps $\mathbb{R}^n \to \mathbb{R}^m$:

Definition
Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be linear, and $A \in M_{m \times n}$ its matrix. Define the operator norm

$$||T||_{op} = \inf\{C \in \mathbb{R}^{>0} : ||T(x)|| \leq C||x|| \ \forall \ x \in \mathbb{R}^n\}$$

Proposition
The operator norm is well-defined and is a norm.
Proof:
It is well-defined since the set $E := \{C \in \mathbb{R}^{>0} : ||T(x)|| \leq C||x|| \ \forall \ x \in \mathbb{R}^n\}$ is bounded from below by 0, and it is nonempty since we can take $C = ||A|| \in E$ (where the desired estimate holds by the above proposition), so the infimum exists. We now prove it meets the desired properties of the norm.
(1) If $T = 0$, then $T(x) = 0$ for all $x \in \mathbb{R}^n$, so $||T(x)|| = 0$ whence $E = \mathbb{R}^{>0}$, so $||T||_{op} = \inf E = 0$. Conversely, suppose $T \neq 0$. Then $0 < ||A|| \in E$, so E has a positive element so its infimum is positive, so $||T||_{op} = \inf E > 0$.
(2) If $\lambda = 0$, then $\lambda T = 0$, so $0 = ||\lambda T||_{op} = |\lambda| \ ||T||_{op}$. On the other hand, if $\lambda \neq 0$ and $x \in \mathbb{R}^n$, then $|\lambda| > 0$, and $||(\lambda T)(x)|| = ||\lambda T(x)|| = |\lambda| \ ||T(x)||$, so we have $||T(x)|| \leq C||x||$ if and only if $||(\lambda T)(x)|| \leq |\lambda|C||x||$, so $|\lambda|E = \{C \in \mathbb{R}^{>0} : ||(\lambda T)(x)|| \leq |\lambda|C||x||\}$, and so taking infimums we get that $||\lambda T||_{op} = |\lambda| \ ||T||_{op}$.
(3) First, we prove $||T(x)|| \leq ||T||_{op}||x||$. By properties of the infimum, there exists a decreasing sequence $(C_k)_{k=1}^{\infty}$ contained in E converging to $||T||_{op}$, with $||T(x)|| \leq C_k||x||$ for all $x \in \mathbb{R}^n$ for each k (by definition of E). Taking the limit as $k \to \infty$, we get $||T(x)|| \leq ||T||_{op} \ ||x||$ for all $x \in \mathbb{R}^n$, as desired. Let $S : \mathbb{R}^n \to \mathbb{R}^m$ be another linear map. Then, by the triangle inequality for $|| \cdot ||$, we have

$$||(T + S)(x)|| \leq ||T(x)|| + ||S(x)|| \leq ||T||_{op} \ ||x|| + ||S||_{op} \ ||x|| = (||T||_{op} + ||S||_{op})||x||$$

and hence $||T||_{op} + ||S||_{op}$ is in the corresponding set E for the linear map $T + S$, so taking infimums, we get that $||T + S||_{op} \leq ||T||_{op} + ||S||_{op}$. ■

Topology of $\mathbb{R}^n$

Topology is the study of "closeness" in a space, and is how we'll approach notions of limits and continuity in $\mathbb{R}^n$.

Open and Closed Sets

Definition
The open ball of radius $r \in \mathbb{R}^{>0}$ centered at $x \in \mathbb{R}^n$ is the set $B_r(x) := \{y \in \mathbb{R}^n : ||y - x|| < r\}$ of all points a distance less than r from x. The closed ball of radius r centered at x is the set $\overline{B_r(x)} := \{y \in \mathbb{R}^n : ||y - x|| \leq r\}$ of all points a distance less than or equal to r from x.

Example
In $\mathbb{R}^1$, $B_r(x) = (x - r, x + r)$, $\overline{B_r(x)} = [x - r, x + r]$.

Definition
Let $U \subseteq \mathbb{R}^n$. We say U is open if and only if for all $x \in U$, there exists $\varepsilon > 0$ such that $B_\varepsilon(x) \subseteq U$. In essence, U is open if all points sufficiently close to any $x \in U$ are also in U.

Proposition
Open balls are open sets.
Proof:
Let $y \in B_r(x)$ for some $x \in \mathbb{R}^n, r > 0$. Define $d := ||y - x|| < r$, and take $\varepsilon := r - d > 0$. We claim $B_\varepsilon(y) \subseteq B_r(x)$. Indeed, let $z \in B_\varepsilon(y)$. Then $||z - x|| \leq ||z - y|| + ||y - x|| < \varepsilon + d = r$, so $z \in B_r(x)$, completing the proof of the claim and the proof. ■

Example
The empty set $\emptyset$ is vacuously an open set, and $\mathbb{R}^n$ is an open set.

Lemma
(1) If $(U_\alpha)_{\alpha \in A}$ is an arbitrary collection of open subsets of $\mathbb{R}^n$, $\bigcup_{\alpha \in A} U_\alpha$ is open (arbitrary unions of open sets are open).
(2) If $U_1, \ldots, U_k$ are open, then $\bigcap_{i=1}^k U_i$ is open (finite intersections of open sets are open).
Proof:
(1) Let $x \in \bigcup_{\alpha \in A} U_\alpha$. Then there exists $\alpha_0 \in A$ such that $x \in U_{\alpha_0}$. Since U_{α_0} is open, there exists $\varepsilon > 0$ such that $B_\varepsilon(x) \subseteq U_{\alpha_0} \subseteq \bigcup_{\alpha \in A} U_\alpha$, so $\bigcup_{\alpha \in A} U_\alpha$ is open.
(2) Let $x \in \bigcap_{i=1}^k U_i$. Then for all $i = 1, \ldots, k$, there exists $\varepsilon_i > 0$ such that $B_{\varepsilon_i}(x) \subseteq U_i$ since each U_i is open and contains x. Then $\varepsilon = \min_i \varepsilon_i$ satisfies $B_\varepsilon(x) \subseteq B_{\varepsilon_i}(x) \subseteq U_i$ for each $i = 1, \ldots, k$, so $B_{\varepsilon_i}(x) \subseteq \bigcap_{i=1}^k U_i$. ■

Example
Infinite intersections of open sets may not be open, for instance, in $\mathbb{R}^1$, take $U_i = (0, \frac{1}{1+\frac{1}{i}})$ for $i \in \mathbb{N}$. Then $\bigcap_{i=1}^\infty U_i = (0, 1]$, which is not open.

Definition
A subset $F \subseteq \mathbb{R}^n$ is closed if and only if $F^c = \mathbb{R}^n \setminus F$ is open.

Proposition
Closed balls are closed.
Proof:
Let $F = \overline{B_r(x)}$ for some $r > 0, x \in \mathbb{R}^n$. Let $y \in F^c$. Then we have $d := ||x-y|| > r$. Let $\varepsilon = d - r > 0$. We want to show $B_\varepsilon(y) \subseteq F^c$. Let $z \in B_\varepsilon(y)$. Therefore, $||z-y|| < \varepsilon$, so $d = ||x-y|| \leq ||x-z|| + ||z-y|| < ||x-z|| + \varepsilon$, so $||x-z|| > d - \varepsilon = r$, so $z \in F^c$. ■

Lemma
(1) If $F_1, \ldots, F_k$ are closed sets, then $\bigcup_{j=1}^k F_j$ is closed (finite unions of closed sets are closed).
(2) If $(F_\alpha)_{\alpha \in A}$ is an arbitrary collection of closed sets, then $\bigcap_{\alpha \in A} F_\alpha$ is closed (arbitrary intersections of closed sets are closed).
Proof:
Apply DeMorgan's laws $(F_1 \cup \cdots \cup F_k)^c = F_1^c \cap \cdots \cap F_k^c$ and $(\bigcap_{\alpha \in A} F_\alpha)^c = \bigcup_{\alpha \in A} F_\alpha^c$ to the lemma for open sets. ■

Example
$\emptyset$ and $\mathbb{R}^n$ are open and closed. In general, a set may be open, closed, both, or neither (for the last, an example is $[0,1) \subseteq \mathbb{R}$).

Interior, Closure, and Boundary

Definition
Let $A \subseteq \mathbb{R}^n$. Define the interior of A, denoted A° or $\mathrm{Int}(A)$ to be $A^\circ = \bigcup\{V \subseteq A : V \text{ open}\}$.

Remark
By the lemma in the last section, this is an open set, and it is the union of only subsets of A and so $A^\circ \subseteq A$. Moreover, a subset $U \subseteq A$ is open if and only if $U \subseteq A^\circ$, so A° is the "largest open set contained in A". Putting these together, we see $A = A^\circ$ if and only if A is open: $A^\circ \subseteq A$ in general, and if A is open then it is contained in A°, so $A \subseteq A^\circ$, so altogether $A = A^\circ$. Conversely, A° is open so if $A = A^\circ$ then A is open.

Definition
Let $A \subseteq \mathbb{R}^n$. Define the closure of A. denoted $\mathrm{Cl}(A)$ or $\overline{A}$ to be $\overline{A} = \bigcap\{F \supseteq A : F \text{ open}\}$ (notice that this is nonempty since it contains $\mathbb{R}^n$).

Remark
By the lemma in the last section, this is a closed set, and every set in the intersection contains A so $A \subseteq \overline{A}$. Moreover, a subset $F \supseteq A$ is closed if and only if $F \supseteq \overline{A}$, so $\overline{A}$ is the "smallest open set containing A". Putting these together, we have $A = \overline{A}$ if and only if A is closed.

Proposition
The closure of $B_\varepsilon(x)$ is $\overline{B_\varepsilon(x)}$ (so our notation is consistent).
Proof:
Let $A = \mathrm{Cl}\, B_\varepsilon(x)$. We want to show $A = \{y \in \mathbb{R}^n : ||y-x|| \leq \varepsilon\}$ (the right hand side of which we'll denote by $\overline{B_\varepsilon(x)}$). We know $\overline{B_\varepsilon(x)}$ is a closed set and it contains $B_\varepsilon(x)$. So by the remark above, $\overline{B_\varepsilon(x)} \supseteq A$. Thus we need to show $\overline{B_\varepsilon(x)} \subseteq A$. Suppose not. Thus, there exists $y \in \overline{B_\varepsilon(x)}$ such that $y \notin A$. Therefore, $y \notin B_\varepsilon(x) \subseteq A$. Therefore, we must have $||y-x|| = \varepsilon$ by definition of $\overline{B_\varepsilon(x)}$ combined with $y \notin B_\varepsilon(x)$. Since $y \notin A$, we must have $y \in A^c$, which is open since A is closed. Therefore, there exists $\delta > 0$ such that $B_\delta(y) \subseteq A$. Let $0 < t < \min\{\delta, \varepsilon\}$, and set $z = y + \frac{t(x-y)}{||x-y||}$, so $||z-y|| = t < \delta$ (since $\left\|\frac{x-y}{||x-y||}\right\| = 1$), and thus $z \in B_\delta(y) \subseteq A^c$. On the other hand,

$$x - z = x - y - t\frac{(x-y)}{||x-y||} = (x-y)(1 - \frac{t}{||x-y||})$$

so taking norms, we have

$$||x-z|| = ||x-y||\ |1-\frac{t}{||x-y||}| = |||x-y||-t|| = |\varepsilon - t| = \varepsilon - t < \varepsilon$$

and thus $z \in B_\varepsilon(x) \subseteq A$, which is a contradiction. ■

Definition
Let $A \subseteq \mathbb{R}^n$. We define the boundary of A, denoted $\mathrm{Bd}(A)$ or ∂A to be

$$\partial A = \{x \in \mathbb{R}^n : B_\varepsilon(x) \cap A \neq \emptyset \text{ and } B_\varepsilon(x) \cap A^c \neq \emptyset, \forall \varepsilon > 0\}$$

That is, $x \in \partial A$ if and only if there are points arbitrarily close to x that are in A and points arbitrarily close to x that are not in A. Intuitively, ∂A is the "edge" of A.

Proposition
Let $A \subseteq \mathbb{R}^n$. Then $\partial A = \overline{A} \setminus A^\circ$.
Proof:
This follows immediately from the following two claims:
(1) $x \in \overline{A}$ if and only if $B_\varepsilon(x) \cap A \neq \emptyset$ for all $\varepsilon > 0$.
(2) $x \notin A^\circ$ if and only if $B_\varepsilon(x) \cap A^c \neq \emptyset$ for all $\varepsilon > 0$.
To prove (1), in the forward direction let $x \in \overline{A}$, but suppose for the sake of contradiction there exists $\varepsilon_0 > 0$ such that $B_{\varepsilon_0}(x) \cap A = \emptyset$, that is $B_{\varepsilon_0}(x) \subseteq A^c$, so taking complements, $A \subseteq B_{\varepsilon_0}(x)^c$, which is closed, and so $\overline{A} \subseteq B_{\varepsilon_0}(x)^c$, so $\overline{A} \cap B_{\varepsilon_0}(x) = \emptyset$, which is a contradiction since $x \in \overline{A}$ and $x \in B_{\varepsilon_0}(x)$. Conversely, suppose contrapositively that $x \notin \overline{A}$. Therefore, $x \in \overline{A}^c$, which is open, so there exists $\varepsilon_0 > 0$ such that $B_{\varepsilon_0}(x) \subseteq \overline{A}^c$, therefore, $B_{\varepsilon_0}(x) \cap \overline{A} = \emptyset$ and hence $B_{\varepsilon_0}(x) \cap A = \emptyset$, completing the proof of the claim.
To prove (2), in the forward direction, suppose for the sake of contradiction $x \notin A^\circ$ but there exists $\varepsilon_0 > 0$ such that $B_{\varepsilon_0}(x) \cap A^c = \emptyset$, so $B_{\varepsilon_0}(x) \subseteq A$. But $B_{\varepsilon_0}(x)$ is an open set contained in A, so $B_{\varepsilon_0}(x) \subseteq A^\circ$, so $x \in A^\circ$, a contradiction. Conversely, contrapositively suppose $x \in A^\circ$. Since A° is open, there exists $\varepsilon_0 > 0$ such that $B_{\varepsilon_0}(x) \subseteq A^\circ \subseteq A$, hence $B_{\varepsilon_0}(x) \cap A^c = \emptyset$. ■

Sequences

Definition
A sequence of points in $\mathbb{R}^n$ is a map $\mathbb{N} \to \mathbb{R}^n$, denoted by $1, 2, 3, \dots \mapsto x_1, x_2, x_3, \dots$, commonly denoted by $(x_k)_{k\in\mathbb{N}}$ or (x_k) by abuse of notation.

Definition
Let $x \in \mathbb{R}^n$, and let (x_k) be a sequence in $\mathbb{R}^n$. We say (x_k) converges to x if and only if for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $k \geq N$ implies $||x_k - x|| < \varepsilon$, where we then write $\lim_{k\to\infty} x_k = x$.

Lemma
Suppose $\lim_{k\to\infty} x_k = x$ and $\lim_{k\to\infty} x_k = y$. Then $x = y$.
Proof:
Let $\varepsilon > 0$. Then there exists $N_1 \in \mathbb{N}$ such that $k \geq N_1$ implies $||x_k - x|| < \varepsilon/2$, and there exists $N_2 \in \mathbb{N}$ such that $k \geq N_2$ implies $||x_k - x|| < \varepsilon/2$. Let $k \geq \max\{N_1, N_2\}$. Then this implies

$$||x-y|| \leq ||x-x_k|| + ||x_k - y|| < \varepsilon/2 + \varepsilon/2 = \varepsilon$$

so $||x-y|| = 0$ so $x = y$. ■

Relationship between Sequences and Open and Closed Sets

Definition
Let $x \in \mathbb{R}^n$. A subset $U \subseteq \mathbb{R}^n$ is called a neighbourhood of x if and only if there exists $\varepsilon > 0$ such that

$B_\varepsilon(x) \subseteq U$, that is, if and only if U contains an open set containing x. We call U an open neighbourhood of x if it is open.

Lemma
Let (x_k) be a sequence in $\mathbb{R}^n$. Suppose $\lim_{k\to\infty} x_k = x$. Then any neighbourhood U of x contains x_k for all k sufficiently large, that is, there exists $N \in \mathbb{N}$ such that $k \geq N$ implies $x_k \in U$.
Proof:
By definition, there exists $\varepsilon_0 > 0$ such that $B_{\varepsilon_0}(x) \subseteq U$. Since $\lim_{k\to\infty} x_k = x$, there exists $N \in \mathbb{N}$ such that $k \geq N$ implies $||x - x_k|| < \varepsilon_0$, that is, $x_k \in B_{\varepsilon_0}(x) \subseteq U$. ■

Recall that $x \in \overline{A}$ if and only if $B_\varepsilon(x) \cap A \neq \emptyset$ for all $\varepsilon > 0$. Suppose $x \in \overline{A}$. By the above (take $\varepsilon = 1/k$), for all $k \in \mathbb{N}$, there exists $x_k \in A$ with $||x_k - x|| < 1/k$. This gives a sequence (x_k) contained in A satisfying $\lim_{k\to\infty} x_k = x$. Thus, if $x \in \overline{A}$, there exists a (non-unique) sequence in A converging to x (the constant sequence works if $x \in A$). The converse is true as well. Suppose (x_k) is a sequence in A that converges to x. Suppose that $x \notin \overline{A}$. Then $x \in \overline{A}^c$, which is open, so there exists $\varepsilon_0 > 0$ such that $B_{\varepsilon_0}(x) \subseteq \overline{A}^c$, so for k sufficiently large, $x_k \in B_{\varepsilon_0}(x) \subseteq \overline{A}^c$, so $x_k \notin \overline{A}$, which is a contradiction. Altogether we have proved:

Proposition
Let $A \subseteq \mathbb{R}^n$. Then $x \in \overline{A}$ if and only if there exists a sequence (x_k) contained in A converging to x. ■

Corollary
If A is closed and (x_k) is a convergent sequence in A with limit x, then $x \in A$.
Proof:
$\overline{A} = A$. ■

More Sequences

Definition
A sequence (x_k) is bounded if there exists $M > 0$ such that $||x_k|| \leq M$ for all $k \in \mathbb{N}$.

Definition
A sequence (x_k) is Cauchy if for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $k, \ell > N$ implies $||x_k - x_\ell|| < \varepsilon$, i.e., eventually all points in the sequence are as close together as we want.

Lemma
Any convergent sequence is Cauchy.
Proof:
Let (x_k) be a convergent sequence with limit x, and let $\varepsilon > 0$. Therefore, there exists $N \in \mathbb{N}$ such that if $k \geq N$, then $||x_k - x|| < \varepsilon/2$, so if $k, \ell \geq N$, then

$$||x_k - x_\ell|| \leq ||x_k - x|| + ||x - x_\ell|| < \varepsilon/2 + \varepsilon/2 = \varepsilon \quad ■$$

Definition
We say $\mathbb{R}^n$ is complete if every Cauchy sequence converges. In Math 147, you proved $\mathbb{R} = \mathbb{R}^1$ is complete. On Assignment 2, you will prove $\mathbb{R}^n$ is complete.

Lemma
All convergent sequences are bounded.
Proof:
Let (x_k) be a convergent sequences with limit x. With the choice $\varepsilon = 1$, there exists $N \in \mathbb{N}$ such that $k \geq N$ implies $||x_k - x|| < 1$, hence $||x_k|| \leq ||x_k - x|| + ||x|| < 1 + ||x||$. Let $M = \max\{||x_1||, \ldots, ||x_{N-1}||, ||x|| + 1\} > 0$ (this is well-defined since there are only finitely many terms). Then by construction, $||x_k|| \leq M$

for all $k \in \mathbb{N}$. ■

There are many bounded sequences which do not converge, for example $x_k = (-1)^k$ for all $k \in \mathbb{N}$. How badly does this fail? It turns out that any bounded sequence has a convergent subsequence.

Subsequences

Definition
Let (x_k) be a sequence in $\mathbb{R}^n$, and pick indices $1 \leq k_1 < k_2 < k_3 < \dots$. Then the sequence $(x_{k_j})_{j\in\mathbb{N}}$ is called a subsequence of (x_k).

Proposition
Let (x_k) be a convergent sequence with limit x. Any subsequence of (x_k) also converges to x.
Proof:
Let (x_{k_j}) be a subsequence of (x_k), and let $\varepsilon > 0$. There exists $N \in \mathbb{N}$ such that $\ell \geq N$ implies $||x_\ell - x|| < \varepsilon$. But $k_\ell \geq \ell$ for every $\ell \in \mathbb{N}$, so $\ell \geq N$ tells us $k_\ell \geq N$ implies $||x_{k_\ell} - x|| < \varepsilon$. ■

Example
If (x_k) does not converge, then it might not have any convergent subsequences, for example $x_k = k$ for all $k \in \mathbb{N}$, or it could have different subsequences converging to different limits, for example $x_k = (-1)^k$ for all $k \in \mathbb{N}$ has $x_{2k} \to 1$, $x_{2k+1} \to -1$ and x_{3k} does not converge as $k \to \infty$.

Theorem (Bolzano-Weierstrass)
Every bounded sequence has a convergent subsequence.
Remark:
This is highly non-unique.
Proof:
We prove the theorem by induction on n.
We first prove the $n = 1$ base case. Suppose (x_k) is a bounded sequence in $\mathbb{R}$. Then there exists $M > 0$ such that $x_k \in \overline{B_M(0)} = [-M, M]$ for all $k \in \mathbb{N}$. Define $I_1 = [-M, M] = [-M, 0] \cup [0, M]$. Since (x_k) is infinite, at least one of these intervals contains infinitely many elements of the sequence (x_k). Choose I_2 to be one of these intervals with infinitely many of these elements, and write $I_2 = [a_2, b_2] = [a_2, \frac{a_2+b_2}{2}] \cup [\frac{a_2+b_2}{2}, b_2]$. Since I_2 contains infinitely many elements of (x_k), at least one of these subintervals does, so pick one of them and call it I_3, then repeat in this way, getting closed bounded intervals $I_1 \supseteq \dots \supseteq I_k \supseteq \dots$ each containing infinitely many elements of (x_k), where the length of I_ℓ is $\frac{2M}{2^{\ell-1}}$. Now, for each $j \in \mathbb{N}$, pick $x_{k_j} \in I_j$, which is possible since each I_j contains infinitely many elements of (x_k). We claim $\bigcap_{k=1}^\infty I_k$ is nonempty with exactly one point. For each k, it is an interval of the form $[a_k, b_k] \supseteq [a_{k+1}, b_{k+1}]$, so $a_k \leq a_{k+1} < b_1$, $a_1 < b_{k+1} \leq b_k$, so (a_k) is an increasing sequence bounded above by b_1 and (b_k) is a decreasing sequence bounded below by a_1, so by the monotone convergence theorem, there exist a, b such that $a = \lim_{k\to\infty} a_k$ and $b = \lim_{k\to\infty} b_k$. Moreover, $a, b \in \bigcap_{k=1}^\infty I_k$ since the I_k's are nested, so $a_{k+m} \in I_m$ for all $k, m \in \mathbb{N}$. Because I_m is closed (it is a closed interval), this shows $\lim_{k\to\infty} a_{k+m} \in I_m$ for all $m \in \mathbb{N}$, and symmetrically for b, so $\bigcap_{k=1}^\infty I_k$ is nonempty. Moreover, for all $\ell \geq k \in \mathbb{N}$, we know $a_\ell \leq b_\ell \leq b_k$, so taking limits we get $a \leq b_k$ for all $k \in \mathbb{N}$, so taking limits again we have $a \leq b$. Therefore, since $|b_k - a_k|$ is the length of I_k, which is $\frac{2M}{2^{k-1}}$, which goes to 0 as $k \to \infty$, and we have $0 \leq |b - a| \leq |b_k - a_k|$ for all $k \in \mathbb{N}$, altogether by the squeeze theorem we have $a = b$. To show this is the unique point in $\bigcap_{k=1}^\infty I_k$, suppose $x \in \bigcap_{k=1}^\infty I_k$. Then $a_k \leq x \leq b_k$ for all $k \in \mathbb{N}$, so taking the limit $k \to \infty$, we have $a \leq x \leq b$, which since $a = b$, forces $a = x = b$, so $\bigcap_{k=1}^\infty I_k$ has exactly one point, completing the proof of the claim. We claim (x_{k_j}) converges to this point $x \in \bigcap_{k=1}^\infty I_k$. Indeed, since each $x_{k_j} \in I_j$, we have $0 \leq |x - x_{k_j}| \leq \frac{2M}{2^{j-1}} \to 0$, so by the squeeze theorem, $x = \lim_{j\to\infty} x_{k_j}$, completing the proof of the base case.
Inductively, suppose the theorem is true in $\mathbb{R}^n$, and let (x_k) be a bounded sequence in $\mathbb{R}^{n+1}$. Write $x_k = (x_k^1, \dots, x_k^{n+1}) \in \mathbb{R}^{n+1}$ for each $k \in \mathbb{N}$. Then for each k, set $y_k = (x_k^1, \dots, x_k^n) \in \mathbb{R}^n$. This gives a bounded sequence in $\mathbb{R}^n$ (since (x_k) is bounded), so by the induction hypothesis, it has a convergent

subsequence (y_{k_j}), which converges to some $(x^1, \ldots, x^n) \in \mathbb{R}^n$. Now, the sequence (x_k^{n+1}) in $\mathbb{R}$ is also bounded, so therefore so too is its subsequence $(x_{k_j}^{n+1})$, so by the theorem for $n = 1$, it has a convergent subsequence $(x_{k_{j_\ell}}^{n+1})$ converging to some $x^{n+1} \in \mathbb{R}$. Since $(y_{k_j}) \to (x_1, \ldots, x^n)$, its subsequence $(y_{k_{j_\ell}})$ converges to $(x^1, \ldots, x^n)$ as well, so writing $x = (x^1, \ldots, x^n, x^{n+1})$, by assignment 2, $(x_{k_{j_\ell}})$ converges to x. ■

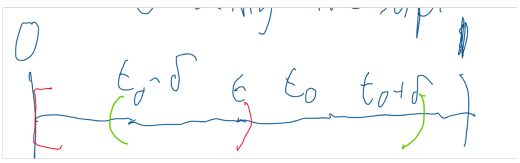
Connectedness

Definition
Let $E \subseteq \mathbb{R}^n$. A separation of E is a pair of open subsets U, V such that $E \cap U \cap V = \emptyset$, $E \subseteq U \cup V$, $E \cap U \neq \emptyset$, $E \cap V \neq \emptyset$, that is, E is the disjoint union of open sets. If there is a separation, we can think of E as "more than one piece".

Definition
A subset $E \subseteq \mathbb{R}^n$ is connected if there is no separation for E.

Example
$\emptyset$ is vacuously connected, and singletons are always connected.

Theorem
A subset $E \subseteq \mathbb{R}$ is connected if and only if it is an interval.
Proof:
Suppose E is not an interval. Then there exist $a, b \in E$ such that $a < b$ with $x \in \mathbb{R}$ such that $a < x < b$ with $x \notin E$. Take $U = (-\infty, x)$, $V = (x, \infty)$. Both are open with empty intersection with $a \in U$, $b \in V$, so $E \cap U \neq \emptyset$, $E \cap V \neq \emptyset$, so $E \cap U, E \cap V$ is a separation.
Conversely, suppose $E \subseteq \mathbb{R}$ is an interval and that there exists a separation U, V. Since $E \cap U \neq \emptyset$, there exists $a \in E \cap U$, and since $E \cap V \neq \emptyset$, there exists $b \in E \cap V$. Since $E \cap U \cap V = \emptyset$, without loss of generality suppose $a < b$. Thus $[a, b] \cap U, [a, b] \cap V \neq \emptyset$. This shows U, V are a separation of $[a, b]$, and so we need only prove the statement for a closed interval $[a, b]$, which without loss of generality we may assume to be $[0, 1]$ by rescaling and translating. Thus we have
(1a) U open,
(1b) V open,
(2) $0 \in U \cap [0, 1]$,
(3) $1 \in V \cap [0, 1]$,
(4) $[0, 1] \subseteq U \cup V$,
(5) $U \cap V \cap [0, 1] = \emptyset$.
By (1a) and (2), there exists $\varepsilon_0 > 0$ such that $(-\varepsilon_0, \varepsilon_0) \subseteq U$. Without loss of generality, suppose $\varepsilon_0 < 1$. So $[0, \varepsilon_0) \subseteq U \cap [0, 1]$, and define $t_0 = \sup\{\varepsilon \in (0, 1) : [0, \varepsilon) \subseteq U \cap [0, 1]\}$, which exists since the set we are taking the supremum over is nonempty and bounded above (it contains ε_0 and is bounded above by 1). Thus t_0 exists and $0 < \varepsilon_0 \leq t_0 \leq 1$, so $t_0 \in [0, 1] \subseteq U \cup V$, so $t_0 \in U$ or $t_0 \in V$. If $t_0 \in U$, then $t_0 \neq 1$ by (3), so $t_0 < 1$. Since U is open, there exists $\delta > 0$ such that $(t_0 - \delta, t_0 + \delta) \subseteq U$. But $t_0 - \delta < t_0$, so by definition of the supremum, there exists $\varepsilon > 0$ such that $t_0 - \delta < \varepsilon \leq t_0$ such that $[0, \varepsilon) \subseteq U \cap [0, 1]$. Therefore, $[0, t_0 + \delta) \subseteq U$, contradicting t_0 being the supremum.



Therefore, we must have $t_0 \notin U$. Therefore, $t_0 \in V$. Since V is open, there exists $\zeta > 0$ such that

$(t_0-\zeta, t_0+\zeta) \subseteq V$. Without loss of generality suppose $0 < t_0 - \zeta$. Since $t_0 - \zeta < t_0$, by properties of the supremum there exists $\varepsilon > 0$ with $t_0 - \zeta < \varepsilon \leq t_0$ and $[0, \varepsilon) \subseteq U \cap [0,1]$. Therefore, $(t_0 - \zeta, \varepsilon)$ is nonempty and contained in $[0,1]$, so there exists $s \in (t_0 - \zeta, \varepsilon) \subseteq [0, \varepsilon) \subseteq U$. Thus, $s \in U$. On the other hand, $s \in (t_0 - \zeta, t_0 + \zeta)$, so $s \in V$, which contradicts (5). ■

Because $\mathbb{R} = (-\infty, \infty)$, this shows $\mathbb{R}$ is connected. What about $\mathbb{R}^n$?
If it isn't connected, then there exists a separation U, V of $\mathbb{R}^n$. Therefore, we have $\mathbb{R}^n = U \cup V$ with U, V disjoint and nonempty, so $U = V^c$, $V = U^c$, so U and V are both open and closed (i.e. clopen) and not the empty set nor $\mathbb{R}^n$. So $\mathbb{R}^n$ is disconnected if and only if there exists a proper nonempty clopen subset of $\mathbb{R}^n$. We'll show $\mathbb{R}^n$ is connected, so a nonempty clopen subset of $\mathbb{R}^n$ is just $\mathbb{R}^n$.

Definition
A subset $A \subseteq \mathbb{R}^n$ is convex if and only if $x, y \in E$ implies $(1-t)x + ty \in E$ for all $t \in [0,1]$, that is, the straight line segment from x to y is contained in E.

Proposition
Any convex set $E \subseteq \mathbb{R}^n$ is connected.
Proof:
Suppose for the sake of contradiction E is not connected. Then there exists a separation U, V of E, so there exists $x \in E \cap U, y \in E \cap V$ with $x \neq y$. Define $U' = \{t \in \mathbb{R} : (1-t)x + ty \in U\} \subseteq \mathbb{R}$ and $V' = \{t \in \mathbb{R} : (1-t)x + ty \in V\}$. We have $0 \in U', 1 \in V'$. We claim U' is open. Indeed, let $t_0 \in U'$. Then $x_0 = (1-t_0)x + t_0 y \in U$, but U is open, so there exists $\varepsilon > 0$ such that $B_\varepsilon(x_0) \subseteq U$. Let $t \in \mathbb{R}$ such that $|t - t_0| < \frac{\varepsilon}{||x|| + ||y||}$. Then,

$$||(1-t)x + ty - x_0|| = ||(1-t)x + ty - (1-t_0)x - t_0 y|| = ||(t_0 - t)x + (t - t_0)y|| \leq |t - t_0| \, ||x|| + |t - t_0| \, ||x|| < \varepsilon$$

Therefore, $(1-t)x + ty \in B_\varepsilon(x_0)$, so $t \in U'$. Therefore, $B_{\frac{\varepsilon}{||x||+||y||}}(t_0) \subseteq U'$, so U' is open. An identical proof shows V' is open. Therefore, $U', V' \subseteq \mathbb{R}$ are open. Moreover, $U' \cap [0,1]] \neq \emptyset$ (since it contains 0), and $V' \cap [0,1] \neq \emptyset$ (since it contains 1), but $U' \cap V' \cap [0,1] = \emptyset$ because if $t \in U' \cap V' \cap [0,1]$, then $(1-t)x + ty \in U \cap V$, and is in E by convexity, so $E \cap U \cap V \neq \emptyset$, contradicting the fact U, V separate E. Finally, $[0,1] \subseteq U' \cup V'$, because if $t \in [0,1]$, then $(1-t)x + ty \in E \subseteq U \cup V$ by convexity, so $t \in U' \cup V'$, so $[0,1] \subseteq U' \cup V'$. This shows all necessary to show that U', V' is a separation of $[0,1]$, a contradiction.
■

Compactness

The last topological property (defined only in terms of open and closed sets) is compactness, which is a measure of "smallness" of a set.

Definition
Let $E \subseteq \mathbb{R}^n$. An open cover of E is a collection of subsets $\{U_a\}_{a \in A}$ such that each U_a is open and $E \subseteq \bigcup_{a \in A} U_a$. A subcover of an open cover of E is an open cover $\{U_b\}_{b \in B}$ of E with $B \subseteq A$.

Definition
A subset $E \subseteq \mathbb{R}^n$ is called compact if and only if every open cover of E admits a finite subcover, that is if $\{U_a\}_{a \in A}$ is an open cover of E, there exists a finite subset $A_0 = \{a_1, \ldots, a_m\}$ such that $E \subseteq \bigcup_{j=1}^m U_{a_j}$. Informally, E is compact if and only if when you can cover it by open sets, you can cover it by finitely many of them, so E is "not too big".

Definition
A subset $E \subseteq \mathbb{R}^n$ is called bounded if and only if there exists $M > 0$ such that $E \subseteq \overline{B_M(0)}$, that is, $||x|| \leq M$ for all $x \in E$.

Proposition
Suppose $E \subseteq \mathbb{R}^n$ is compact. Then it is bounded.
Proof:
For $k \in \mathbb{N}$, let $U_k = B_k(0)$, which is open. Then $U_k \subseteq U_\ell$ if $k \leq \ell$, with $\mathbb{R}^n = \bigcup_{k=1}^\infty U_k$. Hence $E \subseteq \bigcup_{k=1}^\infty U_k$, so $(U_k)_{k=1}^\infty$ is an open cover of E, so by compactness, this open cover admits a finite subcover. Therefore, there exists $1 \leq k_1 < k_2 < \cdots < k_N$ such that $E \subseteq \bigcup_{j=1}^N U_{k_j} = U_{k_N} = B_{k_N}(0) \subseteq \overline{B_{k_N}(0)}$, so E is bounded with the choice $M = k_N$. ■

Example
Since $\mathbb{R}^n$ is not bounded, it is not compact.

Example
A bounded set that is not compact is $E = (0,1) \subseteq \mathbb{R}$. For each $k \in \mathbb{N}$, define $U_k = (1/k, 1)$, which is open. Then $\bigcup_{k=1}^\infty U_k = (0,1) = E$, so (U_k) is an open cover of E. On the other hand it has no finite subcover, since if $N \in \mathbb{N}$ is the largest number such that U_N is included in a finite subcover, then $\frac{1}{N+1} \in (0,1)$ but $\frac{1}{N+1} \notin \bigcup_{k=1}^N U_k$, which contains the union over the finite subcover in question.

Proposition
Let $E \subseteq \mathbb{R}^n$ be compact. Then E is closed.
Proof:
We want to show E is closed, so we'll show E^c is open. We know $E^c \neq \emptyset$ since $\mathbb{R}^n$ is not compact. Thus, let $x \in E^c$. We need to find an open neighbourhood of x contained in E^c. We have $E \subseteq \mathbb{R}^n \setminus \{x\}$ since $x \in E^c$, so $\bigcup_{k=1}^\infty (\overline{B_{1/k}(x)})^c$ is an open cover of E. Write $U_k = (\overline{B_{1/k}(x)})^c = \{y \in \mathbb{R}^n : ||y - x|| > 1/k\}$ for each $k \in \mathbb{N}$. Since $\ell > k$ implies $1/\ell < 1/k$, if $||y - x|| > 1/k$, then $||y - x|| > 1/\ell$, so if $\ell > k$, then $U_k \subseteq U_\ell$. Therefore, by compactness of E, there exists $k_0 \in \mathbb{N}$ such that $E \subseteq (\overline{B_{1/k_0}(x)})^c$. Taking complements, we have $x \in B_{1/k_0}(x) \subseteq \overline{B_{1/k_0}(x)} \subseteq E^c$, so E^c is open, hence E is closed. ■

Thus, we've shown E is compact implies E is closed and bounded. The converse holds in $\mathbb{R}^n$ as well:

Theorem (Heine-Borel)
$E \subseteq \mathbb{R}^n$ is compact if and only if it is closed and bounded.
Remark:
(1) The original definition of compactness is very useful for proving theorems involving compact sets.
(2) The Heine-Borel theorem is true in some more general settings, but not true in general since general topological spaces (with just notions of open sets) don't have a notion of distance or boundedness in general.

Lemma
Let $E \subseteq \mathbb{R}^n$. Then any open cover of E admits a countable subcover.
Proof:
This (crucially) uses the fact that $\mathbb{Q}^n$ is dense in $\mathbb{R}^n$, since $\mathbb{Q}$ is dense in $\mathbb{R}$: let $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$, and $\varepsilon > 0$. For all $j = 1, \ldots, n$, there exists $q_j \in \mathbb{Q}$ such that $(x_j - q_j)^2 < \frac{\varepsilon^2}{n}$ by density, so writing $q = (q_1, \ldots, q_n) \in \mathbb{Q}^n$, we have $||q - x||^2 < \varepsilon^2$, that is, $||q - x|| < \varepsilon$.
Now, for the proof, let $(U_a)_{a \in A}$ be an open cover of E. Let $x \in E$. Then there exists $a(x) \in E$ such that $x \in U_{a(x)}$. But $U_{a(x)}$ is open, so there exists $\varepsilon(x) > 0$ such that $B_{\varepsilon(x)}(x) \subseteq U_{a(x)}$, and so $E \subseteq \bigcup_{x \in E} B_{\varepsilon(x)}(x)$. By density of $\mathbb{Q}^n$ in $\mathbb{R}^n$ and $\mathbb{Q}$ in $\mathbb{R}$, there exists $q(x) \in \mathbb{Q}^n$ and $p(x) \in \mathbb{Q}$ such that $||x - q(x)|| < \frac{\varepsilon(x)}{4}$ and $\frac{\varepsilon(x)}{4} < p(x) < \frac{\varepsilon(x)}{2}$. We claim $B_{p(x)}(q(x)) \subseteq B_{\varepsilon(x)}(x)$. Let $y \in B_{p(x)}(q(x))$. Then,

$$||x - y|| \leq ||x - q(x)|| + ||q(x) - y|| < \varepsilon/4 + p(x) < \varepsilon/4 + \varepsilon/2 < \varepsilon$$

Moreover, we have $x \in B_{p(x)}(q(x))$, since $||x - q(x)|| < \varepsilon/4 < p(x)$. Therefore, we have $E \subseteq \bigcup_{x \in E} B_{p(x)}(q(x))$, but $\mathbb{Q}^n$ and $\mathbb{Q}$ are countable, so for each $j \in \mathbb{N}$, there exists $q_j \in \mathbb{Q}^n$ and $p_j \in \mathbb{Q}$ such that

$$E \subseteq \bigcup_{j=1}^{\infty} B_{p_j}(q_j) \subseteq \bigcup_{j=1}^{\infty} B_{\varepsilon(x_j)}(x_j) \subseteq \bigcup_{j=1}^{\infty} U_{a(x_j)}$$

because $B_{\varepsilon(x)}(x) \subseteq U_{a(x)}$. Therefore, $\{U_{a(x_j)}\}_{j\in\mathbb{N}}$ is a countable subcover for E. ■

Proof of Heine-Borel:
We have already proved one direction, in the other, suppose E is closed and bounded, we must show it is compact. Let $(U_a)_{a\in A}$ be an open cover of E, which by the lemma, we may assume to be a countable subcover $(U_j)_{j\in\mathbb{N}}$. We need to prove there exists a finite subset $\{j_1,\dots,j_N\}$ of $\mathbb{N}$ such that $E \subseteq \bigcup_{\ell=1}^{N} U_{j_\ell}$, which will hold if we show there exists $N \in \mathbb{N}$ such that $E \subseteq \bigcup_{j=1}^{N} U_j$. Suppose for the sake of contradiction that this is not true, so for all $k \in \mathbb{N}$, there exists $x_k \in E \setminus \bigcup_{j=1}^{k} U_j$. This gives a sequence (x_k) in E and E is bounded, so by Bolzano-Weierstrass, (x_k) has a convergent subsequence $(x_{k_\ell})_{\ell\in\mathbb{N}}$, but $x_{k_\ell} \in E$ for all $\ell \in \mathbb{N}$ and E is closed, so $\lim_{\ell\to\infty} x_{k_\ell} \in E$, call it $x = \lim_{\ell\to\infty} x_{k_\ell}$. Since $E \subseteq \bigcup_{j=1}^{\infty} U_j$ and $x \in E$, there exists $j_0 \in \mathbb{N}$ such that $x \in U_{j_0}$. Since $(x_{k_\ell}) \to x$ and U_{j_0} is an open neighbourhood of x, there exists $M \in \mathbb{N}$ such that $\ell \geq M$ implies $x_{k_\ell} \in U_{j_0}$.
On the other hand, by construction, $x_{k_\ell} \notin \bigcup_{j=1}^{m} U_j$ for all $m \leq k_\ell$, that is, x_{k_ℓ} is not in the first k_ℓ U_j's. Now, take ℓ large enough such that $\ell \geq M$ and $k_\ell \geq j_0$. Combining the previous two paragraphs, we have $x_{k_\ell} \in U_{j_0}$ and $x_{k_\ell} \notin U_{j_0}$, a contradiction. ■

Limits of Functions

Definition
Let $V \subseteq \mathbb{R}^n$ be an open set with $x_0 \in V$. Then $V \setminus \{x_0\}$, a so-called punctured neighbourhood of x_0, is an open set. Now, let $f : V \setminus \{x_0\} \to \mathbb{R}^m$ be a function. We say the limit of f as $x \to x_0$ exists and equals $L \in \mathbb{R}^m$ and write $\lim_{x\to x_0} f(x) = L$ if and only if, for all $\varepsilon > 0$, there exists $\delta > 0$ such that $0 < ||x - x_0|| < \delta$ implies $||f(x) - L|| < \varepsilon$. Note that it does not matter what value f takes at x_0 (nor does it even need to be defined there).

Remark
This definition has very different behaviour between $n = 1$ and $n > 1$. When $n = 1$, the limit of a function exists if and only if it exists in both the left and right directions and they are equal. On the other hand, for $n > 1$, there are infinitely many paths to x_0 in $V \setminus \{x_0\}$ (and not just straight lines!), which makes things more difficult.

Example
Let $f : \mathbb{R}^2 \setminus \{(2,3)\} \to \mathbb{R}$ be the function defined by $f(x,y) = \frac{(x-2)^2}{(x-2)^2+(y-3)^2}$. Does $\lim_{(x,y)\to(2,3)} f(x,y)$ exist? Consider the straight line of slope m through $(2,3)$, given by $y - 3 = m(x-2)$. On this line, $f(x,y) = \frac{(x-2)^2}{(x-2)^2+(m(x-2))^2} = \frac{1}{1+m^2}$, which is a constant depending on m, so there is no single limiting value, so $\lim_{(x,y)\to(2,3)} f(x,y)$ does not exist.

Example
Consider $n = 2, m = 1$ with $f(x,y) = \frac{xy^2}{x^2+y^4}$. Consider the point $(0,0)$, and let $V = \mathbb{R}^2 \setminus \{(0,0)\}$. Does $\lim_{(x,y)\to(0,0)} f(x,y)$ exist? Consider the values of f on straight lines through the origin, given by $y = mx$ with slope m. On this line, the values of f are $f(x,mx) = \frac{m^2x^3}{x^2+m^4x^4} = \frac{m^2x}{1+m^4x^2}$, so as $x \to 0$, we have $f(x,mx) \to 0$. On the vertical line $x = 0$, we have $f(0,y) = 0$ for all $y \neq 0$, so we might guess that this limit exists and equals 0. Let's try another path, with $x = ky^2$ for arbitrary $k \in \mathbb{R}$. On this curve, $f(ky^2,y) = \frac{ky^4}{ky^2+y^4} = \frac{k}{1+k^2}$, which is a constant depending on k, so there is no single limiting value and again $\lim_{(x,y)\to(0,0)} f(x,y)$ does not exist.

Example
Let $f(x,y) = \frac{x^4}{x^2+y^2}$. The degree of the numerator is greater than the degree of the denominator, so we might guess $\lim_{(x,y)\to(0,0)} \frac{x^4}{x^2+y^2}$ exists and equals 0. Let's prove this. Let $\varepsilon > 0$. We need to find $\delta > 0$ such that $0 < ||(x,y) - (0,0)|| < \delta$ implies $||f(x,y) - \overbrace{0}^{L}|| < \varepsilon$. Suppose $0 < ||(x,y) - \mathbf{0}|| < \delta$. Then $0 < \sqrt{x^2+y^2} < \delta$, so $0 < x^2 + y^2 < \delta^2$, implying $x^2 < \delta^2$ and $y^2 < \delta^2$ by the triangle inequality, so $|x| < \delta$ and $|y| < \delta$. On the other hand,

$$|f(x,y) - 0| = \left|\frac{x^4}{x^2+y^2}\right| = \frac{x^4}{x^2+y^2} = x^2 \underbrace{\left(\frac{x^2}{x^2+y^2}\right)}_{\leq 1} \leq x^2$$

Thus take $\delta = \sqrt{\varepsilon}$. Then $0 < x^2 + y^2 < \delta^2$ if and only if $0 < ||(x,y) - \mathbf{0}|| < \delta = x^2$, so if $x^2 < \varepsilon$ implies $|f(x,y) - 0| \leq x^2 < \varepsilon$ by the above, so $\lim_{(x,y)\to z} f(x,y) = 0$.

Remark
If a limit exists, it is unique by a similar proof to the one we gave for sequences, especially in light of the following:

Theorem (Sequential Characterization of Functional Limits)
Let $V \subseteq \mathbb{R}^n$ be open containing a point $x_0 \in \mathbb{R}^n$, and consider a map $f : V \setminus \{x_0\} \to \mathbb{R}^m$. The following are equivalent:
(1) $\lim_{x\to x_0} f(x) = L$
(2) For every sequence (x_k) contained in $V \setminus \{x_0\}$ such that $\lim_{k\to\infty} x_k = x_0$, we have $\lim_{k\to\infty} f(x_k) = L$.
Proof:
For (1) implies (2), suppose $\lim_{x\to x_0} f(x) = L$. Let (x_k) be a sequence in $V \setminus \{x_0\}$ such that $\lim_{k\to\infty} x_k = x_0$. We want to show $\lim_{k\to\infty} f(x_k) = L$. Let $\varepsilon > 0$. By our hypothesis, there exists $\delta > 0$ such that if $0 < ||x - x_0|| < \delta$ then $||f(x) - L|| < \varepsilon$. Because $\lim_{k\to\infty} x_k = x_0$, there exists $N \in \mathbb{N}$ such that if $k \geq N$, then $0 < ||x - x_k|| < \delta$, so by definition of δ, we have $||f(x_k) - L|| < \varepsilon$, which shows $\lim_{k\to\infty} f(x_k) = L$. Conversely, suppose (2) holds and for the sake of contradiction that $\lim_{x\to x_0} f(x) = L$ does not hold. Therefore, there exists $\varepsilon_0 > 0$ such that for all $\delta > 0$, there exists x_δ with $0 < ||x_\delta - x_0|| < \delta$ but $||f(x_\delta) - L|| \geq \varepsilon_0$. For each $k \in \mathbb{N}$, take $\delta = 1/k$. Then there exists $x_k := x_\delta$ such that $0 < ||x_k - x_0|| < \delta$ but $||f(x_k) - L|| \geq \varepsilon_0$. Then, (x_k) is a sequence such that $\lim_{k\to\infty} x_k = x_0$ but $\lim_{k\to\infty} f(x_k) \neq L$, contradicting (b). ■

Proposition (Properties of Limits of Functions)
Suppose $f, g : V \setminus \{x_0\} \to \mathbb{R}^m$ are two functions with $V \subseteq \mathbb{R}^n$ open, with $\lim_{x\to x_0} f(x) = L \in \mathbb{R}^m$ and $\lim_{x\to x_0} g(x) = M \in \mathbb{R}^m$. Then,
(1) $\lim_{x\to x_0}(f(x) \pm g(x)) = L \pm M$
(2) $\lim_{x\to x_0}(\lambda f(x)) = \lambda L$ for all $\lambda \in \mathbb{R}$
(3) If $m = 1$, then $\lim_{x\to x_0} f(x)g(x) = LM$, and for any $m \in \mathbb{N}$, then $\lim_{x\to x_0}(f(x) \cdot g(x)) = L \cdot M$.
(4) If $m = 1$, $M \neq 0$, then $\lim_{x\to x_0} \frac{f(x)}{g(x)} = \frac{L}{M}$.
Proof:
Left as an exercise, as they are the exact same proofs for functions on $\mathbb{R}$, only with $|\cdot|$ replaced with $||\cdot||$ (and the general statement of (3) follows from the $m = 1$ case and (1)). ■

Corollary
The same properties as above hold for limits of sequences by the sequential characterization. ■

Lemma
Let $V \subseteq \mathbb{R}^n$ be open containing x_0. Let $f : V \setminus \{x_0\} \to \mathbb{R}^m$ and write $f(x) = (f_1(x), \ldots, f_m(x))$ for functions $f_1, \ldots, f_m : V \setminus \{x_0\} \to \mathbb{R}$, called the component functions of f. Let $L = (L_1, \ldots, L_m) \in \mathbb{R}^m$.

Then $\lim_{x\to x_0} f(x) = L$ if and only if $\lim_{x\to x_0} f_j(x) = L_j$ for all $j = 1, \ldots, m$.
Proof:
First note that

$$||f(x) - L||^2 = (f_1(x) - L_1)^2 + \cdots + (f_m(x) - L_m)^2 = |f_1(x) - L_1|^2 + \cdots + |f_m(x) - L_m|^2 \quad (\star)$$

In the forward direction, let $\varepsilon > 0$. Then there exists $\delta > 0$ such that $0 < ||x - x_0|| < \delta$ implies $||f(x) - L|| < \varepsilon$. By $(\star)$, $|f_j(x) - L_j| \leq ||f(x) - L|| < \varepsilon$ for each $j = 1, \ldots, m$, so $\lim_{x\to x_0} f_j(x) = L_j$ for each j.
Conversely, let $\varepsilon > 0$. For each $j = 1, \ldots, m$, there exists $\delta_j > 0$ such that $0 < ||x - x_0|| < \delta$ implies $|f_j(x) - L_j| < \varepsilon/\sqrt{m}$. Let $\delta := \min\{\delta_1, \ldots, \delta_m\} > 0$. Then if $0 < ||x - x_0|| < \delta$, by $(\star)$ and construction of δ, we have $||f(x) - L|| < \varepsilon$. ■

Proposition (Squeeze Theorem)
Let $V \subseteq \mathbb{R}^n$ be open containing x_0, and let $f, g, h : V \setminus \{x_0\} \to \mathbb{R}$ be functions. If $g(x) \leq h(x) \leq f(x)$ for all $x \in V \setminus \{x\}$ and $\lim_{x\to x_0} g(x) = \lim_{x\to x_0} f(x) = L$, then $\lim_{x\to x_0} h(x) = L$.
Proof:
Let $\varepsilon > 0$. Because $\lim_{x\to x_0} g(x) = L$, there exists $\delta_1 > 0$ such that $0 < ||x - x_0|| < \delta$ implies $|g(x) - L| < \varepsilon$, and because $\lim_{x\to x_0} f(x) = L$, there exists $0 < ||x - x_0|| < \delta$ implies $|f(x) - L| < \varepsilon$. Now, choose $\delta = \min\{\delta_1, \delta_2\}$, and let x satisfy $0 < ||x - x_0|| < \delta$. Then, $g(x) \leq h(x) \leq f(x)$ implies $g(x) - L \leq h(x) - L \leq f(x) - L$, which by construction of δ, tells us

$$-\varepsilon < g(x) - L \leq h(x) - L \leq f(x) - L < \varepsilon$$

and so taking absolute values, we have $|h(x) - L| < \varepsilon$. ■

Proposition
Let $V \subseteq \mathbb{R}^n$ be open containing x_0, and $f : V \setminus \{x_0\} \to \mathbb{R}^m$ be a function. If $\lim_{x\to x_0} f(x)$ exists and equals $L \in \mathbb{R}^m$, then $\lim_{x\to x_0} ||f(x)|| = ||\lim_{x\to x_0} f(x)||$.
Proof:
Let $\varepsilon > 0$. Because $\lim_{x\to x_0} f(x) = L$, there exists $\delta > 0$ such that $0 < ||x - x_0|| < \delta$ implies $||f(x) - L|| < \varepsilon$. By the reverse triangle inequality,

$$|||f(x) - ||L||| \leq ||f(x) - L|| < \varepsilon$$

which tells us $\lim_{x\to x_0} ||f(x)|| = ||L||$. ■

Remark
All of our definitions with limits have been for V an open set containing x_0. What if it isn't open? Given $f : A \to \mathbb{R}^m$ with A not necessarily open, we say $\lim_{x\to x_0} f(x)$ exists and equals $L \in \mathbb{R}^m$ if and only if, for all $\varepsilon > 0$, there exists $\delta > 0$ such that $0 < ||x - x_0|| < \delta$ and $x \in A$ implies $||f(x) - L|| < \varepsilon$. In an open set, the condition $x \in A$ is automatically satisfied for δ sufficiently small, so this is equivalent there, but not necessarily so if the set isn't open.

Continuity

Definition
Let $U \subseteq \mathbb{R}^n$ be open with $f : U \to \mathbb{R}^m$ a function. Let $x_0 \in U$. We say f is continuous at x_0 if and only if $\lim_{x\to x_0} f(x)$ exists and equals $f(x_0)$. That is, f is continuous at x_0 if and only if for all $\varepsilon > 0$, there exists $\delta > 0$ such that $||x - x_0|| < \delta$ implies $||f(x) - f(x_0)|| < \varepsilon$. By the sequential characterization of limits of functions, f is continuous at x_0 if and only if whenever $(x_k) \to x_0$ is a convergent sequence contained in U, $(f(x_k)) \to (f(x_0))$ in $\mathbb{R}^m$.

Remark
(1) Let $f_j : U \to \mathbb{R}$ be the component functions of f for $j = 1, \ldots, m$. The lemma from the last section

shows that f is continuous at x_0 if and only if every f_j is continuous at x_0.
(2) If U is not necessarily open, we define continuity the same way, but note that the definition of the limit has changed here, where the epsilon-delta condition has become: for all $\varepsilon > 0$, there exists $\delta > 0$ such that $||x - x_0|| < \delta$ and $x \in U$ implies $||f(x) - f(x_0)|| < \varepsilon$.

Definition
Let $U \subseteq \mathbb{R}^n$ be open with $f : U \to \mathbb{R}^m$ a function. We say f is continuous on U if and only if it is continuous at each $x_0 \in U$.

Example
Suppose $m = n$. Then the identity function $f : \mathbb{R}^n \to \mathbb{R}^n$ given by $f(x) = x$ for all $x \in \mathbb{R}^n$ is continuous on $\mathbb{R}^n$ (take $\delta = \varepsilon$).

Example
Let $c \in \mathbb{R}^m$ and suppose $f : U \underset{\text{open}}{\subseteq} \mathbb{R}^n \to \mathbb{R}^m$ is the constant function $f(x) = c$ for all $x \in U$. Then f is continuous on U (any δ works provided $B_\delta(x) \subseteq U$), since $||f(x) - L|| = ||c - c|| = 0 < \varepsilon$ regardless of δ.

Example
Let $f : \mathbb{R}^2 \to \mathbb{R}$ be $f(x, y) = xy$ for all $(x, y) \in \mathbb{R}^2$. This is continuous on $\mathbb{R}^2$ because $\lim_{(x_0,y_0)\to(x,y)} f(x, y) = x_0 y_0$ by properties of limits.

Proposition
Let $f : U \underset{\text{open}}{\subseteq} \mathbb{R}^n \to \mathbb{R}^m$ and $g : V \underset{\text{open}}{\subseteq} \mathbb{R}^m \to \mathbb{R}^p$ be functions with $f(U) \subseteq V$.
(1) If f is continuous at $x_0 \in U$ and g is continuous at $f(x_0) \in V$, then $g \circ f : U \to \mathbb{R}^p$ is continuous at x_0.
(2) If f is continuous on U and g is continuous on V, then $g \circ f : U \to \mathbb{R}^p$ is continuous on U. *Proof:* Assignment 4. ■

This gives many examples of continuous functions:
Examples
(1) $(x, y) \mapsto \sin(xy)$ for $(x, y) \in \mathbb{R}^2$ is continuous on $\mathbb{R}^2$.
(2) $e^{\frac{1}{x^2 - y^2}}$ for $(x, y) \in \mathbb{R}^2 \setminus \{\pm 1\}$ is continuous on $\mathbb{R}^2 \setminus \{\pm 1\}$.
(3) Suppose $f = (f_1, \dots, f_m) : U \underset{\text{open}}{\subseteq} \mathbb{R}^n \to \mathbb{R}^m$ and $g = (g_1, \dots, g_m) : U \to \mathbb{R}^m$, and consider the function $f \cdot g : U \to \mathbb{R}^m$ given by

$$(f \cdot g)(x) = f(x) \cdot g(x) = \sum_{i=1}^{m} f_i(x) g_i(x)$$

for $x \in U$. If f and g are continuous on U, their component functions are, so the dot product, which is a sum of their products, is continuous as well.

We can characterize continuity topologically, that is, only in terms of open sets:

Definition
Let $f : U \underset{\text{open}}{\subseteq} \mathbb{R}^n \to \mathbb{R}^m$, and let $B \subseteq \mathbb{R}^m$. The inverse image of B by the map f is $f^{-1}(B) = \{x \in U : f(x) \in B\}$. Note that f need not be invertible, and depends on the domain U: if $V \subseteq U$, and we consider the restriction of f to V, $f|_V : V \to \mathbb{R}^m$ given by $f|_V(x) = f(x)$ for all $x \in V$, then $f|_V^{-1}(B) = V \cap f^{-1}(B)$. Note that the inverse image can be empty.

Remark
The notation for the inverse image is consistent: if $A \subseteq U$, then we have that the image of f under A is

$f(A) = \{f(x) : x \in A\}$, and if f is invertible, then we have

$$\overbrace{f^{-1}(B)}^{\text{image under } f^{-1}} \quad = \quad \overbrace{f^{-1}(B)}^{\text{inverse image by } f}$$

Example
Consider $f : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ given by $f(x) = 1/x$ for all $x \in \mathbb{R} \setminus \{0\}$. Then $f^{-1}((0,1)) = (1, \infty)$ and $f^{-1}(\{0\}) = \emptyset$.

Proposition (Topological Characterization of Continuity)
Let $f : U \underset{\text{open}}{\subseteq} \mathbb{R}^n \to \mathbb{R}^m$. Then f is continuous on U if and only if $f^{-1}(V)$ is open in $\mathbb{R}^n$ for every open subset $V \subseteq \mathbb{R}^m$.
Proof:
In the forward direction, suppose $V \subseteq \mathbb{R}^m$ is open. If $f^{-1}(V) = \emptyset$, it is open. Otherwise, suppose $f^{-1}(V) \neq \emptyset$ and let $x_0 \in f^{-1}(V) \subseteq U$. Then $f(x_0) \in V$ and V is open, so there exists $\varepsilon > 0$ such that $B_\varepsilon(f(x_0)) \subseteq V$. But f is continuous at x_0, so for this ε, there exists $\delta > 0$ such that $||x - x_0|| < \delta$ implies $||f(x) - f(x_0)|| < \varepsilon$, that is, $x \in B_\delta(x_0)$ implies $f(x) \in B_\varepsilon(f(x_0)) \subseteq V$. Therefore, $x \in B_\delta(x_0)$ implies $x \in f^{-1}(V)$, so $B_\delta(x_0) \subseteq f^{-1}(V)$, so $f^{-1}(V)$ is open.
Conversely, let $x_0 \in U$. We want to show f is continuous at x_0. Let $\varepsilon > 0$. Then $V = B_\varepsilon(f(x_0))$ is open in $\mathbb{R}^m$. By hypothesis, $f^{-1}(V)$ is open in $\mathbb{R}^n$, so since $x_0 \in f^{-1}(V)$ because $f(x_0) \in V$, there exists $\delta > 0$ such that $B_\delta(x_0) \subseteq f^{-1}(V)$. Thus, $f(B_\delta(x_0)) \subseteq V = B_\varepsilon(f(x_0))$, which shows f is continuous at x_0. ■

Remark
(1) If we use our definition of continuity for a function f whose domain A is not necessarily open, then f is continuous on A if and only if for every open subset $V \subseteq \mathbb{R}^m$, $f^{-1}(V)$ is the intersection of A with an open subset of $\mathbb{R}^n$.
(2) On the assignments, you'll show that for $f : \mathbb{R}^n \to \mathbb{R}^m$, f is continuous if and only if for every closed set $F \subseteq \mathbb{R}^m$, $f^{-1}(F)$ is closed.

We see that preimages play nicely with open sets. Is it true that the image of an open set under a continuous function is open (or similarly with closed sets?). The answer is no:
Examples
(1) Consider $f : (-1,1) \to \mathbb{R}$ given by $f(x) = x^2$ for all $x \in (-1,1)$, which is continuous with $(-1,1)$ open. Then $f((-1,1)) = [0,1)$, which is not open.
(2) Consider $f : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ given by $f(x) = 1/x$ for all $x \in \mathbb{R} \setminus \{0\}$, which is continuous with $\mathbb{R} \setminus \{0\}$ open. Then $f([1, \infty)) = (0, 1]$ is not closed, but $[1, \infty)$ is.

What about compact or connected sets? Their preimages don't play nicely with continuous functions:
Examples
(1) Consider $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^2$ for all $x \in \mathbb{R}$, which is continuous. Then $E = (1, 9)$ is connected, but $f^{-1}(E) = (-3, -1) \cup (1, 3)$ is disconnected.
(2) Consider $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = \frac{1}{1+x^2}$ for all $x \in \mathbb{R}$. Then $E = [0, 1]$ is compact, but $f^{-1}(E) = \mathbb{R}$ which is not compact.

On the other hand, images of compact or connected sets under continuous functions are compact and connected respectively.

Theorem
Suppose $U \subseteq \mathbb{R}^n$ is open with $f : U \to \mathbb{R}^m$ a continuous function. Suppose $K \subseteq U$ is compact. Then $f(K)$ is compact.
Proof:
Let $\{V_a\}_{a \in A}$ is an open cover of $f(K)$. Then, if $x \in K$, $f(x) \in f(K)$, so there exists $a \in A$ such that

$f(x) \in V_a$, so $x \in f^{-1}(V_a)$. This implies that $K \subseteq \bigcup_{a\in A} f^{-1}(V_a)$. But f is continuous, so for each $a \in A$, $f^{-1}(V_a)$ is open since V_a is open, hence $\{f^{-1}(V_a)\}_{a\in A}$ is an open cover of K. By compactness, there exist $a_1, \dots, a_N \in A$ such that $K \subseteq \bigcup_{j=1}^{N} f^{-1}(V_{a_j})$. However, if $y \in f(K)$, then there exists $x \in K$ such that $f(x) = y$, for which there exists $j \in \{1, \dots, N\}$ such that $x \in f^{-1}(V_{a_j})$ by the previous covering, which shows that $f(K) \subseteq \bigcup_{j=1}^{N} V_{a_j}$, so we have found a finite subcover of $\{V_a\}_{a\in A}$. Therefore, $f(K)$ is compact. ■

Remark

The image of a closed set continuous function under a continuous function isn't necessarily closed, and neither is a bounded set (for example, $f(x) = \log x$ with $U = (0, 1)$ has $f(U) = (-\infty, 0)$ which is not bounded), but this theorem implies the image of closed and bounded sets under continuous functions is closed and bounded by the Heine-Borel theorem.

Corollary (Extreme Value Theorem)

Let $U \subseteq \mathbb{R}^n$ be open, with $f : U \to \mathbb{R}$ continuous on U. Suppose $K \subseteq U$ is compact. Then there exist $x_{min}, x_{max} \in K$ such that $f(x_{min}) \le f(x) \le f(x_{max})$ for all $x \in K$, that is, f attains its global maximum and minimum on K.

Proof:

Since K is compact, $f(K)$ is compact, and therefore closed and bounded by the Heine-Borel theorem. Let $M = \sup_{x\in K} f(x) = \sup f(K)$, which exists and is finite since $f(K)$ is bounded. Let $k \in \mathbb{N}$. Then $M - 1/k < M$, so there exists $x_k \in K$ such that $M - 1/k < f(x_k) \le M$. This gives a sequence (x_k) in K. Since K is bounded, (x_k) is bounded, so by Bolzano-Weierstrass, it has a convergent subsequence $(x_{k_j})_{j\in\mathbb{N}} \to x_{max}$ contained in K. But K is closed, so $x_{max} \in K$. Now, $M - 1/k_j \le f(x_{k_j}) \le M$, and $\lim_{j\to\infty} f(x_{k_j}) = f(\lim_{j\to\infty} x_{k_j}) = f(x_{max})$ by continuity, so by the squeeze theorem, $f(x_{max}) = M$. Therefore, we have $x_{max} \in K$ and $f(x) \le M = f(x_{max})$ for all $x \in K$. A symmetric argument using $L = \inf f(K)$ gives us $x_{min} \in K$ satisfying $f(x_{min}) = L \le f(x)$ for all $x \in K$. ■

Theorem

Suppose $U \subseteq \mathbb{R}^n$ is open with $f : U \to \mathbb{R}^m$ continuous. Let $E \subseteq U$ be connected. Then $f(E) \subseteq \mathbb{R}^m$ is connected.

Proof:

Suppose for the sake of contradiction that $f(E)$ is disconnected, so there exists a separation V_1, V_2 of $f(E)$. Then the properties of separations tell us $E \subseteq f^{-1}(V_1) \cup f^{-1}(V_2)$ with $E \cap f^{-1}(V_1), E \cap f^{-1}(V_2) \neq \emptyset$ (since $y \in V_1 \cap f(E)$, so there exists $x \in E$ such that $y = f(x)$), and $E \cap f^{-1}(V_1) \cap f^{-1}(V_2) = \emptyset$ (if not, there exists $x \in E \cap f^{-1}(V_1) \cap f^{-1}(V_2)$, so $f(x) = y \in f(E) \cap V_1 \cap V_2$, which contradicts V_1 and V_2 being a separation of $f(E)$). Finally, since f is continuous and V_1, V_2 are open, $f^{-1}(V_1)$ and $f^{-1}(V_2)$ are open, so we have made a separation of E, contradicting the fact it is connected. ■

Example

The unit circle S^1 is connected. Indeed, $f : \mathbb{R} \to \mathbb{R}^2$ given by $f(t) = (\cos t, \sin t)$ for all $t \in \mathbb{R}$ is continuous with $\mathbb{R}$ connected, so $f(\mathbb{R}) = S^1$ is connected.

Corollary (Intermediate Value Theorem)

Let $U \subseteq \mathbb{R}^n$ be an open set, with $E \subseteq U$ connected. Suppose $f : U \to \mathbb{R}$ is continuous. Then, if $x, y \in E$ with $f(x) < f(y)$, then for all $w \in (f(x), f(y))$, there exists $z \in E$ such that $f(z) = w$.

Proof:

By the previous theorem, $f(E) \subseteq \mathbb{R}$ is connected, and hence an interval. ■

Uniform Continuity

Definition

Let $U \subseteq \mathbb{R}^n$ be open, with $D \subseteq U$ and $f : U \to \mathbb{R}^m$ a function. We say f is uniformly continuous on D if and only if for all $\varepsilon > 0$, there exists $\delta > 0$ such that $x, y \in D$ and $||x-y|| < \delta$ implies $||f(x)-f(y)|| < \varepsilon$. The key point is we can pick δ without any information about x and y, only ε. Note that if f is uniformly continuous on D, then f is continuous for all $x \in D$.

Remark
The definition of uniform continuity depends on f and the set D. If f is uniformly continuous on D, then it is true that f is uniformly continuous on $\widetilde{D} \subseteq D$, but it is not necessarily true that f is uniformly continuous on $D \subseteq \widehat{D}$.

Example
Consider $f : R \setminus \{0\} \to \mathbb{R}$, $f(x) = 1/x, x \in \mathbb{R} \setminus \{0\}$. Consider the set $D = (0, 1]$. We claim $f(x)$ is not uniformly continuous on D. Indeed, choose $0 < \varepsilon < 1$. Note that $|f(1/n) - f(1/(n+1)) = |n-(n+1)| = 1 > \varepsilon$. On the other hand, $|1/n - 1/(n+1)| = \frac{1}{n(n+1)} < \delta$ for any $\delta > 0$ for n sufficiently large, so no choice of δ depending on just ε will work. Note that in this example, D is bounded but not closed, hence not compact.

Example
Consider $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2, x \in \mathbb{R}$. Take $D = [0, \infty)$. We claim f is not uniformly continuous on D. Let $0 < \varepsilon < 2$, and let $\delta > 0$ be arbitrary. We'll find $x, y \in D$ such that $|x - y| < \delta$, but $|f(x) - f(y)| > 2 > \varepsilon$. Let $x = \frac{2}{\delta}$, $y = \frac{2}{\delta} + \frac{\delta}{2}$. Then $|x - y| = \delta/2 < \delta$, but $|f(x) - f(y)| = |x^2 - y^2| = |x + y||x - y| = (\frac{4}{\delta} + \frac{\delta}{2})\frac{\delta}{2} > 2 > \varepsilon$. Note that in this example, D is cloesd but not bounded, hence not compact.

Theorem
Let $f : U \underset{\text{open}}{\subseteq} \mathbb{R}^n \to \mathbb{R}^m$ be continuous, and let $K \subseteq U$ be compact. Then f is uniformly continuous on K.
Proof:
Let $\varepsilon > 0$. For each $x \in K$, there exists $\delta(x) > 0$ such that $f\left(B_{\delta(x)}(x)\right) \subseteq B_{\varepsilon/2}(f(x))$ by continuity of f at x. Now, $\{B_{\delta(x)/2}(x)\}_{x \in K}$ is an open cover of K, so by compactness there exist $x_1, \ldots, x_N \in K$ such that $K \subseteq \bigcup_{j=1}^{N} B_{\delta(x_j)/2}(x)$. Let $\delta = \min\{\delta(x_1)/2, \ldots, \delta(x_N)/2\}$. We claim this δ works. Suppose $x, y \in K$ with $||x - y|| < \delta$. Because $\{B_{\delta(x_j)/2}(x_j)\}_{j=1}^{N}$ is a cover of K, there exists $j \in \{1, \ldots, N\}$ such that $x \in B_{\delta(x_j)/2}(x_j)$. Then

$$||y - x_j|| \leq ||y - x|| + ||x - x_j|| < \delta + \delta(x_j)/2 \leq \delta(x_j)/2 + \delta(x_j)/2 = \delta(x_j)$$

so $y \in B_{\delta(x_j)/2}(x_j)$. But note that by what we wrote at the start of the proof, $f\left(B_{\delta(x_j)/2}(x_j)\right) \subseteq f\left(B_{\delta(x_j)}(x_j)\right) \subseteq B_{\varepsilon/2}(f(x_j))$. Therefore,

$$||f(x) - f(y)|| \leq ||f(x) - f(x_j)|| + ||f(x_j) - f(y)|| < \varepsilon/2 + \varepsilon/2 = \varepsilon \quad \blacksquare$$

Partial Derivatives

We'll finally use the fact that $\mathbb{R}^n$ and $\mathbb{R}^m$ are vector spaces. For $U \subseteq \mathbb{R}^n$ open and $f : U \to \mathbb{R}^m$ with $x_0 \in U$, we'll eventually define what it means to say f is differentiable at x_0. We'll see that the "derivative" of f at x_0 (when it exists), $(Df)_{x_0}$, is a linear map from $\mathbb{R}^n$ to $\mathbb{R}^m$, that is, an $m \times n$ matrix.

Review of Differentiability when $n = m = 1$:
Let $f : U \underset{\text{open}}{\subseteq} \mathbb{R} \to \mathbb{R}$ be a function with $x_0 \in U$. We say f is differentiable at x_0 if and only if $\lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}$ exists.

Recall that if f is differentiable at x_0, then f is continuous at x_0. To see this,

$$\underbrace{f(x)-f(x_0)}_{\to 0}=\underbrace{\frac{f(x)-f(x_0)}{x-x_0}}_{\to f'(x_0)}\underbrace{(x-x_0)}_{\to 0}$$

However, there are continuous functions that are not differentiable at some (or even all) points. In the general case, $f: U \subseteq \mathbb{R}^n \to \mathbb{R}^m$, whatever ends up being our definition of differentiability, it should imply continuity. For now, we'll try with $m=1$.

Definition

Let $f: U \underset{\text{open}}{\subseteq} \mathbb{R}^n \to \mathbb{R}$, let $a=(a_1,\dots,a_n)\in U$, and let $j\in\{1,\dots,n\}$. We define the partial derivative of f at a in the x_j direction to be

$$\frac{\partial f}{\partial x_j}(a) := \lim_{t\to 0}\frac{f(a+te_j)-f(a)}{t} = \lim_{t\to 0}\frac{f(a_1,\dots,a_{j-1},a_j+t,a_{j+1},\dots,a_n)-f(a_1,\dots,a_n)}{t}$$

if the limit exists, where $\{e_1,\dots,e_n\}$ is the standard basis of $\mathbb{R}^n$, which is the one-dimensional derivative of the function $x_j \mapsto f(a_1,\dots,a_{j-1},x_j,a_{j+1},\dots,a_n)$ at the point $x_j=a_j$.

Remark

To compute or check the existence of a partial derivative in the x_j direction, one sees from the definition that they take an ordinary derivative where all other variables x_i for $i\neq j$ are treated as constant.

Example

Let $f:\mathbb{R}^2\to\mathbb{R}$ be given by $f(x,y)=\sin(xy), (x,y)\in\mathbb{R}^2$. Then for any $(x_0,y_0)\in\mathbb{R}^2$,

$$\frac{\partial f}{\partial x}(x_0,y_0)=y_0\cos(x_0y_0)$$

$$\frac{\partial f}{\partial y}(x_0,y_0)=x_0\cos(x_0,y_0)$$

with $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ existing at any point in $\mathbb{R}^2$.

Example

Let $U=\{(x,y,z)\in\mathbb{R}^3 : y+z>0\}$, and define $g: U\to\mathbb{R}$ by $g(x,y,z)=e^{x^2z}\log(y+z)$ for all $(x,y,z)\in U$. Then for all $(x,y,z)\in U$,

$$\frac{\partial g}{\partial x}(x,y,z)=2xze^{x^2z}\log(y+z)$$

$$\frac{\partial g}{\partial y}(x,y,z)=e^{x^2z}\frac{1}{y+z}$$

and

$$\frac{\partial g}{\partial z}(x,y,z)=x^2e^{x^2z}\log(y+z)+e^{x^2z}\frac{1}{y+z}$$

Notation

Like $\frac{d}{dx}$ for single variable differentiation, we write $\frac{\partial}{\partial x_j}$ to represent the operation of taking partial derivatives with respect to x_j.

Examples

$$\frac{\partial}{\partial y}(y^3\sin(xz)+e^{13z+x^3\log(z^2+1)})=3y^2\sin(xz)$$

$$\frac{\partial}{\partial x}(x^2\sin(y)-\frac{yz}{x}) = 2x\sin(y)+\frac{yz}{x^2}$$
$$\frac{\partial}{\partial x}\sin(xy) = y\cos(xy)$$
$$\frac{\partial}{\partial y}\sin(xy) = x\cos(xy)$$
$$\frac{\partial}{\partial x}(x^2\sin y-\frac{yz}{x}) = 2x\sin y+\frac{yz}{x^2}$$
$$\frac{\partial}{\partial y}(x^2\sin y-\frac{yz}{x}) = x^2\cos y-\frac{z}{x}$$
$$\frac{\partial}{\partial z}(x^2\sin y-\frac{yz}{x}) = -\frac{y}{x}$$

Definition

We can take second order partial derivatives, and more generally partial derivatives of higher orders. For a function f,

$$\frac{\partial^2 f}{\partial x\partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right)$$
$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right)$$
$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right)$$
$$\frac{\partial^2 f}{\partial y\partial x} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)$$

Examples

Let $f(x,y)=\sin(xy)$. Then,

$$\frac{\partial^2 f}{\partial x^2} = -y^2\sin(xy)$$
$$\frac{\partial^2 f}{\partial y^2} = -x^2\sin(xy)$$
$$\frac{\partial^2 f}{\partial x\partial y} = \cos(xy)-xy\sin(xy)$$
$$\frac{\partial^2 f}{\partial y\partial x} = \cos(xy)-xy\sin(xy)$$

Notice in this case the last two “mixed” partial derivatives match. This isn’t special to just this case: let $g(x,y,z)=x^2\sin(y)-\frac{yz}{x}$. Then,

$$\frac{\partial^2 g}{\partial x^2} = 2\sin(y)-\frac{2yz}{x^3}$$
$$\frac{\partial^2 g}{\partial y\partial x} = 2x\cos(y)+\frac{z}{x^2}$$
$$\frac{\partial^2 g}{\partial z\partial x} = \frac{y}{x^2}$$
$$\frac{\partial^2 g}{\partial x\partial y} = 2x\cos y+\frac{z}{x^2}$$
$$\frac{\partial^2 g}{\partial z\partial y} = -\frac{1}{x}$$

$$\frac{\partial^2 g}{\partial y^2} = -x^2 \sin(y)$$
$$\frac{\partial^2 g}{\partial x \partial z} = \frac{y}{x^2}$$
$$\frac{\partial^2 g}{\partial y \partial z} = -\frac{1}{x}$$
$$\frac{\partial^2 g}{\partial z^2} = 0$$

In all of these cases, we see the mixed partials match. This isn't a coincidence.

Definition

Let $f : U \underset{\text{open}}{\subseteq} \mathbb{R}^n \to \mathbb{R}$ be a function. We say that
(1) $f \in C^0(U)$ if f is continuous on U.
(2) $f \in C^1(U)$ if $f \in C^0(U)$ and each $\frac{\partial f}{\partial x_i}$ exists for each $i = 1, \ldots, n$ and is continuous on U (that is, $\frac{\partial f}{\partial x_i} \in C^0(U)$).
(3) $f \in C^k(U)$, read "k times continuously differentiable", for $k \in \mathbb{N}$ if $f \in C^{k-1}(U)$ and each $\frac{\partial^k f}{\partial x_{i_1} \ldots \partial x_{i_k}}$ exists and is continuous on U.
(4) $f \in C^\infty(U)$, also called smooth, if $f \in C^k(U)$ for all $k \in \mathbb{N}$.

Remark

Each $C^0(U), C^1(U), \ldots, C^k(U), \ldots,$ and $C^\infty(U)$ are real vector spaces (all infinite dimensional), with

$$C^0(U) \supsetneq C^1(U) \supsetneq \cdots \supsetneq C^\infty(U)$$

Theorem (Clairaut)

Let $f : U \underset{\text{open}}{\subseteq} \mathbb{R}^n \to \mathbb{R}$ be a function with $a \in U$, and let $j, k \in \{1, \ldots, n\}$. Suppose $\frac{\partial f}{\partial x_j}$ and $\frac{\partial f}{\partial x_k}$ both exist and are continuous in some open neighbourhood of $a \in U$. Moreover suppose $\frac{\partial^2 f}{\partial x_j \partial x_k}$ exists in an open neighbourhood of a and is continuous at a. Then $\frac{\partial^2 f}{\partial x_k \partial x_j}$ exists at a and equals $\frac{\partial^2 f}{\partial x_j \partial x_k}$.

Remark:

In practice, this theorem applies with more than these minimal hypotheses, in particular, if $f \in C^2(U)$, then all hypotheses are satisfied and $\frac{\partial^2 f}{\partial x_j \partial x_k} = \frac{\partial^2 f}{\partial x_k \partial x_j}$ everywhere on U.

Proof:

First, let's show it is enough to prove the case where $n = 2, a = 0$. Let $s, t \in \mathbb{R}$ be small enough in absolute value such that the function $h(s, t) = f(a + se_j + te_k)$ is defined, which is possible since U is open and $a \in U$. Then, for all s_0, t_0 such that $a + s_0 e_j + t_0 e_k \in U$,

$$\frac{\partial h}{\partial s}(s_0, t_0) = \lim_{\varepsilon \to 0} \frac{h(s_0 + \varepsilon, t_0) - h(s_0, t_0)}{\varepsilon} = \lim_{\varepsilon \to 0} \frac{f(a + (s_0 + \varepsilon)e_j + t_0 e_k) - f(a + s_0 e_j + t_0 e_k)}{\varepsilon} = \frac{\partial f}{\partial x_j}(a + s_0 e_j + t_0 e_k)$$

Similarly,

$$\frac{\partial h}{\partial t}(s_0, t_0) = \frac{\partial f}{\partial x_k}(a + s_0 e_j + t_0 e_k)$$

Note that this holds for any f. In particular, applying this argument to $\frac{\partial f}{\partial x_j}$ or $\frac{\partial f}{\partial x_k}$ (which exist by hypothesis), we get

$$\frac{\partial^2 h}{\partial t \partial s}(0, 0) = \frac{\partial^2 f}{\partial x_k \partial x_j}(a)$$

and similarly

$$\frac{\partial^2 h}{\partial s \partial t}(0, 0) = \frac{\partial^2 f}{\partial x_j \partial x_k}(a)$$

In essence, we are just moving the coordinate system and forgetting the other variables. Therefore, we see it suffices to prove the theorem for h, i.e. the case $n = 2, a = 0$.
Choose $(s,t) \in U$ sufficiently close to $(0,0)$ such that $(s,0),(0,t) \in U$ as well. Define

$$H(s,t) = (h(s,t) - h(s,0)) - (h(0,t) - h(0,0))$$

For fixed t, define $K_t(s) = h(s,t) - h(s,0)$. Then $H(s,t) = K_t(s) - K_t(0)$. Then, for $|s|$ sufficiently small for everything to be defined, we have

$$K_t'(s) = \frac{\partial h}{\partial s}(s,t) - \frac{\partial h}{\partial s}(0,0)$$

By hypothesis, we have that K_t is continuous and differentiable near $s = 0$. Therefore, by the Mean Value Theorem applied to K_t on $[0,s]$, there exists $\delta \in (0,1)$ (so $\delta s \in (0,s)$) such that

$$K_t(s) - K_t(0) = K_t'(\delta s)s = s\left(\underbrace{\frac{\partial h}{\partial s}(\delta s,t)}_{\lambda(t)} - \underbrace{\frac{\partial h}{\partial s}(\delta s,0)}_{\lambda(0)}\right)$$

Fix s, and let $\lambda(t) = \frac{\partial h}{\partial s}(\delta s,t)$. Then

$$\lambda'(t) = \frac{\partial^2 h}{\partial t \partial s}(\delta s,t)$$

which exists near $t = 0$ by hypothesis, and so λ is continuous and differentiable near $t = 0$. Therefore, by the Mean Value Theorem applied to λ on $[0,t]$, there exists $\varepsilon \in (0,1)$ (so $\varepsilon t \in (0,t)$) such that

$$\lambda(t) - \lambda(0) = t\lambda'(\varepsilon t) = t\frac{\partial^2 h}{\partial t \partial s}(\delta s,\varepsilon t)$$

Altogether, our work above tells us

$$H(s,t) = st\frac{\partial^2 h}{\partial s \partial t}(\delta s,\varepsilon t)$$

with $\delta s \to 0$ as $s \to 0$ and $\varepsilon t \to 0$ as $t \to 0$, and so

$$\lim_{(s,t)\to(0,0)} \frac{H(s,t)}{st} = \lim_{(s,t)\to(0,0)} \frac{\partial^2 h}{\partial s \partial t}(\delta s,\varepsilon t) = \frac{\partial^2 h}{\partial s \partial t}(0,0)$$

We can also write H as

$$H(s,t) = \underbrace{(h(s,t) - h(0,t))}_{\mu_s(t)} - \underbrace{(h(s,0) - h(0,0))}_{\mu_S(0)}$$

Let $\mu_s(t) = h(s,t) - h(0,t)$ for fixed s. Then

$$\mu_s'(t) = \frac{\partial h}{\partial t}(s,t) - \frac{\partial h}{\partial t}(0,t)$$

exists for all $|t|$ sufficiently small, and is continuous and differentiable by hypothesis. Therefore, by the mean value theorem applied to μ on $[0,t]$, there exists $\theta \in (0,1)$ (so $\theta t \in (0,t)$) such that

$$\mu(t) - \mu(0) = t\mu'(\theta t)$$

and so by definition of H, we have

$$H(s,t) = t\left(\frac{\partial h}{\partial s}(s,\theta t) - \frac{\partial h}{\partial s}(0,\theta t)\right)$$

and so

$$\frac{H(s,t)}{st} = \frac{1}{s}\left(\frac{\partial h}{\partial s}(s,\theta t) - \frac{\partial h}{\partial s}(0,\theta t)\right)$$

Consider the path $(s,t) \to (0,0)$ obtained by first following the vertical line $(s,t) \to (s,0)$ then following the horizontal line $(s,0) \to (0,0)$. Since $\lim_{(s,t)\to(0,0)} \frac{H(s,t)}{st}$ exists as shown above, this path gives

$$\lim_{(s,t)\to(0,0)} \frac{H(s,t)}{st} = \lim_{s\to 0}\left(\lim_{t\to 0} \frac{H(s,t)}{st}\right) = \lim_{s\to 0} \frac{1}{s} \lim_{t\to 0}\left(\frac{\partial h}{\partial s}(s,\theta t) - \frac{\partial h}{\partial s}(0,\theta t)\right)$$

Since $\frac{\partial h}{\partial t}$ is continuous on a neighbourhood of $(0,0)$, as $t \to 0$, $\theta t \to 0$, and so we get $\frac{\partial h}{\partial t}(s,\theta t) \to \frac{\partial h}{\partial t}(s,0)$ and $\frac{\partial h}{\partial t}(0,\theta t) \to \frac{\partial h}{\partial t}(0,0)$. Therefore,

$$\lim_{(s,t)\to(0,0)} \frac{H(s,t)}{st} = \lim_{s\to 0} \frac{1}{s}\left(\frac{\partial h}{\partial t}(s,0) - \frac{\partial h}{\partial t}(0,0)\right) = \frac{\partial^2 h}{\partial s \partial t}(0,0)$$

By uniqueness of limits (we showed above this is $\frac{\partial^2 h}{\partial t \partial s}(0,0)$), we get $\frac{\partial^2 h}{\partial s \partial t}(0,0) = \frac{\partial^2 h}{\partial t \partial s}(0,0)$. ■

Corollary

If $f \in C^k(U)$, the kth partial derivatives don't depend on the order they are taken in.
Proof:
Assignment 5. ■

Definition

Let $U \subseteq \mathbb{R}^n$ be open, and let $f : U \to \mathbb{R}^m$ be a function. Writing the component functions of f as $f = (f_1, \ldots, f_m)$, we define

$$\frac{\partial f}{\partial x_i} = \left(\frac{\partial f_1}{\partial x_i}, \ldots, \frac{\partial f_m}{\partial x_i}\right)$$

That is, for $a \in U$,

$$\frac{\partial f}{\partial x_i}(a) = \lim_{h\to 0} \frac{f(a + he_i) - f(a)}{h} = \lim_{h\to 0} \frac{1}{h}\left(f_1(a + he_i), \ldots, f_m(a + he_i)\right)$$

We define $f \in C^k(U)$ if we have $f_i \in C^k(U)$ for each $i \in \{1, \ldots, m\}$, and Clairaut's theorem above shows that for $f \in C^2(U)$, we have

$$\frac{\partial^2 f}{\partial x_j \partial x_k} = \frac{\partial^2 f}{\partial x_k \partial x_j}$$

for all $1 \leq j, k \leq n$.

Differentiability

Let $f : U \underset{\text{open}}{\subseteq} \mathbb{R}^n \to \mathbb{R}^m$ be a function, with $x_0 \in U$. We want to define what it means for f to be differentiable at x_0. For our definition to make sense, we would expect that f should be:
(1) continuous at x_0
(2) all partial derivatives of f should exist at x_0.
What if we take our definition to be "all partial derivatives exist"?

Example

Define $f : \mathbb{R}^2 \to \mathbb{R}$ by $f(x,y) = x + y$ if $x = 0$ or $y = 0$, and $f(x,y) = 1$ otherwise for all $(x,y) \in \mathbb{R}^2$. Then,

$$\frac{\partial f}{\partial x}(0,0) = \lim_{h\to 0} \frac{f(0+h,0) - f(0,0)}{h} = \lim_{h\to 0} \frac{(0+h) - (0+0)}{h} = 1$$

and similarly $\frac{\partial f}{\partial y}(0,0) = 1$, so f's partial derivatives exist, but $\lim_{h\to 0} f(h,h) = 1 \neq 0 = f(0,0)$, so f is not continuous at the origin. Therefore, we see our guess doesn't work. We'll see on Assignment 5 that the version of this guess where we use directional derivatives instead doesn't work either.

Let's go back to the original definition of the derivative for $f : U \underset{\text{open}}{\subseteq} \mathbb{R} \to \mathbb{R}$ with $x_0 \in U$. We say f is differentiable at x_0 if and only if $\lim_{h\to 0} \frac{f(x_0+h)-f(x_0)}{h}$ exists, and we call this limit $f'(x_0)$, which is a real number, or in other words, a 1×1 matrix! Consider the linear map $T_{x_0} : \mathbb{R} \to \mathbb{R}$ given by $T_{x_0}(h) = f'(x_0)\cdot h$. Then,

$$\lim_{h\to 0} \frac{f(x_0+h)-f(x_0)}{h} = f'(x_0) = \frac{T_{x_0}(h)}{h} = \lim_{h\to 0} \frac{T_{x_0}(h)}{h}$$

which is equivalent to

$$\lim_{h\to 0} \left(\frac{f(x_0+h)-f(x_0)-T_{x_0}(h)}{h} \right) = 0$$

Definition:
Let $f : U \subseteq \mathbb{R}^n \to \mathbb{R}^m$ (U open), with $x_o \in U$. we say f is differentiable at x_o if and only there exists a linear map $T : \mathbb{R}^n \to \mathbb{R}^m$ such that

$$\lim_{h\to \mathbf{0}} \frac{f(x_o+h)-f(x_o)-T(h)}{||h||} = \mathbf{0}$$

.
This says that even though $||h|| \to 0$ as $h \to \mathbf{0}$, the numerator $f(x_o+h)-f(x_o)-T(h)$ goes to zero much faster as $h \to \mathbf{0}$.
Also, we say f is differentiable on U if and only if it is differentiable for all $x_o \in U$.

Remark:
This definition is equivalent to

$$\lim_{h\to \mathbf{0}} \frac{||f(x_o+h)-f(x_o)-T(h)||}{||h||} = 0$$

and

$$\lim_{x\to x_o} \frac{||f(x)-f(x_o)-T(x-x_o)}{||x-x_o||} = 0$$

We'll soon show that if such a linear map T exists, it is necessarily unique. But it need not exist.
If f is differentiable at $x_o \in U$, then this map $T : \mathbb{R}^n \to \mathbb{R}^m$ is denoted $(Df)_{x_o}$ and is called the derivative or differential or linearization of f at x_o, and is an $m\times n$ matrix of real numbers.

Proposition:
If f is differentiable at x_o, it is continuous at x_o.
Proof:
We want to show $\lim_{x\to x_o} f(x) = f(x_o)$. Since f is differentiable at x_o,

$$\lim_{x\to x_o} \frac{\overbrace{||f(x)-f(x_o)-(Df)_{x_o}(x-x_o)||}^{Q(x)}}{||x-x_o||} = 0$$

so $\lim_{x\to x_o} \frac{||Q(x)||}{||x-x_o||} = 0$, so $\lim_{x\to x_o} ||Q(x)|| = 0$.
But now

$$\begin{aligned} ||f(x)-f(x_o)|| &= ||f(x_o)+(Df)_{x_o}(x-x_o)+Q(x)-f(x_o)|| \\ &\leq ||(Df)_{x_o}(x-x_o)|| + ||Q(x)|| \\ &\leq ||(Df)_{x_o}|| ||x-x_o|| + ||Q(x)|| \end{aligned}$$

Now let $x \to x_o$ and use the above limit and the squeeze theorem to get the result. $\blacksquare$

So if f is not continuous at x_o, it can't be differentiable at x_o.

Theorem:
Suppose $f : U \subseteq \mathbb{R}^n \to \mathbb{R}^m$ (U open) is differentiable at $x_o \in U$. Then the derivative $(Df)_{x_o}$ is the $m \times n$ matrix whose (i,j) entry is $\frac{\partial f_i}{\partial x_j}(x_o)$ for $1 \leq i \leq m$, $1 \leq j \leq n$.
In particular, this theorem tells us:
(a) If f is differentiable at x_o, then all its partial derivatives exist at x_o.
(b) That $(Df)_{x_o}$ is uniquely determined.
This $m \times n$ matrix $[(Df)_{x_o}]_{ij} = \frac{\partial f_i}{\partial x_j}(x_o)$ is also called the Jacobian matrix of f at x_o.
Proof:
Suppose f is differentiable at x_o. Let $T = (Df)_{x_o} = [T(e_1) \dots T(e_n)]$. Thus $0 = \lim_{h \to 0} \frac{||f(x_o+h)-f(x_o)-T(h)||}{||h||}$.
Choose a path $h = te_j$ for $t \in \mathbb{R}$ as $t \to 0$. So

$$\lim_{t \to 0} \frac{||f(x_o + te_j) - f(x_o) - T(te_j)||}{||te_j||} = 0$$

$$= \lim_{t \to 0} \left\| \frac{f(x_o + te_j) - f(x_o)}{t} - T(e_j) \right\| = 0$$

$$\iff \frac{\partial f}{\partial x_j}(x_o) := \lim_{t \to 0} \frac{f(x_o + te_j) - f(x_o)}{t} = T(e_j) \quad \blacksquare$$

Lemma:
Let $f : U \subseteq \mathbb{R}^n \to \mathbb{R}^m$ with U open. Let $f_1, \dots, f_m$ be the component functions of f. Let $x_o \in U$. Then f is differentiable if and only if each f_j is differentiable and $(Df)_{x_o} = \begin{bmatrix} (Df_1)_{x_o} \\ \vdots \\ (Df_m)_{x_o} \end{bmatrix}$.
Proof:

$$\lim_{h \to 0} \frac{||f(x_o + h) - f(x_o) - T(h)||}{||h||} = 0$$

$$\iff \lim_{h \to 0} \frac{||f_j(x_o + h) - f(x_o) - T(h)||}{||h||} = 0$$

for all $j = 1, \dots, m$. The rest follows from the previous theorem. $\blacksquare$

Remark:
If f is real-valued and differentiable at x_o, then its derivative (a $1 \times n$ matrix) is often called the gradient of f at x_o and is denoted $(\nabla f)(x_o)$ or $(\operatorname{grad} f)(x_o)$. It is very closely related to directional derivatives (as shown on assignments).

Theorem:
Let $f : U \subseteq \mathbb{R}^n \to \mathbb{R}^m$ (U open) with $a \in U$. Suppose each $\frac{\partial f_j}{\partial x_i}$ exists on an open neighbourhood of a and is continuous at a. Then f is differentiable at a.

Remark:
This is sufficient not necessary.

Corollary:
If $f \in C^1(U)$, then f is differentiable on U.

Proof:
The previous lemma shows without loss of generality we can take $m = 1$. Let $a = (a_1, \dots, a_n)$, $h = (h_1, \dots, h_n) \in \mathbb{R}^n$. Define $v_0 = \mathbf{0}$, $v_1 = (h_1, 0, \dots, 0), \dots, v_j = (h_1, \dots, h_j, 0, \dots, 0), \dots, v_n = h$. Write

$$f(a+h)-f(a) = f(a+v_n)-f(a+v_o) = \sum_{k=1}^{n}(f(a+v_k)-f(a+v_{k-1})) = \sum_{k=1}^{n}(f(a+v_{k-1}+h_ke_k)-f(a+v_{k-1}))$$

Since each $\frac{\partial f}{\partial x_k}$ exists in an open neighbourhood of a for h sufficiently close to 0, $\mu_k(t) = f(a+v_{k-1}+te_k) = f(a_1+h_1, \dots, a_{k-1}+h_{k-1}, a_k+t, a_{k+1}, \dots, a_n)$ is differentiable on $[0, h_k]$ for h_k sufficiently small. Apply the Mean Value Theorem to μ_k on $[0, h_k]$ so there exists $\varepsilon_k \in (0,1)$ so $\varepsilon_k h_k \in (0, h_k)$ such that $\mu_k'(\varepsilon_k h_k)(h_k - 0) = \mu_k(h_k) - \mu_k(0) = f(a+v_k) - f(a+v_{k-1})$, where the left term is $h_k \frac{\partial f}{\partial x_k}(a+v_{k-1}+\varepsilon_k h_k e_k)$ (which is the same as in the proof of Clairaut's Theorem). So

$$f(a+h) - f(a) = \sum_{k=1}^{n}(f(a+v_{k-1}+h_ke_k) - f(a+v_{k-1})) = \sum_{k=1}^{n} h_k \frac{\partial f}{\partial x_k}(a + v_{k-1} + \varepsilon_k h_k e_k)$$

Now, we want to prove f is differentiable at a, and we know that if it is differentiable, the derivative $(Df)_a$ must be the $1 \times n$ matrix, the gradient. So therefore, we need to check that

$$\lim_{h \to \mathbf{0}} \frac{f(a+h) - f(a) - \overbrace{(Df)_a(h)}^{=\sum_{k=1} \frac{\partial f}{\partial x_k}(a)h_k}}{||h||} \stackrel{?}{=} 0$$

$$= \lim_{h \to \mathbf{0}} \frac{1}{||h||} \left(\sum_{k=1}^{n} h_k \left(\frac{\partial f}{\partial x_k}(a + v_{k-1} + \varepsilon_k h_k) - \frac{\partial f}{\partial x_k}(a) \right) \right)$$

Write $L_k = \frac{\partial f}{\partial x_k}(a + v_{k-1} + \varepsilon_k h_k) - \frac{\partial f}{\partial x_k}(a)$, and $L = (L_1, \dots, L_n)$. Then, the above equals

$$= \lim_{h \to \mathbf{0}} \frac{1}{||h||} L \cdot h$$

But now

$$\left| \frac{1}{||h||} L \cdot h \right| = \frac{1}{||h||} |L \cdot h| \underbrace{\leq}_{CS} ||L||$$

so if we have $\lim_{h \to \mathbf{0}} L = \mathbf{0} \iff \lim_{h \to \mathbf{0}} L_k = 0$, then by the squeeze theorem, we will have the result. But $L_k = \frac{\partial f}{\partial x_k}(a + v_{k-1} + \varepsilon_k h_k) - \frac{\partial f}{\partial x_k}(a)$. As $h \to \mathbf{0}$, $v_{k-1} \to \mathbf{0}$ (by definition of its components). $0 < \varepsilon_k h_k < h_k$, so $\varepsilon_k h_k \to 0$, so $a + v_{k-1} + \varepsilon_k h_k \to a$ as $h \to \mathbf{0}$. But since $\frac{\partial f}{\partial x_k}$ is continuous at a, we have

$$\lim_{h \to 0} \frac{\partial f}{\partial x_k}(a + v_{k-1} + \varepsilon_k h_k) = \frac{\partial f}{\partial x_k}(a)$$

so $\lim_{h \to 0} L_k = 0$. ■

Properties of the Derivative:
Suppose $f, g : U \subseteq \mathbb{R}^n \to \mathbb{R}^m$ with U open. Let $a \in U$. Suppose f, g are both differentiable at a. Let $\lambda, \mu \in \mathbb{R}$. Then $\lambda f + \mu g$ is differentiable at a, with $(D(\lambda f + \mu g))_a = \lambda (Df)_a + \mu (Dg)_a$ (proved on assignment).

We know the space of functions $U \to \mathbb{R}^m$ is an infinite dimensional real vector space. This result says that the differentiable functions at $a \in U$ form a (still generally infinite dimensional) linear subspace of the function space and the map bringing $f \mapsto (Df)_a$ is a linear map of vector spaces.

Theorem (Product Rule):
Let $f, g : U \subseteq \mathbb{R}^n \to \mathbb{R}^m$ be differentiable at $a \in U$ (with U open). Then $f \cdot g : \mathbb{R}^n \to \mathbb{R}$ (more generally any inner product) is differentiable at a, and $(D(f \cdot g))_a = \underbrace{f(a)^T}_{1\times m} \underbrace{(Dg)_a}_{m\times n} + g(a)^T (Df)_a$ (more generally whatever this is with the appropriate inner product).
Proof:
Let $h = f \cdot g$. So $h = \sum_{k=1}^m f_k g_k$. Thus, by the product rule for partial derivatives,

$$\frac{\partial h}{\partial x_i} = \sum_{k=1}^n \left(\frac{\partial f_k}{\partial x_i} g_k + f_k \frac{\partial g}{\partial x_i}\right)$$

Thus

$$(Dh)_{a_i} = \sum_{k=1}^n \left(g_k(a) \frac{\partial f_k}{\partial x_i}(a) + f_k(a) \frac{\partial g_k}{\partial x_i}(a)\right) = (g(a)^T (Df)_a)_i + (f(a)^T (Dg)_a)_i$$

so if h is indeed differentiable at a, then $(Dh)_a$ must be as specified in the result.
Now we check

$$\lim_{t\to 0} \frac{||h(a+t) - h(a) - (Dh)_a(t)||}{||t||} \stackrel{?}{=} 0$$

We have

$$h(a+t) - h(a) - (Dh)_a(t)$$

$$= \underbrace{f(a+t) \cdot g(a+t)}_{1} - \underbrace{f(a) \cdot g(a)}_{2} - \underbrace{f(a) \cdot (Dg)_a(t)}_{3} - \underbrace{g(a) \cdot (Df)_a(t)}_{4}$$

$$= (\underbrace{f(a+t)}_{1} - f(a) - (Df)_a(t)) \cdot g(a+t) + f(a) \cdot (g(a+t) - \underbrace{g(a)}_{2} - \underbrace{(Dg)_a(t)}_{3}) + (Df)_a(t) \cdot (g(a+t) - \underbrace{g(a)}_{4})$$

Let $T_1(t) = (f(a+t) - f(a) - (Df)_a(t)) \cdot g(a+t)$, $T_2(t) = f(a) \cdot (g(a+t) - g(a) - (Dg)_a(t))$, $T_3(t) = (Df)_a(t) \cdot (g(a+t) - g(a))$.
We want to show

$$\lim_{t\to 0} \frac{|T_1(t) + T_2(t) + T_3(t)|}{||t||} = 0$$

We have

$$\frac{|T_1(t) + T_2(t) + T_3(t)|}{||t||} \leq \frac{|T_1(t)|}{||t||} + \frac{|T_2(t)|}{||t||} + \frac{|T_3(t)|}{||t||}$$

But let's consider each individual limit:

$$\lim_{t\to 0} \frac{|T_1(t)|}{||t||} \overbrace{\leq}^{CS} \lim_{t\to 0} \frac{||f(a+t) - f(a) - (Df)_a(t)||}{||t||} ||g(a+t)|| \to 0$$

where the left term goes to zero since f is differentiable, and the right term goes to $g(a)$ since g is continuous since it is differentiable.

$$\lim_{t\to 0} \frac{|T_2(t)|}{||t||} \overbrace{\leq}^{CS} \lim_{t\to 0} \frac{||g(a+t) - g(a) - (Dg)_a(t)||}{||t||} ||f(a)|| \to 0$$

since g is differentiable.

$$\lim_{t\to 0} \frac{|T_2(t)|}{||t||} \overbrace{\leq}^{CS} \frac{||(Df)_a(t)|| ||g(a+t) - g(a)||}{||t||} \leq \frac{||(Df)_a|| ||t|| ||g(a+t) - g(a)||}{||t||} \to 0$$

since g is continuous. ■

The $m = 1$ special case is $(D(fg))_a = f(a)(Dg)_a + g(a)(Df)_a$, which is just the product rule.

Theorem (Chain Rule):
Let $f : U \subseteq \mathbb{R}^n \to \mathbb{R}^m$, $g : V \subseteq \mathbb{R}^m \to \mathbb{R}^p$ where U, V open, $f(U) \subseteq V$. If f is differentiable at $a \in U$, and g is differentiable at $f(a) \in V$, then $g \circ f$ is differentiable at a and $(D(g \circ f))_a = (Dg)_{f(a)}(Df)_a$.
Proof:
Let $b = f(a)$. By hypothesis,

$$\lim_{h \to \mathbf{0}} \frac{||\overbrace{f(a+h) - f(a) - (Df)_a(h)}^{Q_1(h)}||}{||h||} = 0$$

$$\lim_{k \to \mathbf{0}} \frac{||\overbrace{g(b+k) - g(b) - (Dg)_b(k)}^{Q_2(k)}||}{||k||} = 0$$

For h small enough so $f(a+h)$ is defined, let $k = f(a+h) - f(a) = f(a+h) - b \to 0$ as $h \to \mathbf{0}$ by continuity of f. We have $f(a+h) = b + k$. So

$$\begin{aligned} g(f(a+h)) - g(f(a)) &= g(b+k) - g(b) \\ &= (Dg)_b(k) + Q_2(k) \\ &= (Dg)_b((Df)_a(h) + Q_1(h)) + Q_2(k) \\ &= ((Dg)_b(Df)_a)(h) + (Dg)_b Q_1(h) + Q_2(k) \end{aligned}$$

hence

$$\begin{aligned} &\lim_{h \to 0} \frac{||g(f(a+h)) - g(f(a)) - (Dg)_b(Df)_a(h)||}{||h||} \\ &= \lim_{h \to 0} \frac{||(Dg)_b Q_1(h) + Q_2(k)||}{||h||} \\ &\leq \lim_{h \to 0} \frac{||(Dg)_b|| ||Q_1(h)||}{||h||} + \frac{||Q_2(k)||}{||h||} \end{aligned}$$

The left term goes to 0 by differentiability. It remains to show $\lim_{h \to 0} \frac{||Q_2(k)||}{||h||} \stackrel{?}{=} 0$. Let $\varepsilon_1 > 0$ be arbitrary. Since $\lim_{h \to 0} \frac{||Q_1(h)||}{||h||} = 0$ by differentiability, there exists $\delta_1 > 0$ such that $0 < ||h|| < \delta_1$ implies $\frac{||Q_1(h)||}{||h||} < \varepsilon_1$. So $||Q_1(h)|| < \varepsilon_1 ||h||$ for $0 < ||h|| < \delta_1$.
But now

$$\begin{aligned} ||k|| &= ||f(a+h) - f(a)|| \\ &= ||(Df)_a(h) + Q_1(h)|| \\ &\leq ||(Df)_a|| ||h|| + ||Q_1(h)|| \\ &\leq ||h|| (||(Df)_a|| + \varepsilon_1) \end{aligned}$$

given $||h||$ sufficiently small. So if $0 < ||h|| < \delta_1$, there exists $C > 0$ such that $k \leq C||h||$.
Since $\lim_{k \to 0} \frac{||Q_2(k)||}{||k||} = 0$, given arbitrary $\varepsilon_2 > 0$, there exists $\delta_2 > 0$ such that $0 < ||k|| < \delta_2$ implies $||Q_2(k)|| < \varepsilon_2 ||k||$. Let $\delta = \min\{\delta_1, \delta_2\}$. Then, if $0 < ||h|| < \delta$, then $||Q_2(k)|| < \varepsilon_2 ||k|| < \varepsilon_2 C ||h||$, so $\frac{||Q_2(k)||}{||h||} < \varepsilon_2 C$, so since $\varepsilon_2 > 0$ was arbitrary, we have $\lim_{h \to 0} \frac{||Q_2(k)||}{||h||} = 0$. ■

Taylor's Theorem and the Mean Value Theorem

Recall Taylor's theorem in 1 dimension:
Theorem (Taylor):

Let I be an open interval, $p \in \mathbb{N}$, and let $h : I \to \mathbb{R}$ be $p+1$ times differentiable on I. Let $t_o, t \in I$ with $t_o < t$. Then there exists $\theta \in (t_o, t)$ such that

$$h(t) = \sum_{k=0}^{p} \frac{h^{(k)}(t_o)}{k!}(t-t_o)^k + \frac{h^{(p+1)}(\theta)(t-t_o)^{p+1}}{(p+1)!}$$

where θ is not in general unique.

Remark:

$$\lim_{t \to t_o} \frac{h(t) - \sum_{k=0}^{p} \frac{h^{(k)}(t_o)}{k!}(t-t_o)^k}{(t-t_o)^p} = 0$$

When $p = 0$, this is the definition of differentiability, and Taylor's Theorem is the Mean Value Theorem in this case.

Proof of Taylor's Theorem in 1-dimension:
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Lemma:
Let $f : U \subseteq \mathbb{R}^n \to \mathbb{R}$. Let $p \in \mathbb{N}$. Suppose $f \in C^p(U)$. Let $a \in U$, let $\xi \in \mathbb{R}^n$ such that $a + t\xi \in U$ for all $t \in [0,1]$ (i.e. the line segment from a to $a+\xi$ is contained in U, which always holds if U is convex and $a + \xi \in U$). Let $g : [0,1] \to \mathbb{R}$ by $g(t) = f(a+t\xi)$. Then g is p times continuously differentiable, and

$$(\frac{d^k}{dt^k} g)(t) = \sum_{j_1,\ldots,j_k=1}^{n} \frac{\partial^k f}{\partial x_{j_1} \ldots \partial x_{j_k}}(a+t\xi)\xi_{j_1} \ldots \xi_{j_k}$$

for $1 \le k \le p$, $\xi = (\xi_1, \ldots, \xi_n)$.
Proof:
By induction on p. Let $p = 1$. Then, we want to show $\frac{dg}{dt}(t) = \sum_{j=1}^{n} \frac{\partial f}{\partial x_j}(a+t\xi)\xi_j$. This is true by the chain rule, f is differentiable (it is C^1), and let $h(t) = a + t\xi$ (for $h : [0,1] \to U$). Then $g(t) = f(a+t\xi) = (f \circ h)(t)$. Then, by the chain rule,

$$\underbrace{\frac{dg}{dt}(t)}_{1\times 1} = \underbrace{(Df)_{h(t)}}_{1\times n} \underbrace{\frac{dh}{dt}(t)}_{n\times 1}$$

$$= (\frac{\partial f}{\partial x_1}(a+t\xi) \ldots \frac{\partial f}{\partial x_n}(a+t\xi)) \begin{pmatrix} \frac{dh_1}{dt}(t) = \xi_1 \\ \vdots \\ \frac{dh_p}{dt}(t) = \xi_p \end{pmatrix}$$

$$= \sum_{j=1}^{n} \frac{\partial f}{\partial x_j}(a+t\xi)$$

Now let $p \ge 2$ and suppose the result holds for $p-1$. By the induction hypothesis, $\frac{d^k}{dt^k} g(t) = \sum_{j_1,\ldots,j_k=1}^{n} \frac{\partial^k f}{\partial x_{k_1} \ldots \partial x_{j_k}}(a+t\xi)\xi_{j_1} \ldots \xi_{j_k}$ for $1 \le k \le p-1$. Thus,

$$(\frac{d^p}{dt^p} g)(t) = \frac{d}{dt}(\frac{d^{p-1}}{dt^{p-1}} g)(t) = \frac{d}{dt}(\sum_{j_1,\ldots,j_{p-1}=1}^{n} \frac{\partial^{p-1} f}{\partial x_{k_1} \ldots \partial x_{j_{p-1}}}(a+t\xi)\xi_{j_1} \ldots \xi_{j_{p-1}})$$

The function $x = (x_1, \ldots, x_n) \mapsto \frac{\partial^{p-1} f}{\partial x_{j_1} \ldots \partial x_{j_{p-1}}}(x)$ is in $C^1(U)$ because $f \in C^p(U)$. Thus by the chain rule again (this function is composed with h in the above sum) this equals

$$\sum_{j_1,\ldots,j_{p-1}=1}^{n} \left(\sum_{j_p=1}^{n} \frac{\partial^p f}{\partial x_{j_p} \partial x_{j_1} \ldots \partial x_{j_{p-1}}}(a+t\xi)\xi_{j_p}\xi_1 \ldots \xi_{j_{p-1}} \right)$$

$$= \sum_{j_1,\ldots,j_p=1}^{n} \frac{\partial^p f}{\partial x_{j_1}\ldots\partial x_{j_p}}(a+t\xi)\xi_{j_1}\ldots\xi_{j_p} \quad \blacksquare$$

Notation:
Let $f \in C^p(U)$, $\xi \in \mathbb{R}^n$, $a \in U$. Let $(D^{(k)}f)_a(\xi) = \sum_{j_1,\ldots,j_k=1}^{n} \frac{\partial^k f}{\partial x_{j_1}\ldots\partial x_{j_k}}(a)\xi_{j_1}\ldots\xi_{j_k} \in \mathbb{R}$ for $1 \le k \le p$. This is a homogeneous degree k polynomial in $\xi_1, \ldots, \xi_n$.

Remark:
$(D^1 f)_\alpha(\xi) = \underbrace{(Df)_a}_{1\times n} \cdot \underbrace{\xi}_{n\times 1}$, and $(D^{(2)}f)_a(\xi) = \sum_{i,j=1} \frac{\partial^2 f}{\partial x_i \partial x_j}\xi_i\xi_j$.
This shows the $n\times n$ symmetric matrix $A_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}(a)$ gives a symmetric bilinear form.
The lemma we proved now says if $f \in C^p(U)$ and $a + t\xi \in U$ for all $t \in [0,1]$, and $g(t) = f(a+t\xi)$, then $\frac{d^k g}{dt^k}(t) = (D^{(k)}f)_a(\xi)$ for $1 \le k \le p$.

Theorem (Taylor Multidimensional):
Let $f : U \subseteq \mathbb{R}^n \to \mathbb{R}$ (U open) be in $C^{p+1}(U)$ (technically this hypothesis can be weakened to $C^p(U)$ with each pth order partial derivative differentiable), $p \ge 0$. Let $a \in U$, $\xi \in \mathbb{R}^n$, and suppose $a + t\xi \in U$ for all $t \in [0,1]$. Then there exists $\theta \in (0,1)$ such that

$$f(a+\xi) = \sum_{k=0}^{p} \frac{(D^{(k)}f)_a(\xi)}{k!} + \frac{1}{(p+1)!}(D^{(p+1)}f)_{a+\theta\xi}(\xi)$$

Proof:
Let $g : [0,1] \to \mathbb{R}$ be $g(t) = f(a+t\xi)$. Then since $f \in C^{p+1}(U)$, and $h(t) = a + t\xi$ is C^∞, by the chain rule, $g = f \circ h$ is in C^{p+1}. Hence by 1-d Taylor's Theorem, there exists $\theta \in (0,1)$ (set $t_o = 0$, $t = 1$) such that

$$f(a+\xi) = g(1) = \sum_{k=0}^{p} \frac{g^{(k)}(0)}{k!}(1-0)^k + \frac{g^{(p+1)}(\theta)}{(p+1)!}(1-0)^{p+1}$$

$$\overbrace{=}^{\text{lemma}} \sum_{k=0}^{p} \frac{(D^{(k)}f)_a(\xi)}{k!} + \frac{(D^{(p+1)}f)_{a+\theta\xi}(\xi)}{(p+1)!} \quad \blacksquare$$

Other forms of Taylor remainder:
If $f \in C^{p+1}(U)$, then

$$f(a+\xi) = \sum_{k=0}^{p} \frac{(D^{(k)}f)_a(\xi)}{k!} + \int_0^1 \frac{1}{p!}(D^{(p+1)}f)_{a+s\xi}ds$$

(to prove this, apply 1-d IBP version of Taylor's Theorem to $g(t) = f(a+t\xi)$, $t_o = 0, t = 1$).
If $f \in C^p(U)$, then

$$\lim_{t\to 0} \frac{f(a+t\xi) - \sum_{k=0}^{p} \frac{(D^{(k)}f)_a(\xi)}{k!}}{t^p} = 0$$

(apply L'Hopitals version of 1d)
We call $f(a+t\xi) \approx \sum_{k=0} \frac{(D^{(k)}f)_a(\xi)}{k!}$ for t small the pth Taylor polynomial of f at a in ξ-direction.

Taking the first form above at $p = 0$, we get:
Theorem (Multivariable Mean Value Theorem)
If f is differentiable on U, $a \in U$, $\xi \in \mathbb{R}^n$ and $a + t\xi \in U$ for all $t \in [0,1]$ then there exists $\theta \in (0,1)$ such that

$$f(a+\xi) - f(a) = (Df)_{a+\theta\xi} \cdot \xi$$

or

$$f(x) - f(x_o) = \underbrace{((Df)_{x_o+\theta(x-x_o)})}_{1\times n}\underbrace{(x - x_o)}_{n\times 1}$$

Example:
Compute the Taylor formula for $p = 1$ for $f(x, y) = \sin(xy)$ at $a = (x_o, y_o)$. Define $(x, y) = (x_o, y_o) + (x - x_o, y - y_o)$ (the latter is our ξ).
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In general, for $f : U \subseteq \mathbb{R}^n \to \mathbb{R}$ with $f \in C^2(U)$, an alternate version of Taylor's Theorem is:
If $a \in U$, $x \in U$, $a + tx \in U$ for all $t \in [0, 1]$, then $\xi = x - a$,

$$f(\underbrace{x}_{a+\xi}) = \underbrace{f(a)}_{(D^0 f)_a(\xi)} + \underbrace{\sum_{k=1}^{n} \frac{\partial f}{\partial x_k}(a)(x_k - a_k)}_{(D^{(1)} f)_a(\xi)} + \underbrace{\frac{1}{2}\sum_{i,j=1}^{n} \overbrace{\frac{\partial^2 f}{\partial x_i \partial x_j}}^{(\text{Hess}(f)_a)_{i,j}}(a)(x_i - a_i)(x_j - a_j)}_{(D^{(2)} f)_a(\xi)} + \text{Remainder}$$

where $\lim_{x\to a} \frac{\text{Remainder}}{||x-a||^2} = 0$.
Define $\text{Hess}(f)_a$ to be the symmetric $n \times n$ matrix whose (i, j) entry is $\frac{\partial^2 f}{\partial x_i \partial x_j}(a)$. This is the Hessian of f at a.
Then the above says

$$f(x) = f(a) + (\nabla f)(a) \cdot (x - a) + \frac{1}{2}(x - a) \cdot (\text{Hess}(f)_a(x - a)) + \text{Remainder}$$

Corollary:
Let $U \subseteq \mathbb{R}^n$ be open, $f : U \to \mathbb{R}$ be in $C^1(U)$, let $K \subseteq U$ be compact, let $E \subseteq K$ be convex. Then there exists $M > 0$ depending on K and f (but not E) such that

$$|f(x) - f(y)| \leq M||x - y||$$

for all $x, y \in E$ (i.e. f is Lipschitz), implying f is uniformly continuous (though stronger). Note that E need not be compact.
Proof:
Since $f \in C^1$, all $\frac{\partial f}{\partial x_i}$ are continuous on U. Since K is compact, the continuous function $||(Df)||^2 = (\frac{\partial f}{\partial x_1})^2 + \cdots + (\frac{\partial f}{\partial x_k})^2$ has a global maximum on K. So there exists $M > 0$ (depending on f and K) such that $||(Df)_a|| \leq M$ for all $a \in K$. By the Multivariable Mean Value Theorem, since E is convex, there exists a on the line segment from x to y such that $f(x) - f(y) = \nabla f(a) \cdot (x - y)$, implying

$$|f(x) - f(y)| = |(\nabla f)(a) \cdot (x - y)| \leq ||(\nabla f)(a)||||x - y|| \leq M||x - y|| \quad \blacksquare$$

Optimization for real-valued functions of several variables

This entire section, as one could guess, we have $f : U \subseteq \mathbb{R}^n \to \mathbb{R}$ with U open.

Definition:
We say f has a local minimum at $a \in U$ if there exists $\varepsilon > 0$ (with $B_\varepsilon(a) \subseteq U$) such that $f(x) \geq f(a)$ for all $x \in B_\varepsilon(a)$, and symmetrically for a local maximum.

Proposition:
Let $f \in C^1(U)$, with a a local extremum of f. Then $(\nabla f)(a) = \mathbf{0}$.
Proof:

Let $a = (a_1, \dots, a_n)$ and fix $k \in \{1, \dots, n\}$. Then $h(t) = f(a_1, \dots, a_k + t, \dots, a_n)$ has a local extremum at $t = 0$. Since $f \in C^1(U)$, the chain rule implies h is differentiable. So by Calc 1, $h'(0) = \frac{\partial f}{\partial x_k}(a) = 0$. Iterating the argument for all k gives the result. $\blacksquare$

Definition:
Let $f \in C^1(U)$. If $(\nabla f)(a) = 0$, then a is called a critical point of f.

We've shown local extrema imply critical points, the converse is not true.
Example:
Let $f(x, y) = x^2 - y^2$. Then $\nabla f = (2x, -2y)$, $(\nabla f)(0, 0) = (0, 0)$ hence $(0, 0)$ is a critical point, but $f(x, 0) > 0$ for x nonzero, $f(0, y) < 0$ for y nonzero, so no local extremum.

Definition
A critical point a of f is called a saddle point if it is not a local extremum.

Digression on symmetric bilinear forms and associated quadratic forms (to talk about the Hessian)

Let H be a symmetric bilinear form on $\mathbb{R}^n$, $H : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$. Bilinear is linear in each argument, symmetric is argument order does not matter. Let $\{e_1, \dots, e_n\}$ be the standard basis. Then,

$$H(x, y) = H(\sum_i x_i e_i, \sum_j x_j e_j) = \sum_{i,j} H(e_i, e_j) x_i y_j = \sum_{i,j} H_{ij} x_i y_j$$
$$= x^T H y = x \cdot (Hy)$$

Given a symmetric bilinear form, define a map $Q : \mathbb{R}^n \to \mathbb{R}$ by $Q(x) = H(x, x)$, called the associated quadratic form to the bilinear form. If $t \in \mathbb{R}$, $Q(tx) = t^2 Q(x)$. If Q is H's quadratic form, we can recover H from Q by $H(x, y) = \frac{1}{2}(Q(x+y) - Q(x) - Q(y))$ (this is a generalization of polarization). By linearity, $Q(0) = 0$ always.

Example:
$H(x, y) = x \cdot y$, then $Q(x) = ||x||^2$.

Definition:
We say a symmetric bilinear form H or equivalently its associated quadratic form Q, is
(a) positive definite if $Q(x) > 0$ for all $x \neq 0$
(b) negative definite if $Q(x) < 0$ for all $x \neq 0$
(c) indefinite if $Q(x) < 0 < Q(y)$ for some $x, y \in \mathbb{R}^n$
(it can be none of these)

Lemma:
Q is positive definite if and only if there exists $M > 0$ such that $Q(x) \geq M||x||^2$ for all $x \in \mathbb{R}^n$. Similarly Q is negative definite if and only if there exists $M > 0$ such that $Q(x) \leq -M||x||^2$ for all $x \in \mathbb{R}^n$.
Proof:
The reverse direction holds immediately because the norm is positive if $x \neq 0$.
Conversely, let $S^{n-1} = \partial(B_1(\mathbf{0})) = \{x \in \mathbb{R}^n : ||x|| = 1\}$ be the unit sphere in $\mathbb{R}^n$. This is closed and bounded, hence compact. Let Q be the quadratic form of a symmetric bilinear form H, $Q(x) = \sum_{i,j} H_{ij} x_i x_j$ is a homogeneous degree 2 polynomial in $x_1, \dots, x_n$ hence continuous.
So by the Extreme Value Theorem, there exists $M > 0$ such that $Q(x) \geq M$ for all $x \in S^{n-1}$ (an absolute minimum is attained). M is strictly positive because Q is positive definite, so $Q(x) > 0$ for x nonzero. Let $x \neq 0 \in \mathbb{R}^n$. Then $\frac{x}{||x||} \in S^{n-1}$ implies

$$Q(\frac{x}{||x||}) \geq M \implies \frac{1}{||x||^2} Q(x) \geq M$$

This completes the proof of the first statement, the negative definite case follows from the first because $-Q$ is then positive definite. ■

Fact (to be proven in linear algebra classes):
If H is a symmetric real $n \times n$ matrix,
H is positive definite if and only if all eigenvalues are positive
H is negative definite if and only if all eigenvalues are negative
H is indefinite if and only if there is at least one positive and negative eigenvalue.

Theorem (Second Derivative Test)
Let $f : U \subseteq \mathbb{R}^n \to \mathbb{R}$ be in $C^2(U)$. Let $a \in U$ be a critical point. Let $H = \text{Hess}(f)_a$.
i) If H is positive definite, then a is a local minimum.
ii) If H is negative definite, then a is a local maximum..
iii) If H is indefinite, then a is a saddle point.
The test fails if it doesn't fall into one of these categories.
Proof:
By Taylor's Theorem (because $f \in C^2(U)$)

$$f(x) = f(a) + (\nabla f)(a)(x-a) + \frac{1}{2}(x-a)H(x-a) + R(x-a)$$

with $\lim_{x\to a} \frac{R(x-a)}{||x-a||^2} = 0$.
Let $y = x - a$. We know $(\nabla f)(a)(x-a) = 0$ because f is a critical point. Therefore $f(a+y) = f(a) + \frac{1}{2}Q(y) + R(y)$.
i) Suppose H is positive definite. Then by the lemma, $Q(x) \geq M||x||^2$ for all x. Moreover, by the limit of the remainder, there exists $\delta > 0$ such that for $||y|| < \delta$, $\frac{||R(y)||}{||y||^2} < \frac{M}{2}$. Therefore,

$$-\frac{M}{2}||y||^2 \leq R(y) \leq \frac{M}{2}||y||^2$$

so

$$\begin{aligned} f(a+y) = f(a) + \frac{1}{2}Q(y) + R(y) &\geq f(a) + \frac{1}{2}M||y||^2 + R(y) \\ &\geq f(a) + \frac{1}{2}M||y||^2 - \frac{1}{2}M||y||^2 = f(a) \end{aligned}$$

so a is a local minimum.
ii) Apply i) to $-f$.
iii) Suppose H is indefinite. Therefore there exists $y, y' \neq 0$ such that $Q(y) = A > 0$, $Q(y') = -A' < 0$. For any open neighbourhood of a, we need to show there are points where f at them is both greater than and less than $f(a)$. Now fix y. We have:

$$\begin{aligned} f(a+ty) &= f(a) + \frac{1}{2}Q(ty) + R(ty) \\ &= f(a) + \frac{t^2}{2}Q(y) + R(ty) \\ f(a+ty) &= f(a) + \frac{t^2}{2}A + R(ty) \end{aligned}$$

Similarly,

$$f(a+ty') = f(a) - \frac{t^2}{2}A' + R(ty')$$

By the limit of the remainder, there exists $\delta > 0$ such that if $||x|| < \delta$, then

$$\frac{|R(x)|}{||x||^2} < \frac{1}{2}\min\{\frac{A}{||y||^2}, \frac{A'}{||y||^2}\}$$

Thus if $||ty|| = |t|||y|| < \delta$, $||ty'|| = |t|||y'|| < \delta$ then

$$|R(ty)| < ||ty||^2(\frac{A}{2||y||^2}) = \frac{t^2}{2}A$$

$$|R(ty')| < ||ty'||^2(\frac{A}{2||y'||^2}) = \frac{t^2}{2}A'$$

Using these estimates,

$$f(a+ty) = f(a) + t^2\frac{A}{2} + R(ty) > f(a)$$

$$f(a+ty') = f(a) + t^2\frac{A'}{2} + R(ty') < f(a) \quad \blacksquare$$

Example:
$f(x,y) = x^4 + y^2$. $\nabla f = (4x^3, 2y)$. Thus, we see $(0,0)$ is a critical point. We have $\text{Hess}(f)_{(x,y)} = \begin{pmatrix} 12x^2 & 0 \\ 0 & 2 \end{pmatrix}$, so $\text{Hess}(f)_{(0,0)} = \begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix}$. This is neither positive definite, nor negative definite, nor indefinite, so the second derivative test fails. But clearly $f(0,0) = 0$ and f is nonnegative, so this is a local minimum.

Example:
$h(x,y) = x^3 + y^2$. $\nabla h = (3x^2, 2y)$, and we see $(0,0)$ is critical. So $\text{Hess}(h)_{(x,y)} = \begin{bmatrix} 6x & 0 \\ 0 & 2 \end{bmatrix}$, so $\text{Hess}(h)_{(0,0)} = \begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix}$. The test fails as before, $(0,0)$ is a saddle because $h(x,0) > 0$ if $x > 0$, $h(x,0) < 0$ if $x < 0$.

Inverse Function Theorem

We look at figuring out the inverses of functions in an open neighborhood of certain points. Suppose $f : U \subseteq \mathbb{R}^n \to \mathbb{R}$, U open, and write $y = f(x)$. We want to solve for x uniquely as a function of y (i.e. invert f).

The simplest case is that f is affine linear, $f(x) = Ax + b$, b fixed, A an $m \times n$ matrix. Then $y - b = Ax$, which has a unique solution if and only if A is invertible only if $m = n$. Thus suppose $m = n$ and $f \in C^1(U)$ (differentiable).
By Taylor's Theorem, for x near x_o, $f(x) \approx f(x_o) + (Df)_a(x - x_o)$ (this is affine linear).
So we would hope that $(Df)_a$ is invertible. So in a sufficiently small neighbourhood of x_o, f should be invertible if and only if $(Df)_a$ is invertible.

Theorem (Inverse Function Theorem)
Let $f : U \subseteq \mathbb{R}^n \to \mathbb{R}^n$ (U open), with $f \in C^k(U)$, $k \geq 1$. Let $a \in U$. If $(Df)_a$ is invertible, then there exists $U' \subseteq U$ open (in $\mathbb{R}^n$) containing a with $V' = f(U')$ open and containing $f(a)$ such that $f|_{U'} : U' \to V'$ is invertible, and $f^{-1} : V' \to U'$ is in $C^k(V')$ for the same k.

Lemma 1:
Let $f : U \subseteq \mathbb{R}^n \to \mathbb{R}^n$ (U open), with $f \in C^k(U)$, $k \geq 1$. Let $a \in U$. If $(Df)_a$ is invertible, then there exists $U' \subseteq U$ open (in $\mathbb{R}^n$) containing a and $M > 0$ such that $||f(x_1) - f(x_2)|| \geq M||x_1 - x_2||$ for all $x_1, x_2 \in U$. In particular this implies $f|_{U'} : U' \to f(U')$ is injective (and thus bijective because it is a restriction).
Proof:
Let $L = (Df)_a$. Then,

$$||x_1 - x_2|| = ||L^{-1}L(x_1 - x_2)|| \leq ||L^{-1}||||L(x_1 - x_2)||$$

Let $M = \frac{1}{||L^{-1}||} > 0$. So

$$||L(x_1 - x_2)|| \geq 2M||x_1 - x_2||$$

for all $x_1, x_2 \in U$. Since $f \in C^k(U)$ for $k \geq 1$, $((Df)_x)_{ij} = \frac{\partial f_i}{\partial x_j}(x)$ is continuous on U for all i, j, so there exists $\varepsilon > 0$ such that $U' := B_\varepsilon(a) \subseteq U$ and

$$x \in U' \implies ||(Df)_x - L|| < M$$

Define $h : U \to \mathbb{R}^n$ by $h(x) = f(x) - Lx$, so $h \in C^k(U)$. Thus, for all $c \in U$, $(Dh)_c = (Df)_c - L$, so $||(Dh)_c|| = ||(Df)_c - L|| < M$. Therefore, if $c \in U'$, by the Mean Value Theorem, for $x_1, x_2 \in U'$, there exists c on the line segment between x_1, x_2 (hence in U')

$$h(x_2) - h(x_1) = (Dh)_c(x_2 - x_1)$$

so

$$||h(x_2) - h(x_1)|| \leq ||(Dh)_c||||x_2 - x_1|| \leq M||x_2 - x_1||$$

so

$$M||x_2 - x_1|| \geq ||h(x_2) - h(x_1)||$$
$$= ||f(x_2) - Lx_2 - (f(x_1) - Lx_1)|| = ||f(x_2) - f(x_1) - (Lx_2 - Lx_1)||$$
$$\geq ||Lx_2 - Lx_1|| - ||f(x_2) - f(x_1)|| \geq 2M||x_2 - x_1|| - ||f(x_2) - f(x_1)||$$

implying

$$||f(x_2) - f(x_1)|| = M||x_2 - x_1|| \quad \blacksquare$$

Lemma 2:
Let $f : U \subseteq \mathbb{R}^n \to \mathbb{R}^n$, with $f \in C^k(U)$, $k \geq 1$. Let $V = f(U)$. If f is injective on U and $(Df)_x$ is invertible for all $x \in U$, then V is open.
Proof:
Let $b \in V$. We need to find $\delta > 0$ such that $B_\delta(b) \subseteq V$ (which will show V is open). Since $V = f(U)$, there exists a unique (by injectivity) $a \in U$ with $b = f(a)$. Since U is open, there exists $\varepsilon > 0$ such that $K = \overline{B_\varepsilon(a)} \subseteq U$. ∂K is closed and bounded hence compact. f is continuous (since it is C^1), so by compact push forward, $f(\partial K)$ is compact, implying $f(\partial K)$ is closed.
Since $a \notin \partial K$ (since it is not a distance ε from itself), and $f(a) = b$, and f is injective on U, $b \notin f(\partial K)$. Since $f(\partial K)$ is closed and $b \notin f(\partial K)$, $\exists \delta > 0$ such that

$$B_{2\delta}(b) \cap f(\partial K) = \emptyset \quad (1)$$

We claim that given $y \in B_\delta(b)$, there exists $x \in K$ such that $y = f(x)$. If we have this claim, $B_\delta(b) \subseteq f(K) \subseteq f(U) = V$, which is what we need. To prove this claim, we use optimization. Let $\phi : U \to \mathbb{R}$ be given by $\phi(x) = ||f(x) - y||^2$ for fixed $y \in B_\delta(b)$. Since $f \in C^k$, and the norm function is C^∞, $\phi \in C^k(U)$ (by the chain rule). In particular ϕ is continuous. $K \subseteq U$ is compact, so by the Extreme Value Theorem, there exists $x_o \in K$ such that $\phi(x_o)$ is the minimum of ϕ on K. Therefore, for $a \in K$,

$$\phi(a) = ||f(a) - y||^2 = ||b - y||^2 < \delta^2$$

(since $y \in B_\delta(b)$), so

$$\phi(x_o) \leq \phi(a) < \delta^2$$

Either $x_o \in \partial K$ or $x_o \in \text{Int}(K)$. If $x_o \in \partial K$, then $f(x_o) \in f(\partial K)$ implies $f(x_o) \notin B_{2\delta}(b)$ by (1). Thus, by the reverse triangle inequality,

$$\sqrt{\phi(x_o)} = ||f(x) - y|| \geq ||f(x_o) - b|| - ||b - y|| \geq 2\delta - \delta = \delta$$

so $\phi(x_o) > \delta^2$, contradicting $\phi(x_o) < \delta^2$.
Thus, we must have $x_o \in \text{Int}(K)$, which is open. Therefore, x_o is a local minimum of ϕ on the open

set Int(K), hence x_o is a critical point. Therefore, $(\nabla\phi)(x_o) = 0$. But $\phi(x) = \sum_{k=1}^n (f_k(x) - y_k)^2$. Computing the gradient, $\nabla\phi = (\frac{\partial\phi}{\partial x_1}, \ldots, \frac{\partial\phi}{\partial x_n})$. Therefore,

$$\frac{\partial\phi}{\partial x_j} = \sum_{k=1}^n 2(f_k(x) - y_k)\frac{\partial f_k}{\partial x_j}$$

Thus

$$0 = (\nabla\phi)(x_o) \iff 2(f(x_o) - y)^T (Df)_{x_o} = 0$$

But $(Df)_{x_o}$ is invertible! Therefore, we must have $f(x_o) = y$ for this equation to hold. ■

Lemma 3:
Let $f : U \subseteq \mathbb{R}^n \to \mathbb{R}^n$, with U open, $f \in C^k(U)$ for $k \geq 1$. Suppose f is injective on U and $(Df)_x$ invertible for all $x \in U$. (same hypotheses as lemma 2). Thus $f : U \to f(U)$ is a bijection, and $V = f(U)$ is open by Lemma 2. Then, $g = f^{-1} : V \to U$ is continuous.
Proof:
Let $W \subseteq \mathbb{R}^n$ be open. We need to show that $g^{-1}(W)$ is open (because V is open). But $g^{-1}(W) = \{y \in W \mid g(y) \in W\} = \{y \in W \mid g(y) \in W \cap U\} = g^{-1}(W \cap U) = f(W \cap U)$ because f is bijective on U. But $W \cap U$ is open. Apply lemma 2 to $W \cap U$ (a subset of U). Therefore, $f(W \cap U)$ is open. ■

Lemma 4:
Under hypotheses of Lemma 3, $g = f^{-1} : V \to U$ is differentiable on V and $(Dg)_b = (Df)_a^{-1}$, where $a = g(b)$.
Proof:
This follows immediately from the chain rule (as proved on our assignment), since $g \circ f = I$, so $(Dg)_{f(a)}(Df)_a = I$, once we've shown g is differentiable on V. Accordingly, Let $b \in V$. Let $L = (Df)_a$ invertible. We need to show that

$$\lim_{h \to 0} \frac{g(b+h) - g(b) - L^{-1}h}{||h||} = 0$$

Let $G(h) = \frac{g(b+h)-g(b)-L^{-1}h}{||h||}$. We need to prove $\lim_{h\to 0} G(h) = 0$. Define $\Delta(h) = g(b+h) - g(b)$ for h sufficiently small.
We claim there exists $\varepsilon > 0$ such that $\frac{\Delta(h)}{||h||}$ is bounded for $0 < ||h|| < \varepsilon$. Notice that $\Delta(h) = ||h||G(h) + L^{-1}h$. Therefore, dividing by $||h||$, $\frac{\Delta(h)}{||h||} = G(h) + L^{-1}\frac{h}{||h||}$. So $\frac{\Delta(h)}{||h||}$ is automatically bounded near 0 if we know $\lim_{h\to 0} G(h) = 0$. But this is what we want to prove: By Lemma 1, there exists a neighbourhood U' of a such that $U' \subseteq U$ and $M > 0$ such that $||f(x_1) - f(x_2)|| \geq M||x_1 - x_2||$ on U' (call this (1)).
By Lemma 2, $V' = f(U')$ is open neighbourhood of b. Choose $\varepsilon > 0$ such that $b + h \in V'$ for all $||h|| < \varepsilon$. If $||h|| < \varepsilon$, $b + h = f(x)$ for $x \in U'$, implying $x = g(b+h)$. Then (1) gives:

$$||h|| = ||\underbrace{b+h}_{f(x)} - \underbrace{b}_{f(a)}|| \geq M||x - a|| = M||g(b+h) - g(b)||$$

Therefore, $\frac{\Delta h}{||h||} = \frac{||g(b+h)-g(b)||}{||h||} \leq M$ if $0 < ||h|| < \varepsilon$ (finishing the claim). Therefore,

$$G(h) = \frac{\Delta(h) - L^{-1}h}{||h||} = -L^{-1}\frac{h - L\Delta(h)}{||\Delta(h)||}\frac{||\Delta(h)||}{||h||}$$

Note that $\Delta(h) = g(b+h) - g(h) \neq 0$ if $h \neq 0$ by bijectivity of g.

Thus, $||G(h)|| \leq \frac{||\Delta(h)||}{||h||} \overbrace{||L^{-1}||}^{\text{constant}} \left\| \frac{h-L\Delta(h)}{||\Delta(h)||} \right\|$, where the left term is bounded for $0 < ||h|| < \varepsilon$. If we show the right term goes to 0, we have the result by the squeeze theorem, since bounded and constant on the

left.
But $b+h=f(g(b+h))=f(g(b)+\Delta(h))=f(a+\Delta(h))$. Therefore $h=f(a+\Delta(h))-f(a)$. Thus,

$$\frac{h-L\Delta(h)}{||\Delta(h)||}=\frac{f(a+\Delta(h))-f(a)-L\Delta(h)}{||\Delta(h)||}$$

as $h\to 0$, $\Delta(h)\to 0$ since g is continuous (Lemma 3). Therefore this expression goes to 0 since as $h\to 0$ since f is differentiable at a, which is what we wanted to show. ■

Lemma 5:
Under the same hypotheses as 2, 3, 4, the inverse $g=f^{-1}:V\to U$ is in $C^k(V)$.
Proof:
We already know from Lemma 4 that g is differentiable on V, and by chain rule, $(Df)_{g(b)}(Dg)_b=I$ (differentiate $f\circ g=I$), so $(Dg)_b=((Df)_{g(b)})^{-1}$, denote as †.
But consider the map $Dg:V\to\mathbb{R}^{n\times n}$ given by $b\mapsto(Dg)_b$, e.g. † shows that Dg is the composition of three maps:

$$\begin{array}{ccccccc} V & \xrightarrow{g} & U & \xrightarrow{Df} & (GL_n(\mathbb{R})) & \xrightarrow{\text{matrix inversion}} & (GL_n(\mathbb{R})) \\ b & \mapsto & g(b) & \mapsto & (Df)_{g(b)} & \mapsto & [(Df)_{g(b)}]^{-1} \end{array}$$

where matrix inversion is defined on the open subset $GL_n(\mathbb{R})$ of $\mathbb{R}^{n\times n}$, consisting of invertible matrices (since the determinant is continuous, so all the set of all points with nonzero determinants forms an open set).
Note that the inversion map i is in C^∞ by Cramer's Rule: $i(A)=A^{-1}=\frac{\text{adj}\,A}{\det A}$, which is an $n\times n$ matrix of rational functions in entries of A.
We want to prove $f\in C^k(U)$ implies $g\in C^k(U)$. We proceed by induction on $k, k\geq 1$. As a base case, we know $f\in C^1(U)$ implies $Df\in C^0$. By Lemma 3, $g=f^{-1}\in C^0$, and $i\in C^\infty$. So $Dg=i\circ(Df)\circ g\in C^0$. So $g\in C^1$. This proves the case $k=1$.
Now, let $k\geq 2$. Suppose Lemma 5 is true for $k-1\geq 1$. Then, $f\in C^k$ implies $f\in C^{k-1}$ implies $g\in C^{k-1}$, the last of which follows by the induction hypothesis and the fact that $Df\in C^{k-1}$. So $Dg=i\circ(Df)\circ g\in C^{k-1}$ implies $g\in C^k$. ■

Proof of Inverse Function Theorem:
Suppose $f:U\subseteq\mathbb{R}^n\to\mathbb{R}^n$ with $f\in C^k(U), k\geq 1$, and $(Df)_a$ is invertible. By Lemma 1, we can shrink U so that f is injective on U. By lemma L, we can further shrink U to $\widetilde{U}$ so that f is injective on $\widetilde{U}$ and $(Df)_x$ is invertible for all $\mathbf{x}\in\widetilde{U}$. So the hypotheses of Lemmas 2, 3, 4, 5 hold, which gives us the conclusion. ■

Example:
Let $f:\mathbb{R}^2\to\mathbb{R}^2$, $(u,v)=f(x,y)=(xy,x^2+y^2)$. $f\in C^\infty(\mathbb{R}^2)$. $u=u(x,y)=xy$. $v=v(x,y)=x^2+y^2$.
Show that there is an open neighbourhood of $(u_o,v_o)=(2,5)$ where we can find an inverse for f.
First, what points $(x,y)\in\mathbb{R}^2$ are sent to $(2,5)$ by f?

$$f(x,y)=(2,5)$$

$$xy=2, x^2+y^2=5$$

Thus $y=2/x, x^2+4/x^2=5$, multiplying through and solving the quartic we get that $x^2=1$ or $x^2=4$, so $x=\pm1,\pm2$. Thus there are 4 points, $(1,2)$, $(-1,-2)$, $(2,1)$, $(-2,-1)$ that f maps to $(2,5)$.
For any of these four points (x_o,y_o), if $(Df)_{(x_o,y_o)}$ is invertible, then the IFT says that near that point, f has a C^∞ inverse. We know $\begin{bmatrix}\frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y}\\ \frac{\partial f_2}{\partial x} & \frac{\partial f^2}{\partial y}\end{bmatrix}=\begin{bmatrix} y & x\\ 2x & 2y\end{bmatrix}$. But at points (x_o,y_o), $\det(Df)_{(x_o,y_o)}=2(y_o^2-x_o^2)$, which is nonzero for these points.
To see this explicitly, we want to invert $(u,v)=f(x,y)$. $u=xy$, $v=x^2+y^2$. Solving this system, we get that $x=\pm\sqrt{\frac{v\pm\sqrt{v^2-4u}}{2}}$, $u^2<v^2/4$, $x\neq 0$.

Implicit Function Theorem

Let $f : W \subseteq R^{n+m} \to \mathbb{R}^3$, W open. Write $z = f(y,x), y \in \mathbb{R}^q, x \in \mathbb{R}^m$. Can we solve the equation $f(y,x) = \mathbf{0}$ for y as a function of x? If so, we say that $f(y,x) = 0$ defines y implicitly as a function of x. The simplest case: if f is left multiplication by a $q \times (n+m)$ matrix: $(\underbrace{A}_{q\times n} \overbrace{B}^{q\times m})$, so

$$f(y,x) = (A\ B)(\underbrace{y}_{n\times 1} \overbrace{x}^{m\times 1})^T = Ay + Bx$$

Thus $(A\ B)(y\ x)^T = \mathbf{0}$. We see this is a homogeneous linear system of equations. From linear algebra, we see that each $y_k, k = 1,\dots,n$ is uniquely determined by each arbitrary choice of $x_1,\dots,x_n \in \mathbb{R}$ if and only if the RREF of A is $I_{n\times n}$, requiring $q = n$ and A to be invertible.
In general, if $f : W \subseteq \mathbb{R}^{n+m} \to \mathbb{R}^n$, and $f \in C^k(W), k \geq 1$ (so f is differentiable). Suppose $f(y_o,x_o) = 0$ for some $(y_o,x_o) \in W$. Then, since $f \in C^k, k \geq 1$, we get that $f(y,x) \approx f(y_o,x_o) + (Df)_{y_o,x_o}\begin{pmatrix} y-y_o \\ x-x_o \end{pmatrix} = (Df)_{y_o,x_o}\begin{pmatrix} y-y_o \\ x-x_o \end{pmatrix}$. By analogy with the linear case, if (y,x) is sufficiently close to (y_o,x_o), then we expect to be able to solve $f(y,x) = 0$ uniquely in terms of x if and only if $(Df)_{(x_o,y_o)} = \begin{pmatrix} \overbrace{\dfrac{\partial f_i}{\partial y_j}(y_o,x_o)}^{A} & \dfrac{\partial f_k}{\partial x_r} \end{pmatrix}$, $1 \leq i,j \leq n$, and $1 \leq k \leq n$, $1 \leq r \leq m$ if A is invertible.

Theorem (Implicit Function Theorem):
Let $f : W \subseteq \mathbb{R}^{n+m} \to \mathbb{R}^n$, W open. $f \in C^k(W), k \geq 1$. Suppose $(y_o,x_o) \in W$ satisfies $f(y_o,x_o) = 0$. Let A be the $n \times n$ matrix such that $A_{ij} = \frac{\partial f_i}{\partial y_j}(y_o,x_o)$. If A is invertible, then there exists an open neighbourhood $\widetilde{W} \subseteq W$ of (y_o,x_o) in $\mathbb{R}^{n+m}$, and an open neighbourhood U of $x_o \in \mathbb{R}^m$ and a map $h : U \subseteq \mathbb{R}^m \to \mathbb{R}^n$, which is $C^k(U)$ for the same k such that $\{(y,x) \in \widetilde{W} : f(y,x) = 0\} = \{(h(x),x) : x \in U\}$. Moreover, we get that $h(x_o) = y_o$. So all points in $\widetilde{W}$ are sent to zero by f are of this form, i.e. the y is a function of x. That is, for all (y,x) sufficiently close to (y_o,x_o) such that $f(y,x) = 0$, we can write y as a C^k function of x.
Proof:
We reformulate to be able to apply the inverse function theorem. Define $F : W \subseteq \mathbb{R}^{n+m} \to \mathbb{R}^{n+m}$ by $F(y,x) = (f(y),x)$. This is clearly $C^k(W)$, since the half we added to f is just the identity which is C^∞. But now differentiate, where the block triangular form is $(DF)_{(y_o,x_o)} = \begin{bmatrix} A & B \\ 0 & I \end{bmatrix}$, where the A block corresponds to the partial derivatives $\frac{\partial f_i}{\partial y_j}$ (size $n \times n$), the B block corresponds to the partial derivatives $\frac{\partial f_i}{\partial x_j}$, the 0 block is $\frac{\partial x_i}{\partial y_j} = 0$, and the I block is $\frac{\partial x_i}{\partial x_j} = \delta_{ij}$, (size $m \times m$). But $\det A \neq 0$, $\det I = 1$, so determinant of block triangular matrix is product of determinants of diagonal blocks, so $(DF)_{(y_o,x_o)}$ is invertible. So we can apply the inverse function theorem.
Therefore, there exists $\widetilde{W} \subseteq W$ open, $(y_o,x_o) \in \widetilde{W}$ such that $\widetilde{V} = f(\widetilde{W})$ is open in $\mathbb{R}^{n+m}$, with $F(y_o,x_o) \in \widetilde{V}$, but $F(y_o,x_o) = (f(y_o,x_o),x_o) = (0,x_o)$ by hypothesis, such that $F : \widetilde{W} \to \widetilde{V}$ is invertible.
Define $U \subseteq \mathbb{R}^n$ by $U = \{x \in \mathbb{R}^m; (0,x) \in \widetilde{V}\}$. We have $x_o \in U$. We claim U is open: Let $\widetilde{x} \in U$, then $(0,\widetilde{x}) \in \widetilde{V}$, but $\widetilde{V}$ is open, so there exists $\varepsilon > 0$ such that if $(y-0)^2 + (x-\widetilde{x})^2 < \varepsilon^2$, then $(y,x) \in \widetilde{V}$. In the case $y = 0$, this means that if $(x-\widetilde{x})^2 < \varepsilon^2$, we have $(0,x) \in \widetilde{V}$, therefore $B_\varepsilon(\widetilde{x}) \subseteq U$. So U is open.
Let $G = F^{-1} : \widetilde{V} \to \widetilde{W}$. Recall that $F(y,x) = (f(y,x),x)$, and write $G(y,x) = (G_1(y,x), G_2(y,x))$. Therefore,

$$(y,x) = (F \circ G)(y,x) = F((G_1(y,x), G_2(y,x))) = (f(G_1(y,x), G_2(y,x)), G_2(y,x))$$

Thus we get $G_2(y,x) = x$ for all $(y,x) \in \widetilde{V}$.
Let $y = 0$. Then we get $F^{-1}(0,x) = (G_1(0,x), x)$. Define $h : U \to \mathbb{R}^n$ by $h(x) = G_1(0,x)$, $h \in C^k(U)$.
Then, we get

$$\{(y,x) \in \widetilde{W} : f(y,x) = 0\} = (F^{-1} \circ F)\{(y,x) \in \widetilde{W} : f(y,x) = 0\}$$
$$= F^{-1}(F(y,x) \in \widetilde{V} : f(y,x) = 0\} = F^{-1}\{(0,x) \in \widetilde{V}\} = \{(h(x), x) : x \in U\}$$

To prove the moreover, note that $(y_o, x_o) = G(F(y_o, x_o)) = G(0, x_o) = G(G_1(0, x_o), x_o) = (h(x_o), x_o)$. ■

Example:
Let $n = 1$, $m = 2$. Define $f : \mathbb{R}^3 \to \mathbb{R}$ by $f(x,y,z) = x^2 + y^z + z^2 - 1$. We have $\{(x,y,z) \in \mathbb{R}^3 : f(x,y,z) = 0\} = S^2$, the unit sphere in $\mathbb{R}^3$. So the implicit function theorem says that if $f(x_o, y_o, z_o) = 0$ and $\underbrace{\frac{\partial f}{\partial z}(x_o, y_o, z_o)}_{1\times 1} \neq 0$, then near (x_o, y_o, z_o), we can write z as a C^∞ function of x, y. We have $\frac{\partial f}{\partial z} = 2z$, so $\frac{\partial f}{\partial z}(x_o, y_o, z_o) = 2z_o$. So if $z_o \neq 0$, then near (x_o, y_o, z_o), we can write $z = h(x,y)$ for all $x, y, z \in S^2$. But let's figure out what this is geometrically (future Alex if you read this, draw a sphere dumbass). We have $x^2 + y^2 + z^2 - 1 = 0$ implies $z = \pm\sqrt{1 - x^2 - y^2}$, $x^2 + y^2 < 1$, so $U = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}$, so $h(x,y) = \pm\sqrt{1 - x^2 - y^2}$. We see that anywhere near a point on the unit sphere, we can describe z in terms of the radius of x and y.

Constrained Optimization: a.k.a the method of Lagrange multipliers

Suppose we want to optimize (i.e. find a global maximum or minimum for) some real valued function f of $x_1, \dots, x_n$ subject to k constraints: $g_1(x_1, \dots, x_n) = 0, \dots, g_k(x_1, \dots, x_n) = 0$, with $1 \leq k < n$.

Example:
Optimize $f(x_1, \dots, x_n) = \sum_{i=1}^n x_i^2$, subject to $x_1 x_2 \dots x_n = 1$, $x_1 + x_2 + \cdots + x_n = 0$.

As motivation, suppose $f : U \subseteq \mathbb{R}^n \to \mathbb{R}$, $f \in C^r(U)$, $r \geq 1$. Let $g_i : U \to \mathbb{R}$ such that $g_i \in C^r(U)$, where $1 \leq i \leq k < n$. Let $a \in U$. Suppose f has a local max or a local min at a (subject to the constraints $g_i(x) = 0$, $i = 1, \dots, k$). It's no longer true that the gradient of f must vanish at 0. This is because we proved it by looking at all the points near our local max/min and taking the gradient/directional derivatives, which we might not be able to do with the constraints.

picture here of level set of one of the constraints, with orthogonal gradient at point

Like on a previous assignment, if $\alpha : (-\varepsilon, \varepsilon) \to \mathbb{R}^n$ is a curve such that $\alpha(0) = a$ and $g_i(\alpha(t)) = 0$ for all t, i.e. we can write $\alpha : (-\varepsilon, \varepsilon) \to g_i^{-1}(0)$, the chain rule tells us $(\nabla g)(a) \cdot \alpha'(0) = 0$.
We saw on a previous assignment, that if $\mathbf{u}$ is a unit vector tangent to this level set $\{x \in U, g_i(x) = 0\}$, then $(\nabla g_i)(a) \cdot \mathbf{u} = 0$. This corresponds geometrically to the fact that we don't change by moving along the level set, so we don't change anything moving in the direction tangent to it (which is the directional derivative dot product).
So since we're constrained to the set $g_i^{-1}(0)$, we can only consider directional derivatives of f at a in directions $\mathbf{u}$ such that $\mathbf{u} \cdot (\nabla g_i)(a) = 0$. In such directions, since f has a constrained local max or min at a, we get $\mathbf{u} \cdot (\nabla f)(a) = 0$. So $(\nabla f)(a) \cdot \mathbf{u} = 0$ for all unit vectors $\mathbf{u}$ such that $(\nabla g_i)(a) \cdot \mathbf{u} = 0$, $i = 1, \dots, k$. We will prove next time that this means $(\nabla f)(a)$ must be a linear combination of the gradients of the g_i's:

$$(\nabla f)(a) = -\lambda_1 (\nabla g_1)(a) - \lambda_2 (\nabla g_2)(a) \cdots - \lambda_k (\nabla g_k)(a)$$

for some $\lambda_1, \dots, \lambda_k \in \mathbb{R}$. Thus we see these equations must hold if a is a constrained local extremum of f:

$$(\nabla f)(a) + \lambda_1 (\nabla g_1)(a) + \cdots + \lambda_k (\nabla g_k)(a) = 0$$
$$g_1(a) = 0, \dots, g_k(a) = 0$$

which are a system of n equations and k scalar equations respectively.

This gives the following:
Recipe:
We want to optimize $f : U \subseteq \mathbb{R}^n \to \mathbb{R}$, U open, $f \in C^r(U)$, $r \geq 1$, subject to the constraints $g_1 = 0, \dots, g_k = 0$, where $g_i : U \to \mathbb{R}$, $g_i \in C^r(U)$, $i = 1, \dots, k$.
(1) Argue that a global max or global min must exist. For example, if the constraint set is compact. Then, find all possible points where a constrained local extremum can occur.
(2) Such a candidate point $a \in U$ must satisfy $g_1(a), \dots, g_k(a) = 0$, and $(\nabla f_)(a) + \lambda_1(\nabla g_1)(a) + \cdots + \lambda_k(\nabla g_k)(a) = 0$ for some $\lambda_1, \dots, \lambda_k$. We have k equations from the g_i's, and n equations from the vector equation, but $n + k$ unknowns $a_1, \dots, a_n, \lambda_1, \dots, \lambda_k$. Our system is determined if its linear, but usually these equations are very nonlinear.

Theorem (Lagrange):
Let $1 \leq k < n$ (our number of constraints). Let $W \subseteq \mathbb{R}^n$ open, $f : W \to \mathbb{R}$, $g : W \to \mathbb{R}^k$, $g = (g_1, \dots, g_k)$, and $f, g \in C^1(W)$. Then let $S = \{x \in W : g(x) = \mathbf{0}\}$. Then suppose that f has a local max or local min at $a \in S$, and $\operatorname{rank}(Dg)_a = k$ (maximal rank). Then there exists $\lambda \in \mathbb{R}^k$ such that $(\nabla f)(a) + \sum_{i=1}^k \lambda_i (\nabla g_i)(a) = 0$.
Proof:
We have $(Dg)_a = (\dfrac{\partial g}{\partial x_j}(a))_{ij}$. Since $(Dg)_a$ is $k \times n$, it is rank k if and only if there exists a $k \times k$ minor with nonzero determinant. By reordering the coordinates, we can assume that $(Dg)_a = [A\ B]$, where A is $k \times k$, $\det A \neq 0$, and B is $k \times (n-k)$. By the implicit function theorem applied to $g : W \to \mathbb{R}^k$, $W \in C^1$, there exists $U \subseteq \mathbb{R}^{n-k}$ open, $\overline{a} = (a_{k+1}, \dots, a_n) \in U$, and a C^1 function $h : \mathbb{R}^{n-k} \to \mathbb{R}^k$ such that $(x_1, \dots, x_k) = h(x_{k+1}, \dots, x_n)$ on S near a. i.e. on some neighbourhood of a, all the points in S can be written with the first k coordinates as C^1 functions of the last $n-k$ coordinates. Note $(a_1, \dots, a_k) = h(a_{k+1}, \dots, a_n)$ and $\{x \in \widetilde{W} : g(x) = \mathbf{0}\} = \{h(x_{k+1}, \dots, x_n), x_{k+1}, \dots, x_n) : (x_{k+1}, \dots, x_n) \in U\}$.
Define $H : U \subseteq \mathbb{R}^{n-k} \to \mathbb{R}^k$ by $H(x_{k+1}, \dots, x_n) = (h(x_{k+1}, \dots, x_n), x_{k+1}, \dots, x_n)$. Letting $\overline{x} = (x_{k+1}, \dots, x_n)$ for any $\overline{x} \in U$, we have $g(H(\overline{x})) = \mathbf{0}$ (since we are essentially just restricting to domain where g vanishes), i.e, $g \circ H$ is essentially the local restriction of g to S.
Similarly let $\widetilde{f} = f \circ H : U \subseteq \mathbb{R}^{n-k} \to \mathbb{R}$. $\widetilde{f}(\overline{x}) = f(H(\overline{x}))$.Then $\widetilde{f}$ is the local restriction of f to S (near a). As $\overline{x}$ varies near U, we get all points in S close to a (the points in $\widetilde{W} \cap S$). So since $a \in S$ is a constrained local extremum for f, this means $\overline{a} = (a_{k+1}, \dots, a_n)$ is an unconstrained local extremum for $\widetilde{f}$. Hence we must have $(\nabla \widetilde{f})(\overline{a}) = 0$, also $\widetilde{g} = 0$ (identically), so $(D\widetilde{g})_{\overline{a}} = 0$ too, where $\widetilde{g} = g \circ H$. But now using the fact that $H(\overline{x}) = (h(\overline{x}), \overline{x})$, $H(\overline{a}) = (h(\overline{a}, \overline{a}) = a$, we have $(DH)_{\overline{a}} = \begin{bmatrix} (Dh)_{\overline{a}} \\ I \end{bmatrix}$, where the top block is $k \times (n-k)$, and the bottom block is $(n-k) \times (n-k)$.

But now use the chain rule to differentiate $\widetilde{g} = g \circ H$. We have $(D\widetilde{g})_{\overline{a}} = (Dg)_{H(\overline{a})}(DH)_a = 0$, $(Dg)a = [A\ B]$, so $0 = [AB] \begin{bmatrix} (Dh)_{\overline{a}} \\ I \end{bmatrix}$, so $(Dh)_{\overline{a}} = -A^{-1}B$. Now differentiating $\widetilde{f} = f \circ H$ with the chain rule, we get that $0 = (D\widetilde{f})_a = (Df)_{H(\overline{a})}(DH)_{\overline{a}} = [CD] \begin{bmatrix} (Dh)_{\overline{a}} \\ I \end{bmatrix}$, where C is the $1 \times k$ entries of $(Df)_{H(\overline{a})}$ so $D = -C(Dh)_{\overline{a}} = CA^{-1}B$. But now write

$$[0\ \ 0] = [C\ \ CA^{-1}B] = [CAA^{-1}\ \ CA^{-1}B] = (Df)_a - \underbrace{CA^{-1}}_{1 \times k}(Dg)_a$$

Let $\lambda = -CA^{-1} \in \mathbb{R}^k$. This satisfies $(\nabla f)(a) = \sum_{i=1}^k \lambda_i (\nabla g_i)(a) = 0$, which proves the theorem. ■

Example:
Find the global extreme values of $f(x, y) = 4x^2 - 3xy$ subject to the constraint $g(x, y) = x^2 + y^2 - 1 = 0$. S (where the constraint is satisfied) is the unit circle, which is compact. f is continuous so by the Extreme Value Theorem, there exists a global maximum and minimum. We need to solve $\nabla f + \lambda \nabla g = 0$. $\nabla f = (8x - 3y, -3x)$, $\nabla g = (2x, 2y)$, giving
(1) $8x - 3y + 2x\lambda = 0$.
(2) $-3x + 2y\lambda = 0$.
(3) $x^2 + y^2 = 1$.
From (2), we have $x = \frac{2}{3}y\lambda$. Plugging into (1), $8(\frac{2}{3}\lambda y) - 3y = -2\frac{2}{3}(\lambda y)\lambda$, giving $\frac{16}{3}\lambda y - 3y = -\frac{4}{3}\lambda^2 y$. Therefore, $(4\lambda^2 + 16\lambda - 9)y = 0$. If $y = 0$, by (2), this implies $x = 0$, contradicting (3). So $y \neq 0$. So $4\lambda^2 + 16\lambda - 9 = (2\lambda - 1)(2\lambda + 9) = 0$, so $\lambda = \frac{1}{2}$, $\lambda = -\frac{9}{2}$. If $\lambda = \frac{1}{2}$. (2) becomes $-3x = -y$, so $y = 3x$, which by (3) says that $10x^2 = 1$, so $x = \pm\frac{1}{\sqrt{10}}$, $y = 3x$. If $\lambda = -\frac{9}{2}$, an identical calculation gives $x = 3y$, $y = \pm\frac{1}{\sqrt{10}}$.
Evaluating f at these points, we get the points $(1/\sqrt{10}, 3/\sqrt{10}), (-1/\sqrt{10} - 3/\sqrt{3})$, $(-3/\sqrt{10}, 1/\sqrt{10}), (3/\sqrt{10}, -1/\sqrt{10})$. At the first two, f has value $-1/2$, At the second two, f has value $9/2$. Thus global min $= -1/2$, global max $= 9/2$.

Example
Find all global extreme values of $f(x, y, z) = x^2 + y^2 + z^2$ subject to the constraints $x - y = 1$ and $y^2 - z^2 = 1$ (plane and hyperbola surfaces, draw it Alex). This constrained set S is unbounded (not compact), so f has no global max on S, but it does have a global min, which must be a (constrained) global min. At the global min, $\nabla f + \lambda \nabla g_1 + \mu \nabla g_2 = 0$. We have $\nabla f = (2x, 2y, 2z)$, $\nabla g_1 = (1, -1, 0)$, and $\nabla g_2 = (0, 2y, -2z)$. This gives the equations

$$2x + \lambda = 0 \quad (1)$$
$$2y - \lambda + 2y\mu = 0 \quad (2)$$
$$2z - 2z\mu = 0 \quad (3)$$
$$x - y = 1 \quad (4)$$
$$y^2 - z^2 = 1 \quad (5)$$

By (3), $z(1 - \mu) = 0$. If $\mu = 1$, then $\lambda = 4y$ by (2), so (1) becomes $2x + 4y = 0$, implying $x = -2y$, so (4) gives $y = -\frac{1}{3}$, which tells us by (5) that $z^2 = -\frac{8}{9}$, which is impossible. Therefore, we must have $z = 0, \mu \neq 1$. Therefore, by (5), $y = \pm 1$. By 4, $x = 1 + y$. This gives the points $(0, -1, 0)$ (which f equals 1 at) and $(2, 1, 0)$ (which f equals 5 at), the former of which must be a global minimum.

Riemann Integration Theory

Our goal: Given a subset $D \subseteq \mathbb{R}^n$ and a function $f : D \to \mathbb{R}$, we want to define $\int_D f \in \mathbb{R}$. This will not always be possible. It depends both on the structure of D and properties of the function f.

Remark:
If $f = (f_1, \dots, f_m)$ then $\int_D f \in \mathbb{R}^m$ and we define $(\int_D f)_k = \int_D f_k$, so without loss of generality we can just consider functions $f : D \to \mathbb{R}$.

We'll start by considering for which subsets $D \subseteq \mathbb{R}^n$ we can attach a meaningful notion of "size" or "volume".

Definition:
Let $I = [a_1, b_1] \times [a_2, b_2] \times \cdots \times [a_n, b_n] \subseteq \mathbb{R}^n$. We have $x \in I$ if and only if $a_k \leq x_k \leq b_k$ for all $k = 1, \dots, n$. Such a subset I is called a box.

Remark
A box is compact since it's closed and bounded. It's obviously bounded, and it's closed if we take the sequential definition of compactness and then split the sequences componentwise (since coordinatewise, boxes are closed sets).

Definition
For a box I, we define its size (or Riemann volume) to be $\mu(I) = (b_1 - a_1)(b_2 - a_2)\dots(b_n - a_n) = \prod_{i=1}^n (b_i - a_i) > 0$.
If $n = 1, \mu(I)$ is its length.
If $n = 2, \mu(I)$ is its area.
If $n = 3, \mu(I)$ is its volume.
Some authors call μ the "Jordan content".
We want to determine which subsets D of $\mathbb{R}^n$ can be given a "size". We'll see that many, but not all subset of $\mathbb{R}^n$ are "sizeable".

First, we'll consider what it should mean for a subset to have zero size.

Definition:
We say $D \subseteq \mathbb{R}^n$ has size zero, if for all $\varepsilon > 0$, there exist finitely many boxes $I_1, \dots, I_N$, such that $D \subseteq \cup_{j=1}^N I_j$ and $\sum_{j=1}^N \mu(I_j) < \varepsilon$. These $I_1, \dots, I_n$ need not be disjoint (they can overlap). Clearly, if D has zero size, then any subset $E \subseteq D$ also has zero size.

Example:
A singleton $D = \{x\}$ has size zero. Let $\varepsilon > 0$. Choose $\delta > 0$ such that $2^n\delta^n < \varepsilon$. Let $I_\delta = [x_1 - \delta, x_1 + \delta] \times \cdots \times [x_n - \delta, x_n + \delta]$. We have $D \subseteq I_\delta$, and $\mu(I_\delta) = (2\delta)^n < \varepsilon$, so any singleton has zero size.

Proposition:
Let $D_1, \dots, D_m$ be subsets of size zero. Then for any finite m, $\cup_{j=1}^m D_j$ has size zero.
Proof:
Let $\varepsilon > 0$. Since each D_j has zero size, there are boxes $I_1^j, \dots, I_{N_j}^j$ with $D_j \subseteq \cup_{k=1}^{N_j} I_k^j$ and $\sum_{k=1}^{Nj} \mu(I_k^j) < \frac{\varepsilon}{m}$. We have $\cup_{j=1}^m D_j \subseteq \cup_{j=1}^m \cup_{k=1}^{Nj} I_k^j$ and $\sum_{j=1}^m \sum_{k=1}^{Nj} \mu(I_k^j) < \varepsilon < \frac{\varepsilon}{m}$. ■

Example
In particular, any finite set has size zero since singletons have size zero.

Proposition:
Let $K \subseteq \mathbb{R}^n$ be compact, and let $f : K \to \mathbb{R}$ be continuous on K (so f is necessarily uniformly continuous on K). Let $\Gamma_f = \{(x, f(x)) \in \mathbb{R}^{n+1}; x \in K\}$ be the graph of f over K. Then Γ_f has size zero.
Proof:
Let $\varepsilon > 0$. K is compact, hence bounded, so there exists $M > 0$ such that $K \subseteq [-M, M] \times \cdots \times [-M, M]$ (since $|x_k| \leq ||x|| \leq M$). By uniform continuity of f on K, there exists $1 > \delta > 0$ such that $||x - x'|| < \delta$ for $x, y \in K$, then $|f(x) - f(x')| < \frac{\varepsilon}{(2M+1)^n}$. Let $0 < \delta' < \frac{\delta}{\sqrt{n}}$ (so $\delta' < 1$ also), and choose $N \in \mathbb{N}$ such that $N\delta' < 2M \leq (N + 1)\delta'$.
For $k_1, \ldots, k_n \in \{0, 1, \ldots, N\}$ define the box $I_{k_1,\ldots,k_n} = [-M + k_1\delta', -M + (k_1 + 1)\delta'] \times \cdots \times [-M + k_n\delta', -M + (k_n + 1)\delta']$, so $x \in I_{k_1,\ldots,k_n}$ if and only if $-M + k_j\delta' \leq x_j \leq -M + (k_j + 1)\delta'$, for all $j = 1, \ldots, n$.
By construction, $\mu(I_{k_1,\ldots,k_n}) = \delta'^n$, and by construction, $K \subseteq \cup_{k_1,\ldots,k_n=0}^N I_{k_1,\ldots,k_n}$ (we might not need all of the boxes to cover K).
Given $k_1, \ldots, k_n \in \{0, \ldots, N\}$ such that $I_{k_1,\ldots,k_n} \cap K \neq \emptyset$, choose $x_{k_1,\ldots,k_n} \in I_{k_1,\ldots,k_n} \cap K$ (if the intersection is empty, we can ignore this box).
For this case, define the box

$$J_{k_1,\ldots,k_n} = I_{k_1,\ldots,k_n} \times [f(x_1, \ldots, x_n) - \frac{\varepsilon}{2(2M + 1)^n}, f(x_1, \ldots, x_n) + \frac{\varepsilon}{2(2M + 1)^n}] \subseteq \mathbb{R}^{n+1}$$

with $\mu(J_{k_1,\ldots,k_n}) = \mu(I_{k_1,\ldots,k_n}) \cdot \frac{\varepsilon}{(2M+1)^n} = \left(\frac{\delta'}{(2M+1)}\right)^n \varepsilon$.
Let $x \in K$. Then $x \in I_{k_1,\ldots,k_n}$ for some $k_1, \ldots, k_n$, so $||x - x_{k_1,\ldots,k_n}|| < n\delta'^2 < \delta^2$ so $|f(x) - f(x_{k_1,\ldots,k_n})| < \frac{\varepsilon}{(2M+1)^n}$, implying $(x, f(x)) \in J_{k_1,\ldots,k_n}$.
Hence $\Gamma_f \subseteq \cup_{k_1=0}^N \cdots \cup_{k_n=0}^N J_{k_1,\ldots,k_n}$, which is finitely many boxes.
Moreover, $\sum_{k_1=0}^N \cdots \sum_{k_n=0}^N \mu(J_{k_1,\ldots,k_n}) = \sum_{k_1=0}^N \cdots \sum_{k_n=0}^N (\frac{\delta'}{2M+1})^n \varepsilon = \frac{(N+1)^n \delta'^n}{(2M+1)^n} \varepsilon < \varepsilon$, where the left factor is less than 1 since $N\delta' < 2M, \delta' < 1$, implying $(N + 1)\delta' < 2M + 1$. ■

Example
Take $S^1 \subseteq \mathbb{R}^2$. We have $S^1 = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\} = S^1_+ \cup S^1_-$, i.e. the top and bottom halves of the circle, given by $y = \pm\sqrt{1 - x^2}$. These top and bottom halves are graphs of these continuous functions over $[-1, 1] \subseteq \mathbb{R}$ (which is compact). Each has size zero by the above proposition, hence S^1 is a union of two sets of size zero, hence has size zero.

Let $D \subseteq \mathbb{R}^n$, and suppose D does not have size zero. Hence negating the definition, there exists $\varepsilon_0 > 0$ such that for any collection $I_1, \ldots, I_N$ of finitely many boxes with $D \subseteq \cup_{j=1}^N I_j$, we have $\sum_{j=1}^N \mu(I_j) \geq \varepsilon_0$.

Lemma:
$D \subseteq \mathbb{R}^n$ does not have size zero, if and only if there exists $\varepsilon' > 0$, such that whenever $I_1, \ldots, I_N$ are finitely many boxes with $D \subseteq \cup_{j=1}^N I_j$, then $\sum_{j=1,\mathrm{Int}(I_j)\cap D\neq\emptyset}^N \mu(I_j) \geq \varepsilon'$. Let $(*)$ denote this bound.
Proof:
It's clear that $(*)$ implies that D does not have size zero (as if we add the rest of the sets with empty intersection with D, then adding their lengths will still be above ε').
Conversely, suppose D does not have size zero. Then, there exists $\varepsilon_0 > 0$ such that if $I_1, \ldots, I_n$ cover D, then $\sum_{j=1}^N \mu(I_j) \geq \varepsilon_0$. By reordering the boxes if necessary, we can assume $\mathrm{Int}(I_j) \cap D \neq \emptyset$ for $i = 1, \ldots, m$, and $\mathrm{Int}(I_j) \cap D = \emptyset$ for $m + 1, \ldots, N$.
Because they have empty intersection with the interior, if $m + 1 \leq j \leq N$, $I_j \cap D \subseteq \partial I_j$ (could be empty, or consist only of boundary points).
But notice that ∂I_j has size zero, because it is a finite union of graphs of continuous functions over compact sets, as its edges are just straight lines on closed intervals.
So $I_j \cap D$ has size zero for $j = m + 1, \ldots, N$, hence so does $\cup_{j=m+1}^N I_j \cap D$. So there exist finitely many

boxes $J_1, \dots, J_k$ such that $\cup_{j=m+1}^N I_j \cap D \subseteq \cup_{i=1}^k J_i$ with $\sum_{i=1}^k \mu(J_i) < \frac{\varepsilon_o}{2}$. But now notice that

$$
\begin{aligned}
D &\subseteq \bigcup_{j=1}^m I_j \cup \bigcup_{j=m+1}^N (I_j \cap D) \\
&\subseteq \bigcup_{j=1}^m I_j \cup \bigcup_{i=1}^k J_i
\end{aligned}
$$

so we've covered D by finitely many boxes. Hence $\sum_{j=1} \mu(I_j) + \sum_{i=1}^k \mu(J_i) \geq \varepsilon_o$ (since D does not have size zero.)
Therefore,

$$\sum_{\substack{j=1 \\ \mathrm{Int}(I_j)\cap D \neq \emptyset}}^N \mu(I_j) = \sum_{j=1}^m \mu(I_j) \geq \varepsilon_o - \sum_{i=1}^k \mu(J_i) \geq \varepsilon_o/2 =: \varepsilon' \quad \blacksquare$$

The Riemann Integral

Definition
Let $I = [a_1, b_1] \times \cdots \times [a_n, b_n] \subseteq \mathbb{R}^n$ be a box. For each $j = 1, \dots, n$, choose a partition P_j of the interval $[a_j, b_j]$ into N_j pieces:

$$a_j = t_{j,0} < t_{j,1} < \cdots < t_{j,N_{j-1}} < t_{j,N_j} = b_j$$

Let $\mathsf{P} = P_1 \times \cdots \times P_n$. This is called the partition of I into sub-boxes.
The subboxes are of the form $[t_{1,k_1}, t_{1,k_1+1}] \times \cdots \times [t_{n,k_n}, t_{n,k_n+1}]$ for $0 \leq k_1 \leq N_1 - 1, \dots, 0 \leq k_n \leq N_n - 1$. There are $N_1, N_2, \dots, N_n$ subboxes generated by the partition P.

Definition:
Let I be a box in $\mathbb{R}^n$. Let $f : I \to \mathbb{R}$. Let P be a partition of I that generates a subdivsion $I = \cup_\alpha I_\alpha$ for I_α's being the subboxes created by the partition.
For each subbox I_α, choose a point $x_\alpha \in I_\alpha$. Then, define

$$S(f, \mathsf{P}) = \sum_\alpha f(x_\alpha)\mu(I_\alpha)$$

This is called a Riemann sum of f respect to the partition P of I.

Remark
Even given a partition P, this is not unique, it depends on the choices of the tags x_α.

Definition:
Let P, Q be two partitions of the box I. We say that Q is a refinement of P if $P_j \subseteq Q_j$ for all $j = 1, \dots, n$. So if Q is a refinement of P, then each subbox of the partition Q is completely contained inside a subbox of the partition P. More precisely, each subbox of P is a union of subboxes of Q.

Remark:
If $P, \widetilde{P}$ are partitions of I, then there exists a partition Q of I that is a refinement of both (called a common refinement of $P, \widetilde{P}$) given by $Q_j = P_j \cup \widetilde{P}_j$, $j = 1, \dots, n$.

Definition:
Let $I \subseteq \mathbb{R}^n$ be a box, $f : I \to \mathbb{R}$. We say f is Riemann integrable over I with Riemann integral $y = \int_I f \in \mathbb{R}$ if and only if:
For all $\varepsilon > 0$, there exists a partition P_ε of I, such that for any refinement P of P_ε, and any Riemann

sum $S(f, P)$ of f with respect to P, we have $|y - S(f, P)| < \varepsilon$.

Remark:
You should think about this intuitively as giving rigorous meaning to the statement "the Riemann sums of f over I converge to y".

Proposition:
If f is Riemann integrable over I, then its Riemann integral $y = \int_I f$ is unique.
Proof:
Assignment 10. ■

Theorem (Cauchy criterion for Riemann integrability):
Let $I \subseteq \mathbb{R}^n$ be a box, and $f : I \to \mathbb{R}$. The following are equivalent:
(a) f is Riemann integrable on I
(b) For all $\varepsilon > 0$, there exists a partition P_ε of I such that, for any refinements P, Q of P_ε, and any two Riemann sums $S(f, P)$ and $S(f, Q)$ with respect to P and Q respectively, we have

$$|S(f, P) - S(f, Q)| < \varepsilon$$

(think of this as saying the Riemann sums are Cauchy)
(c) For all $\varepsilon > 0$, there exists a partition P_ε of I suchc that whenever $S_1(f, \mathsf{P}_\varepsilon), S_2(f, P_\varepsilon)$ are any two Riemann sums with respect to P, we have $|S_1(f, P_\varepsilon) - S_2(f, P_\varepsilon)| < \varepsilon$ (this is a weaker form of (b)).
Proof:
Proving (a) implies (b): Let $y = \int_A f$. Given $\varepsilon > 0$, there exists a partition P_ε of I such that $|S(f, P) - y| < \frac{\varepsilon}{2}$ whenever P is a refinement of P_ε and $S(f, P)$ is a Riemann sum with respect to P.

Let P, Q be refinements of P_ε, and let $S(f, P)$ and $S(f, Q)$ be Riemann sums with respect to P and Q respectively. So

$$|S(f, P) - y| < \frac{\varepsilon}{2}, \ \ |S(f, Q) - y| < \frac{\varepsilon}{2}$$

so by the triangle inequality,

$$|S(f, P) - S(f, Q)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

proving this direction.
Proving (b) implies (a), for each $k \in \mathbb{N}$, there exists a partition P_ε of I such that $|S(f, \mathsf{P}) - S(f, \mathsf{Q})| < \frac{1}{2^k}$, where $S(f, \mathsf{P})$ and $S(f, \mathsf{Q})$ are any Riemann sums of f with respect to refinements P, Q of P_k. Without loss of generality, by replacing P_{k+1} by the common refinement of P and P_{k+1} we can assume P_{k+1} is a refinement of P_k for all k. Now for each $k \in \mathbb{N}$, fix a Riemann sum $S(f, \mathsf{P}_k) = S_k \in \mathbb{R}$. Then, if $\ell > k$, by the triangle inequality, $|S_\ell - S_k| \leq \sum_{j=k}^{l-1} |S_{j+1} - S_j| < \sum_{j=k}^{l} \frac{1}{2^j}$ because $|S_{j+1} - S_j| = |S(f, \mathsf{P}_{j+1}) - S(f, \mathsf{P}_j)| < \frac{1}{2^j}$.
But $\sum_{j=k}^{l-1} \frac{1}{2^j}$ goes to zero as $k \to \infty$, as it is the tail of a convergent power series. So the sequence of real numbers (S_k) is Cauchy, and thus converges to some limit $y \in \mathbb{R}$. We need to show $y = \int_I f$.
Let $\varepsilon > 0$. Choose $k_0 \in \mathbb{N}$ large enough so that $\frac{1}{2^{k_0}} < \frac{\varepsilon}{2}$ and $|S_{k_0} - y| < \frac{\varepsilon}{2}$. Let $\mathsf{P}_\varepsilon = \mathsf{P}_{k_0}$, let P be any refinement of P_ε. Then, $|S(f, \mathsf{P}) - y| \leq |S(f, \mathsf{P}) - S_{k_0}| + |S_{k_0} - y| < \frac{1}{2^{k_0}} + \frac{\varepsilon}{2} < \varepsilon$.
(b) implies (c) is clear by taking $\mathsf{P} = \mathsf{Q} = \mathsf{P}_\varepsilon$.
Proving (c) implies (b), let $\varepsilon > 0$. Let P_ε be the partition given by (c), and let P and Q be arbitrary refinements of P_ε, and let $S(f, \mathsf{P})$ and $S(f, \mathsf{Q})$ be Riemann sums. We need to show $|S(f, \mathsf{P}) - S(f, \mathsf{Q})| < \varepsilon$.
Let $I = \cup_\alpha I_\alpha$ be the decomposition with respect to P_ε.
Let $I = \cup_\beta J_\beta$ be the decomposition with respect to P.
Let $I = \cup_\gamma K_\gamma$ be the decomposition with respect to Q.
Then,

$$S(f, \mathsf{P}) - S(f, \mathsf{Q}) = \sum_\beta f(x_\beta)\mu(J_\beta) - \sum_\gamma f(y_\gamma)\mu(K_\gamma)$$

where $x_\beta \in J_\beta$, $y_\gamma \in K_\gamma$ for all β and γ. But this equals

$$\sum_\alpha \left(\sum_{\substack{\beta,\\ J_\beta \subseteq I_\alpha}} f(x_\beta)\mu(J_\beta) \right) - \sum_\alpha \left(\sum_{\substack{\gamma,\\ K_\gamma \subseteq I_\alpha}} f(y_\gamma)\mu(K_\gamma) \right)$$

For each α, pick z_α such that $f(z_\alpha) = \max\{f(x_\beta), f(y_\gamma)\}$ for all β such that $J_\beta \subseteq I_\alpha$ and for all γ such that $K_\gamma \subseteq I_\alpha$. Similarly, choose $f(w_\alpha) = \min\{f(x_\beta), f(y_\gamma)\}$ for all β such that $J_\beta \subseteq I_\alpha$ and for all γ such that $K_\gamma \subseteq I_\alpha$. These choices are well-defined, since $\{f(x_\beta), f(y_\gamma) : J_\beta \subseteq I_\alpha, K_\gamma \subseteq I_\alpha\}$ is a finite set.
By construction, $f(w_\alpha) \leq f(x_\beta) \leq f(z_\alpha)$ for $J_\beta \subseteq I_\alpha$, and $-f(z_\alpha) \leq -f(y_\gamma) \leq -f(w_\alpha)$ whenever $K_\gamma \subseteq I_\alpha$.
Fixing α, this tells us that

$$(f(w_\alpha) - f(z_\alpha))\mu(I_\alpha) = f(w_\alpha) \sum_{\substack{\beta,\\ J_\beta \subseteq I_\alpha}} \mu(J_\beta) - f(z_\alpha) \sum_{\substack{\gamma,\\ K_\gamma \subseteq I_\alpha}} \mu(K_\gamma)$$

$$\leq \sum_{\substack{\beta,\\ J_\beta \subseteq I_\alpha}} f(x_\beta)\mu(J_\beta) - \sum_{\substack{\gamma,\\ K_\gamma \subseteq I_\alpha}} f(y_\gamma)\mu(K_\gamma) \leq f(z_\alpha) \sum_{\substack{\beta,\\ J_\beta \subseteq I_\alpha}} \mu(J_\beta) - f(w_\alpha) \sum_{\substack{\gamma,\\ K_\gamma \subseteq I_\alpha}} \mu(K_\gamma)$$

$$= f(z_\alpha)\mu(I_\alpha) - f(w_\alpha)\mu(I_\alpha) = (f(z_\alpha) - f(w_\alpha))\mu(I_\alpha)$$

Therefore,

$$\left| \sum_{\substack{\beta:\\ J_\beta \subseteq I_\alpha}} \mu(J_\beta) - \sum_{\substack{\gamma:\\ K_\gamma \subseteq I_\alpha}} \mu(K_\gamma) \right| \leq |f(z_\alpha) - f(w_\alpha)|\mu(I_\alpha) \quad \dagger$$

hence,

$$|S(f, \mathsf{P}) - S(f, \mathsf{Q})| = |\sum_\alpha \sum_{\substack{\beta:\\ J_\beta \subseteq I_\alpha}} f(x_\beta)\mu(J_\beta) - \sum_\alpha \sum_{\substack{\gamma:\\ K_\gamma \subseteq I_\alpha}} f(y_\gamma)\mu(K_\gamma)| \underbrace{\leq}_{\dagger} \sum_\alpha |f(z_\alpha) - f(w_\alpha)|\mu(I_\alpha)$$

$$= |\sum_\alpha f(z_\alpha)\mu(I_\alpha) - \sum_\alpha f(w_\alpha)\mu(I_\alpha)| = |S_1(f, \mathsf{P}_\varepsilon), S_2(f, \mathsf{P}_\varepsilon)| \underbrace{<}_{(c)} \varepsilon \quad \blacksquare$$

Theorem:
$I \subseteq \mathbb{R}^n$ a box, $f : I \to \mathbb{R}$. Suppose f is bounded, and suppose the set of discontinuities of f, S, has size zero. Then f is integrable on I.
Remark: If $S = \emptyset$ (f is continuous on I), then it's automatically bounded by the Extreme Value Theorem.
Proof:
We'll show (c) of the above theorem holds.
Let $\varepsilon > 0$. Since S has size zero, there exists a partition P of I such that $\displaystyle\sum_{\substack{\alpha\\ I_\alpha \cap S \neq \emptyset}} \mu(I_\alpha) < \frac{\varepsilon}{4C}$, where C is the bounded constant of f.
Let $K = \displaystyle\bigcup_{\substack{\alpha\\ I_\alpha \cap S = \emptyset}} I_\alpha$. K is compact, since it is the finite union of compact sets.
By construction, $f|_K : K \to \mathbb{R}$ is continuous, hence by compactness of K, is uniformly continuous. So there exists $\delta > 0$ such that if $x, y \in K$ and $||x - y|| < \delta$ then $|f(x) - f(y)| < \frac{\varepsilon}{2\mu(I)}$.
Now choose a partition Q of I (which without loss of generality can be taken to be a refinement of P)

such that for each subbox J_β of Q, we have $J_\beta = [a_1^\beta, b_1^\beta] \times \cdots \times [a_n^\beta, b_n^\beta]$ where $\max_{j=1,\ldots,n} |b_j^\beta - a_j^\beta| < \frac{\delta}{\sqrt{n}}$.
Therefore $x, y \in J_\beta$ implies $||x-y|| < \delta$.
Now let $S_1(f, \mathsf{Q})$, $S_2(f, \mathsf{Q})$ be any two Riemann sums of f with respect to Q.

$$S_1(f, \mathsf{Q}) = \sum_\beta f(x_\beta)\mu(J_\beta)$$

$$S_2(f, \mathsf{Q}) = \sum_\beta f(y_\beta)\mu(J_\beta)$$

So

$$|S_1(f, \mathsf{Q}) - S_2(f, \mathsf{Q})| \leq \sum_\beta |f(x_\beta) - f(y_\beta)|\mu(J_\beta)$$

But now we can get our desired estimate:

$$\sum_{\substack{\beta \\ J_\beta \not\subseteq K}} |f(x_\beta) - f(y_\beta)|\mu(J_\beta) + \sum_{\substack{\beta \\ J_\beta \subseteq K}} |f(x_\beta) - f(y_\beta)|\mu(J_\beta)$$

in the first sum, we use the fact S is small (since f is bounded), in the second sum, we use uniform continuity.
Thus, we have

$$\begin{aligned}
&\sum_{\substack{\beta \\ J_\beta \not\subseteq K}} |f(x_\beta) - f(y_\beta)|\mu(J_\beta) + \sum_{\substack{\beta \\ J_\beta \subseteq K}} |f(x_\beta) - f(y_\beta)|\mu(J_\beta) \\
&\leq \sum_{\substack{\beta \\ J_\beta \not\subseteq K}} 2C\mu(J_\beta) + \sum_{\substack{\beta \\ J_\beta \subseteq K}} \frac{\varepsilon}{2\mu(I)}\mu(J_\beta) \\
&\leq 2C \underbrace{\sum_{J_\beta \subseteq K} \mu(J_\beta)}_{< \frac{\varepsilon}{4C}} + \frac{\varepsilon}{2\mu(I)} \underbrace{\sum_\beta \mu(J_\beta)}_{\mu(I)} < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon
\end{aligned}$$

so by (c), f is integrable on I. ■

(Riemann) integration over general bounded domains

Definition
Let $D \subseteq \mathbb{R}^n$ be a bounded subset. Let $f : D \to \mathbb{R}$. Choose a box I such that $D \subseteq I$ (which is possible since D is bounded). Define $\widetilde{f} : I \to \mathbb{R}$ by $\widetilde{f}(x) = f(x)$ if $x \in D$, $\widetilde{f}(x) = 0$ if $x \notin D$, i.e. $\widetilde{f}$ is the extension by by zero from D to I.
We say f is integrable over D if $\widetilde{f}$ is integrable over I, and in this case define $\int_D f = \int_I \widetilde{f}$.

Fact
This does not depend on the choice of I (exercise).

Theorem:
Let D be bounded, and suppose ∂D has size zero ($\mu(\partial D) = 0$). Let $f : D \to \mathbb{R}$ be bounded and continuous on D. Then f is integrable on D.
Proof:
Let I be a box with $D \subseteq I$. Define $\widetilde{f}$ as before. $\widetilde{f}$ is bounded on I, since f is. Let S be the set of discontinuities of $\widetilde{f}$ on I. If $x \in D^\circ$, then $\widetilde{f}$ is continuous at x by hypothesis. If $x \in (I \setminus D)^\circ$, then $\widetilde{f}$ is identically zero and thus continuous at x. Therefore, $S \subseteq (\partial D) \cup (\partial I)$. Both of these have size zero,

the first by hypothesis, the latter as seen previously (it is a box, so its boundary is the union of straight lines, which are graph of functions).
Hence, by the previous Theorem, $\widetilde{f}$ is integrable on I, so f is integrable on D. ■

We can now define the size or Riemann volume of an arbitrary bounded subset D of $\mathbb{R}^n$, when it makes sense.

Definition:
Let $D \subseteq \mathbb{R}^n$ be bounded. Let $f : D \to \mathbb{R}$ be the constant function $f(x) = 1$ for all $x \in D$. We say D is sizeable (i.e. has a well-defined volume) if and only if this constant function f is integrable over D, and in this case, we define the size of D, denoted $\mu(D)$, to be $\mu(D) = \int_D 1$.

Remark:
D being not sizeable is not the same as D having size zero. A bounded subset D may or may not be sizeable, and if it is, its size $\mu(D)$ may be zero or positive.

Remark:
If D has size zero, then it is bounded. We'll see soon that $\mu(D) = 0$ in the previous sense agrees with $\mu(D) = 0$ in the sense we just defined. We'll also show that $\mu(D) \geq 0$ whenever defined.

Definition:
Let $S \subseteq \mathbb{R}^n$ be any subset. The indicator function of S, denoted χ_S is the function $\chi_S : \mathbb{R}^n \to \mathbb{R}$ defined by $\chi_S(x) = 1$ if $x \in S$, $\chi_S(x) = 0$ if $x \notin S$.
So if D is bounded and sizeable, then $\mu(D) = \int_D \chi_D = \int_I \chi_D$.

Next, we characterize those bounded subsets $D \subseteq \mathbb{R}^n$ that are sizeable.

Theorem:
Let $D \subseteq \mathbb{R}^n$ be bounded. Then D is sizeable if and only if ∂D has size zero.
Proof:
In the forward direction: Suppose ∂D has size zero. Then $f \equiv 1$ on D is bounded and continuous, let $\widetilde{f} = \chi_D$ be the extension of f to a box I containing D. Then χ_D is continuous except on ∂D, and $\mu(\partial D) = 0$, so by the previous theorem, χ_D is integrable on I, hence χ_D is integrable over D, so D is sizeable.
Conversely, suppose D is sizeable. By contradiction, suppose $\mu(\partial D)$ does not have size zero. By the technical lemma from last week, there exists some $\varepsilon_o' > 0$ such that whenever $I_1, \ldots, I_n$ are finitely many boxes that cover ∂D, then $\sum_{\substack{j=1 \\ I_j \cap \partial D \neq \emptyset}}^N \mu(I_j) \geq \varepsilon_o'$.
Since D is sizeable, that means that χ_D is integrable over any box I containing D, and $\int_I \chi_D = \mu(D)$. Thus there exists a partition P of I such that $|S(\chi_D, \mathsf{P}) - \mu(D)| < \frac{\varepsilon_o'}{2}$ for any Riemann sum $S(\chi_D, \mathsf{P})$.
Let $I = \cup_\alpha I_\alpha$ be the decomposition of I with respect to P. Let's define two Riemann sums $S_1(\chi_D, \mathsf{P})$, $S_2(\chi_D, \mathsf{P})$.
For S_1, if $\text{Int}(I_\alpha) \cap \partial D = \emptyset$, let $x_\alpha \in I_\alpha$ be arbitrary. If $\text{Int}(I_\alpha) \cap \partial D \neq \emptyset$, let $x_\alpha \in D \cap I_\alpha$ (which exists by definition of the boundary, since $\text{Int}(I_\alpha)$ is open).
For S_2, if $\text{Int}(I_\alpha) \cap \partial D = \emptyset$, let $x_\alpha \in I_\alpha$ be arbitrary. If $\text{Int}(I_\alpha) \cap \partial D \neq \emptyset$, let $x_\alpha \in D^c \cap I_\alpha$ (with same justification as above).
Therefore,

$$|S_1(\chi_D, \mathsf{P}) - S_2(\chi_D, \mathsf{P})| = \sum_{\substack{\alpha \\ I_\alpha \cap \partial D \neq \emptyset}} \mu(I_\alpha) \geq \varepsilon_o'$$

But then by the Triangle Inequality,

$$|S_1(\chi_D, \mathsf{P}) - S_2(\chi_D, \mathsf{P})| \leq |S_1(\chi_D, \mathsf{P}) - \mu(D)| + |S_2(\chi_D, \mathsf{P}) - \mu(D)| < \frac{\varepsilon_o'}{2} + \frac{\varepsilon_o'}{2} = \varepsilon_o'$$

contradicting what we have above. $\blacksquare$

Properties of Riemann integration:
Let $D \subseteq \mathbb{R}^n$ be bounded.
(0) If $f : D \to \mathbb{R}$ is integrable on D, then f is bounded. Exercise: Assignment 11.
(1) Let $f, g : D \to \mathbb{R}$ both be integrable on D. Let $\lambda, \mu \in \mathbb{R}$. Then $\lambda f + \mu g : D \to \mathbb{R}$ is integrable on D, and $\int_D (\lambda f + \mu g) = \lambda \int_D f + \mu \int_D$. Exercise: A10Q3. This says integrable functions on D are a subspace and integration over D is a linear map from this subspace to $\mathbb{R}$.
(2) Let $f : D \to \mathbb{R}$ be integrable on D, and nonnegative. Then $\int_D f \geq 0$.
Proof:
Suppose not. Then $\int_D f < 0$. Let $\varepsilon = -\int_D f > 0$. There exists a partition P of a box I containing D such that $|S(f, \mathsf{P}) - \int_D f| < \frac{\varepsilon}{2}$.
But by definition of ε, this tells us that $-\frac{\varepsilon}{2} < S(f, \mathsf{P}) + \varepsilon < \frac{\varepsilon}{2}$, or in other words, $S(f, \mathsf{P}) < -\frac{\varepsilon}{2} < 0$.
But $f(x) \geq 0$ for all $x \in D$, therefore $S(f, \mathsf{P}) \geq 0$ for any Riemann sum, showing the contradiction. $\blacksquare$

Corollary:
$\mu(D) \geq 0$ if D is sizeable. $\blacksquare$

(3) Let $f : D \to \mathbb{R}$ be integrable on D (so f is bounded by (0)), and suppose D is sizeable. There exist $m, M \in \mathbb{R}$ such that $m \leq f(x) \leq M$ for all $x \in D$. Then, $m\mu(D) \leq \int_D f \leq M\mu(D)$.
Proof:
Because $m \leq f(x) \leq M$, the function $g : D \to \mathbb{R}$ given by $g(x) = M - f(x)$ satisfies $g(x) \geq 0$ for all x, and, is integrable on D by property (1) since it is a linear combination of 1 (multipled by M) and f, and 1 is integrable on D since D is sizeable. So by (2) and (1),

$$0 \leq \int_D g = \int_D (M \cdot 1 - f) = M \int_D 1 - \int_D f = M\mu(D) - \int_D f$$

implying $\int_D f \leq M\mu(D)$. The other inequality is the same proof with $h(x) = f(x) - m \geq 0$. $\blacksquare$

Corollary
Suppose D has size zero and $f : D \to \mathbb{R}$ is bounded. Then $f : D \to \mathbb{R}$ is integrable over D, and $\int_D f = 0$ (Exercise, A11). $\blacksquare$

The second corollary of these properties so far is:

Theorem (Mean Value Theorem for Integration):
Let $D \subseteq \mathbb{R}^n$ be compact, connected, and sizeable. Let $f : D \to \mathbb{R}$ be continuous. Then, f is integrable on D and there exists $x_o \in D$ such that $\int_D f = f(x_o)\mu(D)$.
Remark:
If D is sizeable with $\mu(D) > 0$ and f is integrable on D, we define the average (or mean) value of f over D to be $\frac{1}{\mu(D)} \int_D f$. This theorem says such a mean always exists given the hypotheses.
Proof:
Since D is sizeable, $\mu(\partial D) = 0$. Since f is continuous on D, and $\mu(\partial D) = 0$, f is integrable on D. Since D is compact, and f is continuous, by the Extreme Value Theorem, f attains a global minimum and maximum on D. Therefore, there exist $x_1, x_2 \in D$ such that $m = f(x_1) \leq f(x) \leq f(x_2) = M$ for all $x \in D$.
Case 1: Suppose $\mu(D) = 0$. By (3), $\underbrace{m\mu(D)}_{0} \leq \int_D f \leq \underbrace{M\mu(D)}_{0}$, implying $\int_D f = 0$, so our desired equality holds since $\int_D f = 0 = \mu(D)f(x_o)$ for all $x_o \in D$ (both sides vanish).
Case 2: Suppose $\mu(D) > 0$. The inequality used in case 1 says $f(x_1) = m \leq \frac{1}{\mu(D)} \int_D f \leq M = f(x_2)$, so by the Intermediate Value Theorem (since f is continuous and real-valued, with D connected), there

exists $x_o \in D$ such that $f(x_o) = \frac{1}{\mu(D)} \int_D f$. ■

(4) Let $D_1, D_2 \subseteq \mathbb{R}^n$ be bounded with $\mu(D_1 \cap D_2) = 0$. Suppose $f : D_1 \cup D_2 \to \mathbb{R}$ is integrable over D_1 and integrable over D_2. Then, f is integrable over their union, and $\int_{D_1 \cup D_2} f = \int_{D_1} f + \int_{D_2} f$.
Proof:
Let I be a box containing $D_1 \cup D_2$. f being integrable over D_1 says that $f\chi_{D_1}$ is integrable over I and $\int_{D_1} f = \int_I f\chi_{D_1}$. But this is also equal to $\int_{D_1 \cup D_2} f\chi_{D_1}$, since $f\chi_{D_1} \equiv 0$ outside of $D_1 \cup D_2$.
Exactly the same way, f being integrable on D_2 tells us that $f\chi_{D_2}$ is integrable on $D_1 \cup D_2$ and $\int_{D_1 \cup D_2} f\chi_{D_2} = \int_{D_2} f$.
Moreover, since f is integrable on D_1, D_2, we have that f is bounded on D_1, D_2, so f is bounded on $D_1 \cup D_2$. Also, $\mu(D_1 \cap D_2) = 0$, and our above argument also shows f is bounded on $D_1 \cap D_2$. So by the corollary above, f is integrable on $D_1 \cap D_2$, and $\int_{D_1 \cap D_2} f = 0$.
Therefore, $0 = \int_{D_1 \cap D_2} f = \int_{D_1 \cap D_2} f\chi_{D_1 \cap D_2} = \int_{D_1 \cup D_2} f\chi_{D_1 \cap D_2}$, where the last step follows similarly to what we did at the start.
Finally, we know that $f = f\chi_{D_1} + f\chi_{D_2} - f\chi_{D_1 \cap D_2}$ on $D_1 \cup D_2$ (this identity just holds in general). Therefore, by (1), f is integrable on $D_1 \cup D_2$ and $\int_{D_1 \cup D_2} f = \int_{D_1 \cup D_2} f\chi_{D_1} + \int_{D_1 \cup D_2} f\chi_{D_2} - \int_{D_1 \cup D_2} \chi_{D_1 \cap D_2} = \int_{D_1} f + \int_{D_2} f - 0$. ■

(5) Let $D \subseteq \mathbb{R}^n$ be bounded. Let $f : D \to \mathbb{R}$ be integrable on D. Define $|f| : D \to \mathbb{R}$ to be $|f(x)|$ for all $x \in D$. Then f is integrable on D, and $|\int_D f| \leq \int_D |f|$.
Proof:
We'll use the simplified Cauchy criterion for integrability. Since f is integrable over D, there exists a partition $\mathcal{P}_\varepsilon$ of a box I such that $|S_1(\chi_D f, \mathcal{P}_\varepsilon) - S_2(\chi_D f, \mathcal{P}_\varepsilon)| < \varepsilon$ for any two Riemann sums of f with respect to $\mathcal{P}_\varepsilon$.
Let $I = \cup_\alpha I_\alpha$ with respect to $\mathcal{P}_\varepsilon$. Choose x_α, $\mu_\alpha \in I_\alpha$ arbitrarily to get Riemann sums for $|f|$:

$$S_1(|f|, \mathcal{P}_\varepsilon) = \sum_\alpha |f(x_\alpha)|\mu(I_\alpha)$$

$$S_2(|f|, \mathcal{P}_\varepsilon) = \sum_\alpha |f(y_\alpha)|\mu(I_\alpha)$$

We'll show $|S_1(|f|, \mathcal{P}_\varepsilon) - S_2(|f|, \mathcal{P}_\varepsilon)| < \varepsilon$.
For each α, choose z_α, $w_\alpha \in \{x_\alpha, y_\alpha\}$ such that $f(z_\alpha) = \max\{f(x_\alpha), f(y_\alpha)\}$, and $f(w_\alpha) = \min\{f(x_\alpha), f(y_\alpha)\}$.
Then, $|f(x_\alpha) - f(y_\alpha)| = f(z_\alpha) - f(w_\alpha)$.
Therefore,

$$\sum_\alpha |f(x_\alpha) - f(y_\alpha)|\mu(I_\alpha) = \sum_\alpha f(z_\alpha)\mu(I_\alpha) - \sum_\alpha f(w_\alpha)\mu(I_\alpha) < \varepsilon$$

where the inequality follows by the integrability of f (Cauchy criterion).
Hence, $|S_1(|f|, \mathcal{P}_\varepsilon) - S_2(|f|, \mathcal{P}_\varepsilon)| = |\sum_\alpha |f(x_\alpha)|\mu(I_\alpha) - \sum_\alpha |f(y_\alpha)|\mu(I_\alpha)| \leq \sum_\alpha |f(x_\alpha) - f(y_\alpha)|\mu(I_\alpha)$ (by the reverse triangle inequality), but this is just the inequality above, so it is less than ε. So $|f|$ is integrable over D.
Now let $\varepsilon > 0$. Choose a partition $\mathcal{P}_\varepsilon$ of I (using integrability of both f and $|f|$ and taking a common refinement) such that $|\sum_\alpha f(x_\alpha)\mu(I_\alpha) - \int_I f| < \frac{\varepsilon}{2}$ and $|\sum_\alpha |f(x_\alpha)|\mu(I_\alpha) - \int_I |f|| < \frac{\varepsilon}{2}$.
Hence, $|f| \leq |\sum_\alpha f(x_\alpha)\mu(I_\alpha)| + \frac{\varepsilon}{2}$, so by the triangle inequality, $|f| \leq \sum_\alpha |f(x_\alpha)|\mu(I_\alpha) + \frac{\varepsilon}{2} < \int_I |f| + \frac{\varepsilon}{2} + \frac{\varepsilon}{2}$ (the last inequality follows by the second one in the above paragraph), so $|\int_I f| \leq \int_I |f| + \varepsilon$ for any $\varepsilon > 0$, so $|\int_I f| \leq \int_I |f|$. ■

Fubini's Theorem

Fubini's Theorem gives a method for computing integrals in higher dimensions.

Theorem (Fubini):
Let $I \subseteq \mathbb{R}^n$ and $J \subseteq \mathbb{R}^m$ be boxes, and let $f : \underbrace{I \times J}_{\text{box in } \mathbb{R}^n \times \mathbb{R}^m \cong \mathbb{R}^{m+n}} \to \mathbb{R}$ be a function. Suppose that
(1) f is Riemann integrable on $I \times J$ and
(2) for all $x \in I$, the function $y \mapsto f(x, y)$ on J is integrable over J (i.e. $F(x) = \int_J f(x, \cdot)$ exists for all $x \in I$, $f(x, \cdot) : J \to \mathbb{R}$).
Then, $F : I \to \mathbb{R}$ is integrable over I and $\int_I F = \int_{I\times J} f$.
Remark:
Fubini extends by finite induction to any finite Cartesian product of boxes.
Remark:
This tells us that to compute a multidimensional integral,s first integrate over all $y \in J$ (this gives a function of $x \in I$), then integrate over all $x \in I$.

We'll prove this later, for now:
Corollary:
Let $f : [a, b] \times [c, d] \to \mathbb{R}$. Suppose $\int_a^b f(x, y)dx$ exists for all $y \in [c, d]$, and $\int_{[a,b]\times[c,d]} f$ exists. Then, $\int_{[a,b]\times[c,d]} f = \int_c^d \int_a^b f(x, y)dxdy$. In particular if it's also true that $\int_c^d f(x, y)dy$ exists for all $x \in [a, b]$, then $\int_{[a,b]\times[c,d]} f = \int_a^b \int_c^d f(x, y)dydx$. So we can change the order of integration if the single-variable integrals exist. So in particular, if f is continuous, we can change the order of integration. Also, in the same way, if $f : [a_1, b_1] \times [a_2, b_2] \times [a_3, b_3] \to \mathbb{R}$, and $f(x_1, x_2, x_3)$ is integrable on $[a_i, b_i]$ if we fix i and hold constant $j, k \neq i$, and f is integrable over I, then we can iterate the integral in an arbitrary order. ■

Example:
Choose $D = [0, 1] \times [0, 1]$, take $f(x, y) = y^3 e^{xy^2}$. f is integrable on D, and $f(x, \cdot)$ and $f(\cdot, y)$ are integrable on $[0, 1]$, so Fubini applies, so $\int_D f = \int_0^1 \int_0^1 \overbrace{y^3 e^{xy^2}}^{\text{can't find antiderivative w.r.t. to y}} dydx = \int_0^1 \int_0^1 y^3 e^{xy^2} dxdy$. Integrating, this equals $\int_0^1 ye^{xy^2}|_{x=0}^{x=1} dy = \int_0^1 (ye^{y^2} - y)dy = (\frac{e^{y^2}}{2} - \frac{y^2}{2})|_{y=0}^{y=1} = \frac{e}{2} - 1$.

Corollary:
Suppose $\phi, \psi : [a, b] \to \mathbb{R}$ are continuous with $\psi(x) \geq \phi(x)$ for all $x \in [a, b]$. Let $D = \{(x, y) \in \mathbb{R}^2 : a \leq x \leq b,\ \phi(x) \leq y \leq \psi(x)\}$. Let $f : D \to \mathbb{R}$ be bounded, and suppose the set of discontinuities of f on D, D_o, has size zero, and suppose $E_x = \{y : \phi(x) \leq y \leq \psi(x), (x, y) \in D_o\}$ also has size zero. Then, f is integrable over D, and $\int_D f = \int_a^b \int_{\phi(x)}^{\psi(x)} f(x, y)dydx$.
Proof:
Choose $c, d \in \mathbb{R}$ such that $c \leq \phi(x) \leq \psi(x) \leq d$ for all $x \in [a, b]$. Extend f to $\widetilde{f} : [a, b] \times [c, d] \to \mathbb{R}$ as the extension by zero of f from D to $[a, b] \times [c, d]$. Since f is continuous except on D_o, $\widetilde{f}$ is continuous everywhere in $[a, b] \times [c, d]$ except on $D_o \cup \partial D$. This has size zero, since D_o has size zero by hypothesis, and ∂D is the finite union of graphs of continuous functions on bounded domains, and thus $\widetilde{f}$ is integrable over $[a, b] \times [c, d]$, and so f is integrable over D. By Fubini, $\int_D f = \int_{[a,b]\times[c,d]} \widetilde{f} = \int_a^b \int_c^d \widetilde{f}(x, y)dydx$. Note that this also requires our assumption about E_x to know that the inner iterated integral with respect to y exists (since E_x is size zero). But this equals $\int_a^b \int_{\phi(x)}^{\psi(x)} f(x, y)dydx$. ■
Remark:
This also works with horizontal versus vertical lines.

Example:
$D = \{(x, y) \in \mathbb{R}^2 : 1 \leq x \leq 3, x^2 \leq y \leq x^2 + 1$. D is sizeable, since $\mu(\partial D) = 0$ since its boundary is clearly the union of graphs of continuous functions on compact sets. We then have by the previous corollary,

$$\mu(D) = \int_D = \int_1^3 \int_{x^2}^{1+x^2} 1dydx = \int_1^3 y|_{y=x^2}^{y=x^2+1} dx = \int_1^3 1dx = 2$$

Example:
Let $D \subseteq \mathbb{R}^3$ be the set bounded by the planes $x = 0$, $y = 0$, $z = 0$, $x + y + z = 1$. Let $f(x, y, z) = y$. We wish to find $\int_D f$. (Picture of domain, which is a region above xy-plane, in front of yz-plane, to the right of xz-plane, and cut off by triangle formed by fourth plane's intersection).
In the xy-plane, we get the triangle in the first quadrant with $0 \le x \le 1$, $0 \le y \le 1-x$. Solving for z, we then get $0 \le z \le 1-x-y$. This function meets all hypotheses of continuity/size zero boundary, so by Fubini's theorem, $\int_D f = \int_0^1 \int_0^{1-x} \int_0^{1-x-y} y dz dy dx = \int_0^1 \int_0^{1-x} yz|_0^{1-x} dy dx = \int_0^1 \int_0^{1-x} ((1-x)y - y^2) dy dx = \int_0^1 \frac{((1-x)y^2}{2} - \frac{y^3}{3})|_{y=0}^{y=1-x} dx = \int_0^1 (\frac{(1-x)^3}{2} - \frac{(1-x)^3}{3}) dx = \int_0^1 \frac{(1-x)^3}{6} dx = -\frac{(1-x)^4}{24}|_{x=0}^1 = \frac{1}{24}$.

Proof of Fubini's Theorem:
Let $F(x) = \int_J f(x, \cdot)$.
Let $\varepsilon > 0$. We need to show that there exists a partition of I such that for any Riemann sum of F with respect to any refinement of this partition, it is within ε of $\int_{I\times J} f$.
Since f is integrable over $I \times J$, there exists a partition $\mathcal{P}_\varepsilon$ of $I \times J$ such that

$$|S(f, \mathcal{P}) - \int_{I\times J} f| < \frac{\varepsilon}{2} \quad (1)$$

for any refinement $\mathcal{P}$ of $\mathcal{P}_\varepsilon$ and any Riemann sum $S(f, \mathcal{P})$ of f with respect to $\mathcal{P}$.
Split $\mathcal{P}_\varepsilon$ into $\mathcal{P}_\varepsilon = \mathcal{Q}_\varepsilon \times \mathcal{R}_\varepsilon$, where $\mathcal{Q}_\varepsilon$ is a partition of I, and $\mathcal{R}_\varepsilon$ is a partition of J.
Let $\mathcal{Q}$ be any refinement of $\mathcal{Q}_\varepsilon$, and let $S(F, \mathcal{Q})$ be a Riemann sum of F over $\mathcal{Q}$. We want to show $|S(F, \mathcal{Q}) - \int_{I\times J} f| < \varepsilon$. Let $\cup_\alpha I_\alpha$ be an arbitrary decomposition of I with respect to $\mathcal{Q}$. For all α, pick $x_\alpha \in I_\alpha$ for this Riemann sum $S(F, \mathcal{Q})$.
Since $f(x_\alpha, \cdot) : J \to \mathbb{R}$ is integrable over J, there exists a partition J_α of J such that for any refinement $\mathcal{T}$ of J_α with corresponding subdivision $J = \cup_\beta J_\beta$,

$$\left| \sum_\beta f(x_\alpha, y_\beta)\mu(J_\beta) - F(x_\alpha) \right| < \frac{\varepsilon}{2\mu(I)} \quad (2)$$

where this is the Riemann sum of $f(x_\alpha, \cdot) : J \to \mathbb{R}$ with respect to $\mathcal{T}$ of J and $y_\beta \in J_\beta$, and the right is $\int_J f(x_\alpha, \cdot)$.
Let J_ε be a common refinement of R_ε and J_α for all α. Then $Q_\varepsilon \times J_\varepsilon$ is a refinement of P_ε. Let $\cup_{\alpha,\beta} I_\alpha \times J_\beta = I \times J$ be the arbitrary decomposition of $I \times J$ with respect to this partition $Q_\varepsilon \times J_\varepsilon$.
Let $x_\alpha \in I_\alpha$, $y_\beta \in J_\beta$. So, by (1),

$$\left| \sum_{\alpha,\beta} f(x_\alpha, y_\beta)\mu(I_\alpha)\mu(J_\beta) - \int_{I\times J} f \right| < \frac{\varepsilon}{2} \quad (1')$$

since $\mathcal{Q} \times J_\varepsilon$ is a refinement of $\mathcal{Q}_\varepsilon \times \mathcal{R}_\varepsilon = \mathcal{P}_\varepsilon$, and $\mu(I_\alpha)\mu(J_\beta) = \mu(I_\alpha \times J_\beta)$.
Then

$$|\underbrace{\sum_\alpha \mu(I_\alpha)}_{S(F,\mathcal{Q})} - \int_{I\times J} f| \le |\sum_\alpha F(x_\alpha)\mu(I_\alpha) - \sum_{\alpha,\beta} f(x_\alpha, y_\beta)\mu(I_\alpha)\mu(J_\beta)| + |\sum_{\alpha,\beta} \mu(I_\alpha)\mu(J_\beta) - \int_{I\times J} f|$$

$$\underbrace{<}_{(1')} |\sum_\alpha (F(x_\alpha) - \sum_\beta f(x_\alpha, y_\beta)\mu(J_\beta))\mu(I_\alpha)| + \frac{\varepsilon}{2}$$

$$\le \sum_\alpha |F(x_\alpha) - \sum_\beta f(x_\alpha, y_\beta)\mu(J_\beta)|\mu(I_\alpha) + \frac{\varepsilon}{2}$$

$$\underbrace{<}_{(2)}\sum_\alpha\left(\frac{\varepsilon}{2\mu(I)}\right)\mu(I_\alpha)+\frac{\varepsilon}{2}=\frac{\varepsilon}{2}+\frac{\varepsilon}{2}=\varepsilon$$

So we found a partition $\mathcal{Q}_\varepsilon$ of I such that for any refinement $\mathcal{Q}$ of $\mathcal{Q}_\varepsilon$ and any Riemann sum $S(F,\mathcal{Q})$ of F with respect to $\mathcal{Q}$, we have $|S(F,\mathcal{Q})-\int_{I\times J}f|<\varepsilon$. ■

Example:
Find the values of the region D lying inside the elliptical cylinder $x^2+4y^2=4$ above the xy plane, and below the plane $z=2+x$.
We have $\text{vol}(D)=\mu(D)=\int_D 1$. D is clearly sizeable (its boundary is the union of graphs of continuous functions over compact sets).
On the ellipse, we have $-2\le x\le 2$, and $-\sqrt{1-\frac{x^2}{4}}\le y\le\sqrt{1-\frac{x^2}{4}}$, and $0\le z\le 2+x$.
Thus,

$$\mu(D)=\int_{-2}^{2}\int_{-\sqrt{1-\frac{x^2}{4}}}^{\sqrt{1-\frac{x^2}{4}}}\int_0^{2+x}1dzdydx=\int_{-2}^{2}\int_{-\sqrt{1-\frac{x^2}{4}}}^{\sqrt{1-\frac{x^2}{4}}}(2+x)dydx$$
$$=\int_{-2}^{2}(2+x)2\sqrt{1-\frac{x^2}{4}}dx=8\int_0^2\sqrt{1-\frac{x^2}{4}}dx=4\pi$$

Example:
$\int_0^1\int_z^1\int_0^x e^{x^3}dydxdz=\int_0^1\int_z^1 xe^{x^3}dxdz$, which is not particularly easy (read: impossible). But let's draw the region: (draw lines $x=z$, $x=1$, $x=0$, it's a triangle in the first quadrant). We have $0\le z\le 1$, $z\le x\le 1$, which change-of-variable'd is $0\le x\le 1$, $0\le z\le x$. So our integral equals $\int_0^1\int_0^x xe^{x^3}dzdx=\int_0^1 x^2e^{x^3}dx=\frac{e-1}{3}$.

Example:
Rewriting a triple integral over a region in $\mathbb{R}^3$ in other orders of integration: $\int_0^1\int_0^{\sqrt{1-y^2}}\int_{y^2+z^2}^1 h(x,y,z)dxdzdy$.
We have $0\le y\le 1$, $0\le z\le\sqrt{1-y^2}$, $y^2+z^2\le x\le 1$. Let's try to rewrite this the 5 other ways.
Draw in the yz-plane, $0\le z\le\sqrt{1-y^2}$ and $0\le y\le 1$ (quarter of a circle in the quadrant). Then, add the paraboloid $x=y^2+z^2$ and bound it by the plane $x=1$ above it. So overall, our region is the quarter paraboloid (caused by the quarter of a circle in the quadrant of y and z) bounded at the top by the plane $x=1$.
This gives us the reorderings: First we can just reconsider the quarter circle in the yz-plane as $0\le z\le 1$, $0\le y\le\sqrt{1-z^2}$ instead, x staying the same, giving $\int_0^1\int_0^{\sqrt{1-z^2}}\int_{y^2+z^2}^1 h(x,y,z)dxdydz$.
Draw the half-parabola in the xy plane, the region bounded by $0\le x\le 1$, $0\le y\le\sqrt{x}$, and then using the paraboloid equation we get $0\le z\le\sqrt{x-y^2}$. This gives $\int_0^1\int_0^{\sqrt{x}}\int_0^{\sqrt{x-y^2}}dzdydx$.
We can do this same breakdown as $0\le y\le 1$, $y^2\le x\le 1$, $0\le z\le\sqrt{x-y^2}$, giving $\int_0^1\int_{y^2}^1\int_0^{\sqrt{x-y^2}}h(x,y,z)dzdxdy$.
There are 2 other orders, $dydzdx$ and $dydxdz$ (exercise).

Change of Variables Theorem

Theorem (Change of Variables):
Let $U\subseteq\mathbb{R}^n$ be nonempty and open. Let $K\subseteq U$ be nonempty, compact, and sizeable, and let $h:U\to\mathbb{R}^n$ be in $C^1(U)$ and suppose there exists $D\subseteq K$ with $\mu(D)=0$ such that $h|_{K\setminus D}$ is injective and $\det(Dh)_x\ne 0$ for all $x\in K\setminus D$.
Then:
(1) $h(K)$ is sizeable (it is compact since K is and h is continuous)
(2) $\int_{h(K)}f=\int_K(f\circ h)|\det(Dh)|$ for any continuous function $f:h(U)\to\mathbb{R}$.
Sketch:
$f\circ h$ is continuous on K since f and h are, and $|\det(Dh)|$ is, since the absolute value is, and the

determinant of continuous functions is a polynomial in the entries of the matrix, which are continuous since $h \in C^1$. Intuitively, what this is saying is

$$\int_{h(K)} f \text{``}=\text{''} \lim \sum_{\alpha} f(h(x))\Delta \underbrace{h_1(x_1)\dots \Delta h_n(x_n)}_{\mu(I_\alpha)}$$

and

$$\int_K f \circ h \text{``}=\text{''} \lim \sum_{\alpha} f(x)\Delta x_1 \dots \Delta x_n$$

we have

$$\Delta h_1(x_1)\dots \Delta h_n(x_n) \approx |\det(\frac{\partial h_i}{x_j}|\Delta x_1 \dots \Delta x_n|$$

This is just the change in volume of the determinant, which is exactly true for linear functions, and given that h is nice enough on a nice enough set (D being the only set where it is possibly bad, but size zero so it doesn't matter too much), this approximation is good enough for the limit of Riemann sums to work.

Cylindrical Coordinates

Cylindrical coordinates on $\mathbb{R}^3$, with $r \geq 0$, $0 \leq \theta \leq 2\pi$, and $-\infty < z < \infty$. Then, $x = r\cos\theta$, $y = r\sin\theta$, $z = z$. Consider

$$h : \overbrace{[0, 2\pi] \times [0 \times \infty) \times (-\infty, -\infty)}^{\subseteq \mathbb{R}^3_{\theta,r,z}} \to \mathbb{R}^3$$

given by $h(\theta, r, z) = (r\cos\theta, r\sin\theta, z)$. $h \in C^\infty$ clearly, and only has problematic points at $h(0, r, z) = h(2\pi, r, z)$, and θ not being well-defined at $r = 0$ (the z-axis). These sets where h is not defined or not injective have size zero when intersected with any compact set we actually want to integrate over. In particular, $h \in C^1$ and injective on any compact set $K \subseteq [0, 2\pi] \times [0, \infty) \times \mathbb{R}$ except on a set of size zero. What is $|\det(Dh)|$?

$$Dh = \begin{bmatrix} \cos\theta & -r\sin\theta & 0 \\ \sin\theta & r\cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

so $\det(Dh) = r$, so $|\det(Dh)| = r > 0$ outside of a set of size zero when intersected with a compact set. This says that $\Delta x\Delta y\Delta z \text{``}=\text{''} r\Delta r\Delta\theta\Delta z$ (consider the standard picture drawn of cylindrical coordinates). So if we want to integrate in cylindrical coordinates, we replace the "$dxdydz$" by "$rdrd\theta dz$", where r is our volume stretching factor.
When to use cylindrical coordinates? If either the region of integration $D \subseteq \mathbb{R}^3$ is simpler to describe in cylindrical coordinates, and/or if the function f is much simpler in cylindrical coordinates. For example, if $x^2 + y^2$ is simpler with f, then its simpler with $r^2 = x^2 + y^2$.

Example:
Compute $\int_D f$ where $f(x, y, z) = x^2 + y^2$, and $D \subseteq \mathbb{R}^3$ is the region bounded by $x^2 + y^2 = 1$, $x^2 + y^2 = 4$, $z = 0$, $z = 1$, $y = 0$, $y = x$ in the first octant. In cylindrical coordinates, this region is $0 \leq z \leq 1$, $0 \leq \theta \leq \frac{\pi}{4}$, $1 \leq r \leq 2$, also $f(x, y, z) = r^2$, so

$$\int_D f = \int_0^{\frac{\pi}{4}} \int_0^1 r^2 rdzdrd\theta = \frac{15\pi}{16}$$

Example:
Find the volume of the region D above the paraboloid $z = x^2+y^2$, and inside the sphere $x^2+y^2+z^2 = 12$. We know $z = r^2$ and $r^2 + z^2 = 12$ by the constraints, so $z + z^2 = 12$, implying $(z - 3)(z + 4) = 0$ so $z = 3$ or $z = -4$, the last is extraneous since $z = r^2 \geq 0$, so we must have $z = 3$, so $r = \sqrt{3}$. Thus, our region is from $0 \leq \theta \leq 2\pi$, $0 \leq r \leq \sqrt{3}$, $r^2 \leq z \leq \sqrt{12 - r^2}$, so

$$\mu(D) = \int_0^{2\pi} \int_0^{\sqrt{3}} \int_{r^2}^{\sqrt{12-r^2}} rdzdrd\theta = 2\pi(\frac{1}{3}((12)^{\frac{3}{2}} - (9)^{\frac{3}{2}}) - \frac{9}{4})$$

Spherical Coordinates

"Imagine the Earth is the center of the universe like they used to believe ... Actually if you go back far enough, they knew the Earth was round, but then people got stupid and believed it was flat, then got smart again".

Let ρ be the distance to the origin. Drawing the right-triangle for the distance, we get that $\rho^2 = x^2+y^2+z^2$, $x = r\cos\theta = \rho\sin\phi\cos\theta$, $y = r\sin\theta = \rho\sin\phi\sin\theta$, $z = \rho\cos\phi$, where θ is the longitude (the same angle measured in cylindrical coordinates), going from 0 to 2π, and where ϕ is the latitude (ranging from 0 to π), the angle between the z-axis and our point, increasing from the positive z-axis.

The map F given by $(\rho,\phi,\theta) \mapsto (\rho\sin\phi\cos\theta, \rho\sin\phi\sin\theta, \rho\cos\phi)$ is C^∞ everywhere but not a bijection, because θ is not defined when $x^2+y^2 = r^2 = 0$, and θ and ϕ if $\rho = 0$, also, $F(\rho,\phi,0) = F(\rho,\phi,2\pi)$ (it's periodic). The bad parts of the domain are graphs of continuous functions, hence if $K \subseteq [0,\infty)\times[0,\pi]\times[0,2\pi]$ is a compact set, then F injective on $K\setminus D$ where D is a set of size zero. We need to compute $dxdydz = |\det(DF)|d\rho d\phi d\theta$, the "volume stretching factor". Consider the standard picture of spherical coordinates.

Explicitly,

$$DF = \begin{bmatrix} \sin\phi\cos\theta & \rho\cos\phi\cos\theta & -\rho\sin\phi\sin\theta \\ \sin\phi\sin\theta & \rho\sin\phi\cos\theta & \\ \cos\phi & -\rho\sin\phi & 0 \end{bmatrix}$$

Expanding on the bottom row, we have $\det DF = \cos\phi(\rho^2\sin\phi\cos\phi\cos^2\theta + \rho^2\sin\phi\cos\phi\sin^2\theta) + \rho\sin\phi(\rho\sin^2\phi\cos^2\theta + \rho\sin^2\phi\sin^2\theta) = \rho^2\sin\phi$ after doing some Pythagorean factoring. But based on our choice of ϕ, sin is positive, so this absolute value is $|\det DF| = \rho^2\sin\phi$.

You should use spherical coordinates if either the region of integration is easier to describe in spherical coordinates, or the function we are integrating is simpler in spherical coordinates. For example, if $x^2+y^2+z^2$ appears, this is just ρ^2. Similarly, spheres are easy to describe (they are the case when ρ is constant). Also, cones are easy to describe since they are the case when ϕ is a constant.

Example:

Compute $\int_D g$ where $g(x,y,z) = 1-\sqrt{x^2+y^2+z^2}$ and D is the region above the cone $z = \frac{1}{\sqrt{3}}\sqrt{x^2+y^2}$ and inside the sphere $x^2+y^2+z^2 = 1$. Solving for ϕ from the cone, we have $z = \frac{1}{\sqrt{3}}r \implies \frac{r}{z} = \sqrt{3} \implies \tan\phi = \sqrt{3} \implies \phi = \frac{\pi}{3}$. Thus, we have $0 \le \phi \le \frac{\pi}{3}$, $0 \le \theta \le 2\pi$, and $0 \le \rho \le 1$, giving

$$\int_D g = \int_0^{2\pi}\int_0^{\frac{\pi}{3}}\int_0^1 (1-\rho)\rho^2\sin\phi d\rho d\phi d\theta = 2\pi\int_0^{\frac{\pi}{3}}\int_0^1(\rho^2-\rho^3)\sin\phi d\rho d\phi = 2\pi\int_0^{\frac{\pi}{3}}[\frac{\rho^3}{3}-\frac{\rho^4}{4}]_0^1\sin\phi d\phi$$

$=$

$$\frac{2\pi}{12}\int_0^{\frac{\pi}{3}}\sin\phi d\phi = \frac{2\pi}{12}(-\cos\phi)_0^{\frac{\pi}{3}}$$

Example:

Compute $\int_D (x^2+y^2+z^2)^{\frac{3}{2}}$ where D lies between the spheres $x^2+y^2+z^2 = 1$ and $x^2+y^2+z^2 = 9$ in the region $z \le 0, x \ge 0, y \ge 0$. We are outside the sphere of radius 1, and inside the sphere of radius 3, so we will have $1 \le \rho \le 3$, our bounds on x, y, z tell us which octant we are in, which says $0 \le \theta \le \frac{\pi}{2}$ and $\frac{\pi}{2} \le \phi \le \pi$. Thus, our integral equals

$$\int_0^{\frac{\pi}{2}}\int_{\frac{\pi}{2}}^{\pi}\int_1^3 \rho^3\rho^2\sin\phi\rho d\rho d\phi d\theta = \frac{\pi}{2}[-\cos\rho]_{\frac{\pi}{2}}^{\pi}\int_1^3\rho^5 d\rho = \frac{\pi}{2}[\frac{\rho^6}{6}]_1^3 = \frac{\pi}{12}(3^6-1)$$